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Differential Equation

Differentiating Extra Ordinary Situation!

(Section-01)

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ORDINARY DIFFERENTIAL EQUATION

SECTION-01

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CHAPTER 1

INTRODUCTION AND PRELIMINARIES

1.1 Variables

We live in a world of interrelated changing entities like, the position $p(t)$ of earth changes with time t , the velocity of a falling body $v(d)$ changes with distance d , the area of circle $A(r)$ changes with the size of radius r . In the language of mathematic changing entities are called variable

1.2 Example

Position of earth $p(t)$, time t , velocity of falling body $v(d)$, distance d , Area of circle $A(r)$, radius of circle r , all are variable's

1.3 Independent and Dependent Variable

The variables to which the value is assigned is called independent variables and the variable whose value is obtained corresponding to the assigned value is called dependent variable.

If $f: A \rightarrow B$ be a function, then for each assigned value $x \in A \exists$ unique $y \in B$ s.t. $y = f(x)$. The variable x to which value is assigned is called independent variable and the variable y which takes value corresponding to assigned value of x is called dependent variable.

(Independent variable) causes a change in **(Dependent variable)** and it isn't possible that **(Dependent variable)** could cause a change in **(Independent variable)**.

Example: (Time Spent Studying) causes a change in (Test Score) and it isn't possible that (Test Score) could cause a change in (Time Spent Studying).

We see that "Time Spent Studying" must be the independent variable and "Test Score" must be the dependent variable because the sentence doesn't make sense the other way around.

1.4 Derivatives

The concept of derivative is at the core of Calculus and modern mathematics. The definition of the derivative can be approached in two different ways. One is geometrical (as a slope of a curve) and the other one is physical (as a rate of change).

The Physical Concept of the Derivative: This approach was used by Newton in the development of his Classical Mechanics. The rate of change of one variable with respect another variable is called derivative.

The Geometrical Concept of the Derivative: Consider a function $y = f(x)$ and its graph of one variable x then the derivative of y at a point $p(x, y)$ is the slope of tangent to the curve $y = f(x)$ at $p(x, y)$ and denoted by $\frac{dy}{dx}$ and called total derivative.

Partial Derivatives: If $z = z(x, t)$, where x, t are independent variable. Then $z = z(x, t)$ denotes the surface in 3-D and for any point $P(x, t, z(x, t))$, the slope of tangent in x -direction is defined by $\frac{\partial z}{\partial x}$ and slope of tangent in t -direction is defined by $\frac{\partial z}{\partial t}$.

1.5 Differential Equation

The equation which expresses the relationship between dependent variables, independent variables and the derivatives of dependent variables with respect to independent variables is called differential Equation.

1.6 Classification of Differential Equation

When you study differential equations, you learn to look at an equation and classify it into a certain group. The reason is that the techniques for solving differential equations are common to these various classification groups. And sometimes you can transform an equation of one type into an equivalent equation of another type, so that you can use easier solution techniques. Here there are some of the major classifications of differential equations:

Ordinary Differential Equation: A differential equation which expresses relationship between one or more dependent variable which are function of a single independent variable and the total derivatives of dependent variables with respect to independent variable, i.e., the differential equation involving only total derivatives. That is “**An ordinary differential equation (ODE) contains differentials with respect to only one variable**”

Partial Differential Equation: A differential equation which contains the derivatives of one or more dependent variables with respect to more than one independent variables i.e., the differential equation involving partial derivatives. That is “**Partial differential equations (PDE) contain differentials with respect to several independent variables**”.

Classification of Ordinary Differential Equations:

- I. **Simple/ordinary differential Equations:** An ordinary differential equation (ODE) which contains one dependent variable and derivatives of dependent variable with respect to independent variable.

Example: Let $y = y(x)$, x is independent variables then

$$\frac{d^3 y}{dx^3} + \sin x \frac{dy}{dx} = \cos y \text{ is a simple / ordinary differential equation.}$$

- II. **System of Differential Equations:** A differential equation which contains one independent variable, more than one dependent variables and derivatives of dependent variables with respect to independent variables.

Example: let $y = y(x)$ and $z = z(x)$ are two dependent variables of a single independent variable x then $\frac{dz}{dx} + \frac{dy}{dx} = \sin x$ and $\frac{dy}{dx} + x \frac{dz}{dx} = \cos x$ together is a system of the differential equations.

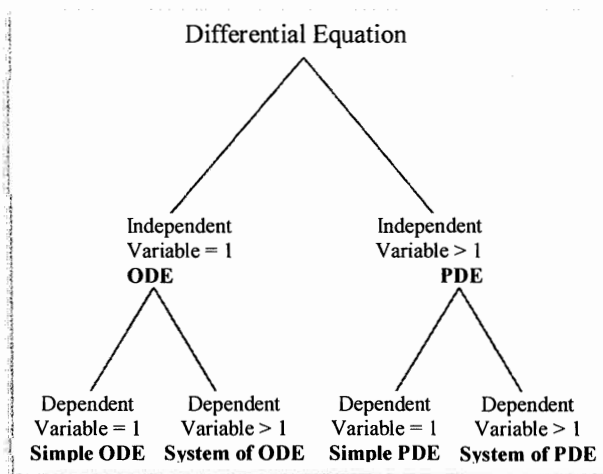
Partial Differential Equations:

- I. A differential equation with one dependent variable which is the function of more than one independent variables.

Example: let $z = z(x, t)$, where x, t independent variables are then

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial t^2} = 0 \text{ is a partial differential equation.}$$

- II. **System of Partial Differential Equations:** A differential equation which contains more than one independent variables with more than one dependent variables and their derivatives with respect to independent variables.



Example: Let $y = y(x, t)$ and $z = z(x, t)$, where x, t are independent

variables then $\frac{\partial^2 y}{\partial x^2} + \frac{\partial^2 y}{\partial t^2} = 0$, $\frac{\partial^2 y}{\partial t^2} + \frac{\partial^2 z}{\partial x^2} = 0$ together is a system of partial differential equation

1.7 Order and Degree of Differential Equation

Order of Differential Equation: The order of the highest ordered derivative occurs in a differential equation is called the order of the differential equation.

Degree of Differential Equation: The degree of the differential equation is the highest exponent of the highest derivative which occurs in it, after the differential equation has been made free from radicals and fractions as per the derivatives are concerned.

Note: The order of the differential equations is always exist but the degree of the differential equations need not exist.

Example:

(a) $y = \sin \frac{dy}{dx}$, order is one and the degree does not exist.

(b) $\log \left(1 + \frac{dy}{dx} \right) = c$, order is one and the degree does not exist.

(c) $e^{y'''} - y'' + xy = 0$ order is three and the degree does not exist

(d) The degree of the equation $\left(\frac{d^3 y}{dx^3} \right)^{2/3} + \left(\frac{d^3 y}{dx^3} \right)^{3/2} = a$ is nine

Example: The order and degree of the differential equation $\frac{d^2}{dx^2} \left(\frac{d^2 y}{dx^2} \right)^{-3/2} = 0$ is

- (a) 1, 4
- (b) 4, 1
- (c) 4, 4
- (d) 1, 1

Ans: b

Examples: Write down the order and degree of the following differential equations

✓ (a) $\left(\frac{dy}{dx} \right)^3 - 4 \left(\frac{dy}{dx} \right) + y = 3e^x$

✓ (b) $\left(\frac{d^2 y}{dx^2} \right)^3 + 7 \left(\frac{dy}{dx} \right)^4 = 3 \sin x$

✓ (c) $\frac{d^2 x}{dy^2} + a^2 x = 0$

✓ (d) $\left(\frac{dy}{dx} \right)^2 - 3 \frac{d^3 y}{dx^3} + 7 \frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} - \log x = 0$

✓ (e) $\sqrt{1 + \left(\frac{dy}{dx} \right)^2} = 4x$

✓ (f) $\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{\frac{2}{3}} = \frac{d^2 y}{dx^2}$

✓ (g) $\frac{d^2 y}{dx^2} - \sqrt{\frac{dy}{dx}} = 0$

(h) $\sqrt{1+x^2} = \frac{dy}{dx}$

Answers: The order and the degree respectively are,

- (a) 1 ; 3
- (b) 2 ; 3
- (c) 2 ; 1
- (d) 3 ; 1
- (e) 1 ; 2
- (f) 2 ; 3
- (g) 2 ; 2
- (h) 1 ; 1

Put Your Own Notes

1.8 Family of Curve's

A family of curves is a set of curves, each of which is given by a function or parameterization in which one or more of the parameters are variable. That is the n -parameters family of curves is a set of relations of the form $\{(x, y) | g(x, y, c_1, c_2, \dots, c_n)\}$, where g is real valued function of $x, y, c_1, c_2, \dots, c_n$ and each c_i is parameter. Families of curves appear frequently in solutions of differential equations.

Example: Family of circle whose centre is (c_1, c_2) and radius is 3 i.e., $\{(x, y) : (x - c_1)^2 + (y - c_2)^2 = 9\}$ is two-parameter family of curves.

Example: $\{(x, y) : y = mx\}$ represents the equation of a family of straight lines through the origin, where m is the parameter.

Example: $\{(x, y) : x^2 + y^2 = c^2\}$ represents the equation of family of concentric circles having the origin as centre and radius c .

Example: $\{(x, y) : y = mx + c\}$ represents the equation of a family of straight lines in a plane, where m and c are parameters.

1.9 Formation of Differential Equation

Let $F(x, y, c_1, c_2, \dots, c_n) = 0$ be n -parameter family of curves, where x is independent variable, y -dependent variable. Then the differential equation whose solution is $F(x, y, c_1, c_2, \dots, c_n) = 0$ is obtained by eliminating n arbitrary constants $c_1, c_2, \dots, c_n$.

Method to eliminate arbitrary constants: Let $F(x, y, c_1, c_2, \dots, c_n) = 0 \dots (i)$ be n -parameter family of curves.

Differentiating (i) w. r. t. x (n -times), we get

$$\phi_1(x, y, y', c_1, c_2, \dots, c_n) = 0 \quad \dots (ii)$$

$$\phi_2(x, y, y', y'', c_1, c_2, \dots, c_n) = 0 \quad \dots (iii)$$

.....

.....

$$\phi_n(x, y, y', \dots, y^{(n)}, c_1, c_2, \dots, c_n) = 0 \dots (n+1), \text{ where } \phi_i \text{ denotes the } i \text{ th derivative of } F(x, y, c_1, \dots, c_n) = 0.$$

Eliminating n unknowns using $(n+1)$ equations, we get the required differential equation.

Example: Consider one parameter family of the equation $y = mx + c \dots \dots$
(1). This represents family of parallel straight lines having same slope m . Then the differential Equation corresponding to the given family is obtained by differentiating (1) with respect to x , we get, $\frac{dy}{dx} = m$ This is the differential equation representing the above family of straight lines.

Example: The differential equation corresponding to the family of circles of radius 'r' whose centre lies on the x - axis is

(a) $y \left(\frac{dy}{dx} \right) + y^2 = r$

(b) $y \left\{ \frac{dy}{dx} + 1 \right\} = r^2$

(c) $y^2 \left\{ \frac{dy}{dx} + 1 \right\} = r^2$

(d) $y^2 \left\{ \left(\frac{dy}{dx} \right)^2 + 1 \right\} = r^2$

Ans: d

Linear Differential Equations: A linear differential equation is any differential equation that can be written in the following form

$a_0(x)y^{(n)} + a_1(x)y^{(n-1)} + \dots + a_{n-1}(x)y' + a_n(x)y = Q(x)$, where $a_i(x) \forall i$ are function of x .

The important thing to note about linear differential equations is that there are no products of the dependent variable and its derivatives or no products of the derivatives and neither the dependent variable nor its derivatives occur to any power other than the first power.

Example: $x^2y'' + xy' + y = e^x$ linear differential Equation of second order

Example: Which of following differential Equation is linear

(a) $(y+x)y' + y = 1$

(b) $3y' + (x+4)y = x^2 + y''$

(c) $y''' = \cos(2ty)$

(d) $y^{(4)} + \sqrt{x}y''' + \cos(x) = e^y$

Ans: b

Example: Let $f(x, y, y', \dots, y^{(n)}) = 0$ be n th order Differential equation then choose the incorrect

(a) If $\deg f = 1 \Rightarrow f = 0$ is linear differential equation

(b) If $f = 0$ is linear $\Rightarrow \deg f = 1$

(c) If $\deg f > 1 \Rightarrow f = 0$ is Non-Linear

(d) If $f = 0$ is Non-Linear $\Rightarrow \deg f > 1$

Ans: a, d

1.10 Solution of Differential Equation

Let $F(x, y, y', \dots, y^{(n)}) = 0$ be n th order differential equation then a real or complex valued function $\phi(x)$ of real variable x is a solution of $F(x, y, y', \dots, y^{(n)})$ if

Put Your Own Notes

- (i) $\phi(x), \phi'(x), \dots, \phi^{(n)}(x)$ exists, where $\phi^{(k)}(x)$ denotes the k^{th} derivative of $\phi(x)$ with respect to x .
- (ii) $\phi(x)$ satisfies the differential equation i.e.,

$$F\left(x, \phi(x), \phi'(x), \dots, \phi^{(n)}(x)\right) = 0$$

Example: If $y(x) = x|x|$ is solution of n^{th} order differential equation defined $\forall x \in \mathbb{R}$, then possible value of n is?

- (a) $n \geq 2$
 (b) $n = 1$
 (c) $n = 2$
 (d) None of these

Ans: b

Example: $y(x)$ has first order derivative, $y(x)$ is a solution of the first order differential equation $y'(x) = 2|x|$

* The solution of differential equation may or may not define for all x in the domain of differential equation.

1.11 Classification of Solution of The Differential Equations

Let $F\left(x, y, y', \dots, y^{(n)}\right) = 0$ be n^{th} order differential equation.

- (i) **General Solution:** Any solution of differential equation which contains n "Independence of arbitrary constants" is called general solution.

Independence of Arbitrary Constants: The arbitrary constants appearing in the general solution must be independent and should not be reducible to a fewer number of equivalent constants.

Example: The solution $y(x) = ce^{x+k}$ apparently contains two arbitrary constants c, k but infact they are equivalent to only one constant, for
 $y(x) = ce^{x+k} = e^x \cdot ce^k = Ae^x$.

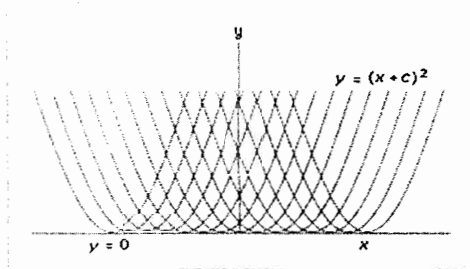
$\therefore$ The two constants c, k are reduced to one constant A and as such c and k are not independent constants.

- (ii) **Particular Solution/Integral Curve:** Any solution which can be obtained from general solution by taking some particular value of arbitrary constants
- (iii) **Singular Solution:** A singular solution in this stronger sense is often given as tangent to every solution from a family of solutions. A singular solution is the envelope of the family of solutions cannot be obtained from general solution by giving particular values of arbitrary constants.

The singular solution is related to the general solution by its being what is called the envelope of that family of curves representing the general solution. An envelope is defined as the curve that is tangent to a given family of curves.

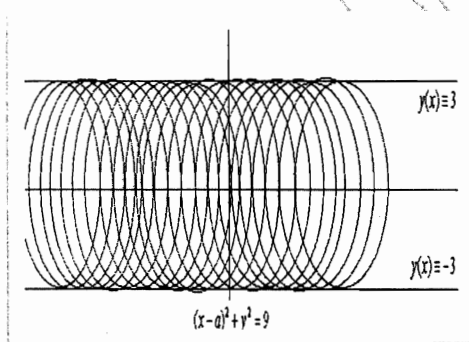
Envelope: An envelope of a family of curves in the plane is a curve that is tangent to each member of the family at some point.

Put Your Own Notes



Example: Example: $y' = 4y$ has the general solution $y = (x+c)^2$, which is a family of parabolas. The line $y(x) \equiv 0$ is also a solution of the differential equation, but it is not a member of the family constituting the general solution. Hence line $y(x) \equiv 0$ is singular solution of the differential equation.

Example: Consider family of circles with centre lying on x-axis and radius 3. This family is given by $(x-a)^2 + y^2 = 9$. And corresponding differential equation can be obtained by eliminating a parameter a .



$$(x-a) + yy' = 0$$

$$\Rightarrow (x-a) = -yy', \Rightarrow (yy')^2 + y^2 = 9,$$

$$\Rightarrow y^2 [y'^2 + 1] = 9$$

Hence $(x-a)^2 + y^2 = 9$ is general solution of the differential equation

$$y^2 [y'^2 + 1] = 9$$

Also we can easily verify that $y(x) = 3$ and $y(x) = -3$ are the solution of $y^2 [y'^2 + 1] = 9$ which cannot be obtained from general solution. These both are envelope to the general solution. Hence $y(x) = 3$ and $y(x) = -3$ are singular solution of the differential equation.

Note: Usually, singular solutions appear in differential equations when there is a need to divide in a term that might be equal to zero. Therefore, when one is solving a differential equation and using division one must check what happens if the term is equal to zero, and whether it leads to a singular solution.

Note: There are solutions, which do not belong to any of the above class; these solutions are not in the scope of this course.

1.12 Some Important Examples

Put Your Own Notes

- ✓ (i) Consider the differential equation, $\left|\frac{dy}{dx}\right| + |y| + a = 0, a \in \mathbb{R}$ then number of real valued solutions of the differential equation are as follows:

Case-I $a = 0$ then $y(x) = 0$ is only real valued solution.

Case-II $a > 0$ differential equation does not have any real valued solutions.

Case-III $a < 0$ differential equation has exactly two real valued solutions.

- (ii) Consider the differential equation, $\left|\frac{dy}{dx}\right| + |y| + a = 0, a \in \mathbb{R}$. Then real valued solution of the differential equation passing through point $(b, c) \in \mathbb{R}^2$.

Case-I $a = 0$

$$\left|\frac{dy}{dx}\right| + |y| = 0, y(b) = c$$

(a) Has unique real solution $\forall (b, 0) \in \mathbb{R}^2$

(b) Has no real solution $\forall (b, c) \in \mathbb{R}^2, c \neq 0$

Case-II $a > 0$ Has no real solution $\forall (b, c) \in \mathbb{R}^2$

Case-III $a < 0$

(c) Has unique real solution

$$y(x) = a \quad \forall (b, a) \in \mathbb{R}^2$$

(d) Has unique real solution

$$y(x) = -a \quad \forall (b, -a) \in \mathbb{R}^2$$

(e) Has no real solution

$$\forall (b, c) \in \mathbb{R}^2, c \neq \pm a$$

Proposition 1.1: Let $y = y(x)$ then considers the differential equation $\frac{dy}{dx} = y^\alpha, y(b) = 0, b \in \mathbb{R} \text{ \& } \alpha \in (0, 1)$. Then number of real valued solutions of the differential equation is infinite and it has infinite number of linearly independent solution.

Done
Example: $\frac{2}{3} \frac{dy}{dx} = y^{\frac{1}{3}}, y(0) = 0, x \geq 0$, if we apply the separation of variable

$$\frac{2}{3} \frac{dy}{y^{\frac{1}{3}}} = dx \Rightarrow y^{\frac{2}{3}} = (x + c)$$

Using initial condition $y(0) = 0 \Rightarrow c = 0$ then solution is $y = (x)^{\frac{3}{2}}$,

Note: If above method we are getting unique solution but actually, it has infinite many solutions all of the below function satisfying the differential equation

$$y(x) = \begin{cases} 0 & 0 \leq x \leq \gamma \\ (x - \gamma)^{\frac{3}{2}} & x > \gamma \end{cases} \quad \text{for each } \gamma \geq 0 \text{ } y(x) \text{ is solution of the differential equation.}$$

CHAPTER 2

FIRST ORDER FIRST DEGREE DIFFERENTIAL EQUATION

2.1. First Order First Degree Differential Equation

The most general form of a first order first degree differential equation is $\frac{dy}{dx} = f(x, y)$ where $f(x, y)$ is function of two variables defined on a region in the xy -plane or it can be written as $M(x, y)dx + N(x, y)dy = 0$, where $M(x, y)$ & $N(x, y)$ is function of two variables.

Example:

- (a) $\frac{dy}{dx} = \frac{x-3y}{2y-x}$ it can be written as $(2y-x)dy = (x-3y)dx$
- (b) $(1+x^2)dy - xydx = 0$
- (c) $(x^2 - y^2)dy + 2xydx = 0$
- (d) $(3x^2y - y)dx + (x^3 - x + 2y)dy = 0$
- (e) $(x^2 + y^2 + y)dx - xdy = 0$
- (f) $\frac{dy}{dx} + \frac{2y}{x} = 6x^3$

2.2. Separable Differential Equation

A differential equation of the form $\frac{dy}{dx} = f(y)g(x)$ is called separable form of the differential equation with separation of variables. It can be express as follow $f_1(x)dx = f_2(y)dy$

i.e. if an equation in x and y can be put in such a way that dx and all the terms containing x can be put on one side, and dy and all the terms containing y on the other side then we say that the equation is variable separable. Its general solution can be obtained by directly integrating terms of x and y and adding an arbitrary constant to any one side. That is solution obtained by integrating $\int f_1(x)dx = \int f_2(y)dy + c$, where c is arbitrary constant of integration.

Example: Solve $(1-x)dy + (1-y)dx = 0$.

Solution: The given differential equation can be written as $\frac{dx}{1-x} + \frac{dy}{1-y} = 0$.

Integrating both side.

We have, $-\log(1-x) - \log(1-y) = -\log c \Rightarrow (1-x)(1-y) = c_1$

Example: Solve $(1+x^2)dy = (1+y^2)dx$.

Solution: The given equation can be written as $\frac{dy}{1+y^2} = \frac{dx}{1+x^2}$. Then integrating both sides, we have $\tan^{-1}(y) - \tan^{-1}(x) = \tan^{-1}(c)$, where c is an arbitrary constant.

Example: Solve $(xy^2 + x)dx + (yx^2 + y)dy = 0$

Solution: The given equation can be rewritten as $x(y^2 + 1)dx + y(x^2 + 1)dy = 0$

$$\Rightarrow \frac{x dx}{(x^2 + 1)} + \frac{y dy}{(y^2 + 1)} = 0 \Rightarrow \frac{2x dx}{(x^2 + 1)} + \frac{2y dy}{(y^2 + 1)} = 0$$

On integrating, we get $\log(x^2 + 1) + \log(y^2 + 1) = \log(c)$ where c is an arbitrary constant.

$$\Rightarrow (x^2 + 1)(y^2 + 1) = c \text{ is the required solution.}$$

Example: Find the curves passing through $(0,1)$ and satisfying $\sin(dy/dx) = c$.

Solution: Re-writing the given equation, we have

$$dy/dx = \sin^{-1} c \text{ or } dy = (\sin^{-1} c) dx.$$

Integrating, $y = x \sin^{-1} c + c'$, c' being arbitrary constant. ... (1)

Since, (1) must pass through $(0,1)$, we put $x = 0$ and $y = 1$ in (1) and obtain $c' = 1$. Hence, (1) reduces to $y = x \sin^{-1} c + 1$ or $(y-1)/x = \sin^{-1} c$ or $\sin\{(y-1)/x\} = c$, which gives the desired curves.

Example: If $dy/dx = e^{x+y}$ and it is given that for $x=1, y=1$; find y when $x=-1$.

Solution: Rewriting the given equation, we get $e^{-y} dy = e^x dx$. Integrating it, $-e^{-y} = e^x + c$... (1)

Put $y(1) = 1$ so that $c = -e^{-1} - e$.

Hence (1) reduces to $-e^{-y} = e^x - e^{-1} - e$ (2)

Putting $x = -1$ in (2), we obtain $-e^{-y} = e^{-1} - e^{-1} - e$ so that $y = -1$.

2.3. Equation Reducible into Separable Form

If the first order first degree differential equation is of the form $\frac{dy}{dx} = f(ax + by + c)$. Then it can be reducible into separable form in variable u, x by using transformation $u = ax + by + c$ and corresponding reduced

differential equation is $a + bf(u) = \frac{du}{dx}$ and solution is given by

$$\int \frac{du}{a + bf(u)} = \int dx + c.$$

Example: $\frac{dy}{dx} = (x+y)^2$

Let $v = x + y$ then equation reducible into $\frac{dy}{dx} = (v)^2$ and $\frac{dv}{dx} = 1 + \frac{dy}{dx}$ compare both equations we get

$\frac{dv}{dx} = 1 + v^2$, it can be written as $\frac{dv}{1+v^2} = dx$ integrate both side we have $\tan^{-1}(v) = x + c$

Hence solution is $x + y = \tan(x + c)$

Homogeneous Function: Let $S \subseteq \mathbb{R}^2$ be a subset of $\mathbb{R}^2$. A function $f: S \rightarrow \mathbb{R}$ is said to be homogeneous function of degree n if for each $\lambda > 0$, $f(\lambda x, \lambda y) = \lambda^n f(x, y)$

Example: $f(x, y) = x^2 + y^2$

Solution: Consider, $f(\lambda x, \lambda y) = (\lambda x)^2 + (\lambda y)^2 = \lambda^2(x^2 + y^2) = \lambda^2 f(x, y)$

So, $f(x, y)$ is a homogeneous function of degree 2.

Example: $f(x, y) = \frac{x^2 + y^2}{xy}$

Solution: Consider, $f(\lambda x, \lambda y) = \frac{(\lambda x)^2 + (\lambda y)^2}{(\lambda x)(\lambda y)}$

$$= \frac{\lambda^2 x^2 + \lambda^2 y^2}{\lambda^2 xy} = \frac{(x^2 + y^2)}{xy} = f(x, y)$$

$\therefore f(x, y)$ is a homogeneous function of degree 0.

2.4. Homogeneous Differential Equation

The first order differential equation $\frac{dy}{dx} = f(x, y)$ is called homogeneous, differential equation if $f(x, y)$ is homogeneous function of degree zero. i.e.

$f(x, y)$ can be expressed as $f(x, y) = \phi\left(\frac{y}{x}\right)$ or $f(x, y) = \phi\left(\frac{x}{y}\right)$, where ϕ is arbitrary function.

The homogeneous differential equation $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$ can be reducible into separable form in new variables x, u using transformation $y = ux$, so $y' = u + xu' \Rightarrow u - f(u) = x \frac{du}{dx} \Rightarrow \frac{du}{u - f(u)} = \frac{dx}{x} \Rightarrow \int \frac{du}{u - f(u)} = x + c$ is general solution of differential equation.

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Proof
 $(x^2y - 2xy^2)dx - (x^3 - 2x^2y)ydy = 0$
 $- (x^3 - 2x^2y)dy = 0$
 $y^2 = x^2 \Rightarrow y = x$

Example: Solve $(x^3 + 3xy^2)dx + (y^3 + 3x^2y)dy = 0$.

Solution: Given $\frac{dy}{dx} = \frac{-x^3 - 3xy^2}{y^3 + 3x^2y} = -\frac{1 + 3(y/x)^2}{(y/x)^3 + 3(y/x)} \dots (1)$

Take $y/x = v$, i.e., $y = vx$ so that $\frac{dy}{dx} = v + x\left(\frac{dv}{dx}\right) \dots (2)$

From (1) and (2), $v + x\frac{dv}{dx} = -\frac{1 + 3v^2}{v^3 + 3v}$

or $x\frac{dv}{dx} = -\frac{1 + 3v^2}{v^3 + 3v} - v = -\frac{v^4 + 6v^2 + 1}{v^3 + 3v}$ or $4\frac{dx}{x} = -\frac{4v^3 + 12v}{v^4 + 6v^2 + 1}dv$.

Integrating, $4\log x = -\log(v^4 + 6v^2 + 1) + \log c$, c being an arbitrary constant.

or $\log x^4 = \log\left[\frac{c}{(v^4 + 6v^2 + 1)}\right]$, i.e., $x^4(v^4 + 6v^2 + 1) = c$

or $y^4 + 6x^2y^2 + x^4 = c$ or $(x^2 + y^2)^2 + 4x^2y^2 = c$, as $y/x = v$.

Example: Solve $(x^2 - 4xy - 2y^2)dx + (y^2 - 4xy - 2x^2)dy = 0$.

Solution: Re-writing the given differential equation, we have

$\frac{dy}{dx} = (x^2 - 4xy - 2y^2)/(2x^2 + 4xy - y^2) \dots (1)$

Putting $y = xv$ and $\frac{dy}{dx} = v + x\left(\frac{dv}{dx}\right)$, (1) reduces to

$v + x\frac{dv}{dx} = \frac{1 - 4v - 2v^2}{2 + 4v - v^2}$ or $x\frac{dv}{dx} = \frac{1 - 4v - 2v^2}{2 + 4v - v^2} - v$

or $x\frac{dv}{dx} = \frac{1 - 6v - 6v^2 + v^3}{2 + 4v - v^2}$ or $3\frac{dx}{x} + \frac{3(v^2 - 4v - 2)}{v^3 - 6v^2 - 6v + 1}dv = 0$

Integrating, $3\log x + \log(v^3 - 6v^2 - 6v + 1) = \log c$, c being an arbitrary constant

Or $x^3(v^3 - 6v^2 - 6v + 1) = c$ or $x^3\left\{\left(\frac{y}{x}\right)^3 - 6\left(\frac{y}{x}\right)^2 - 6\left(\frac{y}{x}\right) + 1\right\} = c$

Or $y^3 - 6xy^2 - 6x^2y + x^3 = c$, c being an arbitrary constant.

2.5. Differential Equation Reducible into Homogeneous

Consider the equation $\frac{dy}{dx} = \frac{ax + by + c}{a'x + b'y + c'}$... (i) Where $\frac{a}{b} \neq \frac{a'}{b'}$ If c & c' are both

zero, the equation is homogeneous. If c & c' are not both zero, then we change the variables so that constant terms are no longer present, by the substitutions $x = X + h$, $y = Y + k$... (ii) and reduces (i) to

$\frac{dY}{dX} = \frac{a(X + h) + b(Y + k) + c}{a'(X + h) + b'(Y + k) + c'}$

$$\text{or } \frac{dY}{dX} = \frac{aX+bY+(ah+bk+c)}{a'X+b'Y+(a'h+b'k+c')} \quad \dots(iii)$$

Now, choose h and k such that

$$ah+bk+c=0 \text{ \& } a'h+b'k+c'=0 \quad \dots(iv)$$

Then (iii) reduces to $\frac{dY}{dX} = \frac{aX+bY}{a'X+b'Y}$, which is homogeneous and can be solved by the substitution $Y = vX$. Replacing X & Y in the solution so obtained by $x-h, y-k$ from (ii) respectively we will get the required solution in terms of x and y , the original variables.

A special case when $\frac{a}{a'} = \frac{b}{b'} = k$ in this case we cannot solve the equation given by (iv) above and the differential equation is of the form

$$\frac{dy}{dx} = \frac{ax+by+c}{kax+kby+c'} \quad \dots(v)$$

In this case, the differential equation is solved by putting $v = ax+by$ $\dots(vi)$

Differentiating both sides of (vi) with respect to x , we get $\frac{dv}{dx} = a + b \frac{dy}{dx}$ or

$$\frac{dy}{dx} = \frac{1}{b} \left(\frac{dv}{dx} - a \right)$$

$\therefore$ The equation (v) reduces to $\frac{1}{b} \left(\frac{dv}{dx} - a \right) = \frac{v+c}{kv+c}$, or $\frac{dv}{dx} = a + \frac{b(v+c)}{kv+c}$, the variables are now separable, and we can determine v in terms of x . Replacing v by $ax+by$ in this solution, we can obtain the required solution.

Example: Solve $(2x+y-3)dy = (x+2y-3)dx$

Solution: The given differential equation is $\frac{dy}{dx} = \frac{x+2y-3}{2x+y-3}$ $\dots(i)$

Putting $x = X+h$ and $y = Y+k$ $\dots(ii)$

The equation (i) reduces to $\frac{dY}{dX} = \frac{(X+h)+2(Y+k)-3}{2(X+h)+(Y+k)-3}$ or

$$\frac{dY}{dX} = \frac{X+2Y+(h+2k-3)}{2X+Y+(2h+k-3)} \quad \dots(iii)$$

Choose h and k such that $h+2k-3=0, 2h+k-3=0$ $\dots(iv)$

Solving the equations (iv) we have $h=1, k=1$

$\therefore$ From (ii) $x = X+1, y = Y+1$ $\dots(v)$

Also (iii) reduces to $\frac{dY}{dX} = \frac{X+2Y}{2X+Y}$ solution of this can be obtained by reducing to variable separable form and it is $(x+y-2) = c(x-y)^3$.

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Example: Solve $\frac{dy}{dx} = (x+y+4)/(x-y-6)$.

Solution: Given $\frac{dy}{dx} = (x+y+4)/(x-y-6)$... (1)

Let $x = X + h$, $y = Y + k$ so that $\frac{dy}{dx} = \frac{dY}{dX}$... (2)

Using (2), (1) reduces to $\frac{dY}{dX} = \frac{(X+Y)+(h+k+4)}{(X-Y)+(h-k-6)}$... (3)

We choose h and k , such that $h+k+4=0$, and $h-k-6=0$... (4)

Solving (4), $h=1, k=-5$ and so by (2), $X=x-1, Y=y+5$... (5)

Using (4), (3) reduces to $\frac{dY}{dX} = \frac{X+Y}{X-Y} = \frac{1+(Y/X)}{1-(Y/X)}$... (6)

Putting $Y = Xv$ and $\frac{dY}{dX} = v + X\left(\frac{dv}{dX}\right)$, (6) becomes

$$v + X \frac{dv}{dX} = \frac{1+v}{1-v} \text{ or } \frac{dX}{X} = \frac{1-v}{1+v^2} dv = \frac{dv}{1+v^2} - \frac{v dv}{1+v^2}$$

Integrating, $\log X = \tan^{-1} v - (1/2) \log(1+v^2) + (1/2) \log c$

$$\text{or } 2 \log X + \log \left(1 + \frac{Y^2}{X^2}\right) - \log c = 2 \tan^{-1} \left(\frac{Y}{X}\right), \text{ as } v = \frac{Y}{X}$$

$$\text{or } \log \left\{ \left(X^2 + Y^2 \right) / c \right\} = 2 \tan^{-1} \left(\frac{Y}{X} \right) \text{ or } X^2 + Y^2 = c e^{2 \tan^{-1} (Y/X)}$$

$$\text{or } (x-1)^2 + (y+5)^2 = c e^{2 \tan^{-1} \{(y+5)/(x-1)\}}, c \text{ being an arbitrary constant.}$$

Example: Solve $\frac{dy}{dx} = (x+y-2)/(y-x-4)$.

Solution: Let $x = X + h$ and $y = Y + k$ so that $\frac{dy}{dx} = \frac{dY}{dX}$... (1)

Then given equation gives $\frac{dY}{dX} = \frac{X+Y+(h+k-2)}{Y-X+(k-h-4)}$... (2)

Choose h, k such that $h+k-2=0$ and $k-h-4=0$... (3)

Solving (3); $h=-1, k=3$. Then (1) gives $X=x+1$ and $Y=y-3$... (4)

Using (3), (2) becomes $\frac{dY}{dX} = \frac{X+Y}{Y-X} = \frac{1+(Y/X)}{(Y/X)-1}$... (5)

Let $\frac{Y}{X} = v$, i.e., $Y = vX$ so that $\frac{dY}{dX} = v + X\left(\frac{dv}{dX}\right)$... (6)

$$\text{From (5) and (6), } v + X \frac{dv}{dX} = \frac{1+v}{v-1} \text{ or } X \frac{dv}{dX} = \frac{1+2v-v^2}{v-1}$$

$$\text{or } \frac{(v-1)dv}{1+2v-v^2} = \frac{dX}{X} \text{ or } \frac{(2-2v)dv}{1+2v-v^2} = -2 \frac{dX}{X}$$

$$\text{Integrating, } \log(1+2v-v^2) + 2 \log X = \log C \text{ or } X^2(1+2v-v^2) = C$$

$$\text{or } X^2 \{1 + 2(Y/X) - (Y/X)^2\} = C \text{ or } X^2 + 2XY - Y^2 = C$$

$$\text{or } (x+1)^2 + 2(x+1)(y-3) - (y-3)^2 = C, C \text{ being an arbitrary constant}$$

Example: Solve $(x+y)(dx-dy) = dx+dy$.

Solution: Re-writing the given equation, we get

$$(x+y-1)dx = (x+y+1)dy \text{ or } \frac{dy}{dx} = \frac{x+y-1}{x+y+1} \quad \dots(1)$$

$$\text{Let } x+y=v \quad \dots(2)$$

$$(2) \Rightarrow 1 + dy/dx = dv/dx \text{ so that } dy/dx = (dv/dx) - 1. \quad \dots(3)$$

$$\frac{dv}{dx} - 1 = \frac{v-1}{v+1} \text{ or } \frac{dv}{dx} = \frac{2v}{v+1} \text{ or } 2dx = \left(1 + \frac{1}{v}\right)dv.$$

$\therefore$ Integrating, $2x+c = v + \log v$ or $x-y+c = \log(x+y)$, C being an arbitrary constant

2.6. Exact Differential Equation and Its Solutions

The expression $M(x,y)dx + N(x,y)dy$ is called exact expression if $\exists f(x,y)$ such that the total derivative, $d[f(x,y)]$ of $f(x,y)$ is $Mdx + Ndy$ i.e., $d[f(x,y)] = M(x,y)dx + N(x,y)dy$

$$\Rightarrow \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = M(x,y)dx + N(x,y)dy \Rightarrow \frac{\partial f}{\partial x} = M(x,y), \frac{\partial f}{\partial y} = N(x,y).$$

If we consider the differential equation $M(x,y)dx + N(x,y)dy = 0$ is called, exact if $\exists f(x,y)$ such that the total derivative, $d[f(x,y)]$ of $f(x,y)$ is $Mdx + Ndy$ i.e., $d[f(x,y)] = M(x,y)dx + N(x,y)dy$ and solution of the differential equation.

Proposition 2.1: The Necessary and sufficient condition for a differential equation $M(x,y)dx + N(x,y)dy = 0$ to be exact is $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ i.e., the differential equation $M(x,y)dx + N(x,y)dy = 0$ is exact if and only if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Example: $\cos y dx - (x \sin y - y^2) dy = 0$ is exact differential equation

$$\text{Example: } \left(x - \frac{y}{x^2 + y^2}\right) dx + \left(y + \frac{x}{x^2 + y^2}\right) dy = 0$$

Note: Exactness is a fragile condition in the sense that seemingly minor alterations in an exact equation can destroy its exactness. This is demonstrated in the following example.

Example: $(xy^2 + x)dx + yx^2dy = 0$

$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}[xy^2 + x] = 2xy \quad \text{and} \quad \frac{\partial N}{\partial x} = \frac{\partial}{\partial x}[yx^2] = 2xy$$

And

$$(y^2 + 1)dx + xydy = 0$$

is not exact, even though it is obtained by dividing both sides of the first equation by x .

Example: The differential equation

$\cos y dx + (y^2 - x \sin y)dy = 0$ is exact because

$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}[\cos y] = -\sin y \quad \text{and} \quad \frac{\partial N}{\partial x} = \frac{\partial}{\partial x}[y^2 - x \sin y] = -\sin y$$

But the equation

$$\cos y dx + (y^2 + x \sin y)dy = 0$$

is not exact, even though it differs from the first equation only by a single sign.

Note: Every differential equation of the form $M(x)y' + N(y) = 0$

is exact. In other words, a separable variables equation is actually a special type of an exact equation.

Solution of Exact Differential Equation: Let $M(x, y)dx + N(x, y)dy = 0$ be exact differential equation

$$\Rightarrow \exists f(x, y) \text{ such that } \frac{\partial f}{\partial x} = M, \frac{\partial f}{\partial y} = N.$$

$$M = \frac{\partial f}{\partial x} \Rightarrow f(x, y) = \int M dx + \phi(y) \Rightarrow \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} \int M dx + \phi'(y)$$

$$\Rightarrow \phi'(y) = N - \frac{\partial}{\partial y} \int M dx \Rightarrow \phi(y) = \int \left[N - \frac{\partial}{\partial y} \int M dx \right] dy$$

$$\therefore f(x, y) = \int M dx + \int \left[N - \frac{\partial}{\partial y} \int M dx \right] dy$$

$$\text{Thus, solution is } f(x, y) = \int M dx + \int \left[N - \frac{\partial}{\partial y} \int M dx \right] dy = c \quad \dots(2.1)$$

Similarly, if we start with $\frac{\partial f}{\partial x} = N$ then solution is

$$f(x, y) = \int N dy + \int \left[M - \frac{\partial}{\partial x} \int N dy \right] dx = c \quad \dots(2.2)$$

$$(2.1) \text{ is equivalent to } \int_{y=\text{const}} M dx + \int_{N\text{-free from } x} N dy = c \text{ and}$$

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(2.2) is equivalent $\int_{x=\text{const}} Ndy + \int_{M=\text{free from } y} Mdx = c$.

Example: $(x^2 - ay)dx + (y^2 - ax)dy = 0$ be an exact equation as

$$M = x^2 - ay, N = y^2 - ax$$

Solution: Then solution is $\int Mdx + \int Ndy - \int \left[\frac{\partial}{\partial y} \int Mdx \right] dy = c$

$$\frac{1}{3}x^3 - axy + \frac{1}{3}y^3 - axy - \int \left[\frac{\partial}{\partial y} \left[\frac{1}{3}x^3 - axy \right] \right] dy = c$$

$$\frac{1}{3}x^3 - axy + \frac{1}{3}y^3 - axy - \int -ax dy = c$$

$$\frac{1}{3}x^3 - axy + \frac{1}{3}y^3 - axy + axy = c$$

$$\Rightarrow x^3 - 3axy + y^3 = c$$

2.7. Equation Reducible to Exact Form

Sometimes a differential equation $M(x, y)dx + N(x, y)dy = 0$ which is not exact, we can reduce the differential equation into exact by multiplying a suitable factor $\phi(x, y)$ called an integrating factor. Some of the rules for finding integrating factors of the equation.

1. Integrating Factors Found by Inspection: In a number of cases, the integrating factor can be found after regrouping the terms of the equation and recognizing each group as being a part of an exact differential.

Some of the Important Example:

(a) $xdy + ydx = d(xy)$

(b) $\frac{xdy - ydx}{x^2} = d\left(\frac{y}{x}\right)$

(c) $\frac{xdy - ydx}{xy} = d\left[\log\left(\frac{y}{x}\right)\right]$

(d) $\frac{xdy - ydx}{y^2} = -d\left(\frac{x}{y}\right)$

(e) $\frac{xdy - ydx}{x^2 + y^2} = d\left(\tan^{-1} \frac{y}{x}\right)$

(f) $\frac{xdy - ydx}{x^2 - y^2} = d\left(\frac{1}{2} \log \frac{x+y}{x-y}\right)$

Example: Solve $y(2xy + e^x)dx = e^x dy$

Solution: It is easy to note that terms $ye^x dx, e^x dy$ should be put together $(ye^x dx - e^x dy) + 2xy^2 dx = 0$. Now we observe that the term $2xy^2 dx$ should

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not involve y^2 . This suggests that $\frac{1}{y^2}$ may be an integrating factor

multiplying throughout by $\frac{1}{y^2}$, it follows $\frac{ye^x - e^x dy}{y^2} + 2xdx = 0$ or

$d\left(\frac{e^x}{y}\right) + 2xdx = 0$. Integrating, we get $\frac{e^x}{y} + x^2 = c$ which is the required solution.

Example: Find the particular solution of

$$(\cos x - x \sin x + y^2)dx + 2xy dy = 0$$

That satisfies the initial condition $y(\pi) = 1$

Solution: The differential equation is exact because

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\frac{\partial}{\partial y}[\cos x - x \sin x + y^2] = 2y = \frac{\partial}{\partial x}[2xy]$$

Then general solution is $xy^2 + x \cos x = C$.

Applying the given initial condition we get $C = 0$. Hence, the particular solution is $xy^2 + x \cos x = 0$.

Some of The Important Method for Finding Integrating Factor

Consider the Non-exact differential equation $M(x, y)dx + N(x, y)dy = 0$ then

(i) An integrating factor $\phi(x, y)$ can be function of x only

Example: $xdy - ydx = 0$ be a non exact differential equation and $\frac{1}{x^2}$ is an integrating factor

(ii) An integrating factor $\phi(x, y)$ can be function of y only

Example: $xdy - ydx = 0$ be a non exact differential equation and $\frac{1}{y^2}$ is an integrating factor

(iii) An integrating factor $\phi(x, y)$ can be function of x, y only

Example: $xdy - ydx = 0$ be a non exact differential equation and $\frac{1}{x, y}$ is an integrating factor

(iv) An integrating factor $\phi(x, y)$ can be function of $x^2 + y^2$ only

Example: $xdy - ydx = 0$ be a non exact differential equation and $\frac{1}{x^2 + y^2}$ is an integrating factor

(v) An integrating factor $\phi(x, y)$ can be function of $x + y$ only

Example: $xdy - ydx = 0$ be a non exact differential equation and

$\frac{1}{(x + y)^2}$ is an integrating factor

(vi) An integrating factor $\phi(x, y)$ can be function of $x - y$ only

Example: $xdy - ydx = 0$ be a non exact differential equation and

$\frac{1}{(x - y)^2}$ is an integrating factor

That is we can assume in general an integrating factor $\phi(x, y)$ can be function of v , where v is known function of x, y

Now $\phi(v)$ is an integrating factor if and only if $\phi(v)(M(x, y)dx + N(x, y)dy) = 0$ is exact differential equation that is

$\phi(v)$ is an integrating factor if and only if $(\phi(v)M)_y = (\phi(v)N)_x$

$\phi(v)$ is an integrating factor if and only if $\frac{M_y - N_x}{N.v_x - M.v_y} = \frac{\phi'(v)}{\phi(v)} = f(v)$

That if $\phi(v)$ is an integrating factor then $\phi(v) = e^{\int f(v)dv}$

Proposition 2.1:

(i) If the integrating factor $\phi(x, y) = \phi(v)$ is function of v then

$\frac{M_y - N_x}{N.v_x - M.v_y} = f(v)$ is function of v only or at most constant. And

$\phi(v) = e^{\int f(v)dv}$

(ii) If $\frac{M_y - N_x}{N.v_x - M.v_y}$ is function of v only or constant then there exist an

integrating factor $\phi(v)$ which is function of v only and it is given by

$\phi(v) = e^{\int f(v)dv}$

(iii) If $\frac{M_y - N_x}{N.v_x - M.v_y}$ is not function of v or constant. Then there does not

exist any integrating factor which is function of v only

Proposition 2.2:

(i) If the integrating factor $\phi(x, y)$ is function of x then $\frac{M_y - N_x}{N} = f(x)$ is function of x only or constant.

(ii) If $\frac{M_y - N_x}{N} = f(x)$ is function of x only or constant then there exist an integrating factor $\phi(x)$ which is function of x only and it is given by

$\phi(x) = e^{\int f(x)dx}$

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- (iii) If $\frac{M_y - N_x}{N}$ is not free from y . Then there does not exist any integrating factor which is function of x only

Example: Solve $(xy^2 - e^{1/x^3})dx - x^2ydy = 0$

Solution: Here $M = xy^2 - e^{1/x^3}$ and $N = -x^2y \Rightarrow \frac{M_y - N_x}{N} = -\frac{4}{x}$ which is a function of x only

$\therefore$ I.F. = $e^{\int \frac{-4}{x} dx} = x^{-4}$ Multiplying throughout by x^{-4} , we get $\left(\frac{y^2}{x^3} - \frac{1}{x^4}e^{1/x^3}\right)dx - \frac{y}{x^2}dy = 0$ which is an exact equation

$\therefore$ the solution is $\int \left(\frac{y^2}{x^3} - \frac{1}{x^4}e^{1/x^3}\right)dx + 0 = c$

$$\text{or } -\frac{y^2x^{-2}}{2} + \frac{1}{3}\int e^{x^{-3}}(-3x^{-4})dx = c \quad \text{or } \frac{1}{3}e^{x^{-3}} - \frac{1}{2}\frac{y^2}{x^2} = c$$

Proposition 2.3:

- (i) If the integrating factor $\phi(x, y)$ is function of y then $\frac{M_y - N_x}{M} = f(y)$ is function of y only or constant.
- (ii) If $\frac{M_y - N_x}{M} = f(y)$ is function of y only or constant. Then there exist an integrating factor $\phi(y)$, which is function of y only and it is given by $\phi(y) = e^{-\int f(y)dy}$.
- (iii) If $\frac{M_y - N_x}{M}$ is not free from x . Then there does not exist any integrating factor which is function of y only.

Proposition 2.4:

- (i) If $\phi(x, y)$ is function of $u = x.y$ then $\frac{M_y - N_x}{yN - xM} = f(u)$ is function of u only or constant and integrating factor $\phi(u)$ is given by $\phi(u) = e^{\int f(u)du}$.
- (ii) $\frac{M_y - N_x}{yN - xM} = f(u)$ is function of $u = x.y$ only or constant then there exist an integrating factor $\phi(u)$ which is function of x only and it is given by $\phi(u) = e^{\int f(u)du}$.

- (iii) $\frac{M_y - N_x}{yN - xM}$ is not function of $u = x.y$ only and non-constant then there does not exist any integrating factor which is function of u only.

Proposition 2.5:

- (i) If $\phi(x, y)$ is function of $u = \frac{x}{y}$ then $\frac{y^2(M_y - N_x)}{yN + xM} = f(u)$ is function of u only or constant and integrating factor $\phi(u)$ is given by $\phi(u) = e^{\int f(u) du}$.
- (ii) $\frac{y^2(M_y - N_x)}{yN + xM} = f(u)$ is function of $u = \frac{x}{y}$ only or constant then there exist an integrating factor $\phi(u)$ which is function of x only and it is given by $\phi(u) = e^{\int f(u) du}$.
- (iii) $\frac{y^2(M_y - N_x)}{yN + xM}$ is not function of $u = \frac{x}{y}$ only and non-constant then there does not exist any integrating factor which is function of u only.

Proposition 2.6:

- (i) If $\phi(x, y)$ is function of $u = \frac{y}{x}$ then $\frac{x^2(M_y - N_x)}{yN + xM} = f(u)$ is function of u only or constant and integrating factor $\phi(u)$ is given by $\phi(u) = e^{-\int f(u) du}$.
- (ii) $\frac{x^2(M_y - N_x)}{yN + xM} = f(u)$ is function of $u = \frac{y}{x}$ only or constant then there exist an integrating factor $\phi(u)$ which is function of x only and it is given by $\phi(u) = e^{-\int f(u) du}$.
- (iii) $\frac{x^2(M_y - N_x)}{yN + xM}$ is not function of $u = \frac{y}{x}$ only and non-constant then there does not exist any integrating factor which is function of u only.

Example: Check whether differential equation is exact or not. If yes, then find solution

(a) $(xe^{xy} + 2y)dy + ye^{xy}dx = 0$

Ans: $e^{xy} + y^2 = c$

(b) $(e^{2y} + 1)\cos x dx + 2e^{2y} \sin x dy = 0$

Ans: $(e^{2y} + 1)\sin x = c$

Put Your Own Notes

Integrating Factor of Some of the First Order First-Degree Standard Differential Equation

Proposition 2.7: In General the homogeneous non-exact differential equation $M(x, y)dx + N(x, y)dy = 0$ is reducible into exact differential equation. It has an integrating factor $\frac{1}{xM + yN}$, provided $xM + yN \neq 0$.

Proposition 2.8: If differential equation is of the form $y.f_1(xy)dx + x.f_2(xy)dy = 0$ then $\frac{1}{xM - yN}$ is an integrating factor provided $xM - yN \neq 0$

Proposition 2.9: If the differential equation of the form $x^\alpha y^\beta (mydx + nxdy) = 0$. Then $\phi(x, y) = x^{km-1-\alpha} y^{kn-1-\beta}$, $\forall k \in \mathbb{R}$ is an I.F.

Proposition 2.10: If the differential equation of the form $x^\alpha y^\beta (mydx + nxdy) + x^{\alpha'} y^{\beta'} (m'ydx + n'xdy) = 0$. Then an integrating factor is $x^h y^k$, where $\frac{\alpha+h+1}{m} = \frac{\beta+k+1}{n}$, $\frac{\alpha'+h+1}{m'} = \frac{\beta'+k+1}{n'}$.

Example: Solve $xy(ydx + xdy) + x^2 y^2 (2ydx - xdy) = 0$ on comparing with

$x^\alpha y^\beta (mydx + nxdy) + x^{\alpha'} y^{\beta'} (m'ydx + n'xdy) = 0$ we get $h = k = -3 \Rightarrow \frac{1}{(xy)^3}$ is integrating factor.

Multiplying throughout by $\frac{1}{x^3 y^3}$, it becomes

$$\left(\frac{1}{x^2 y} + \frac{2}{x} \right) dx + \left(\frac{1}{xy^2} - \frac{1}{y} \right) dy = 0, \text{ which is an exact equation. The solution is } \frac{1}{y} \left(-\frac{1}{x} \right) + 2 \log x - \log y = c.$$

Proposition 2.11: If the differential equation of the form $M(x, y)dx + N(x, y)dy = 0$ satisfy the equation

$$(i) \quad \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} + \frac{k}{x} N \Rightarrow x^k \text{ is an integrating factor}$$

$$(ii) \quad \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} - \frac{k}{y} M \Rightarrow y^k \text{ is an I.F.}$$

Proposition 2.12: The function $f(x, y)$ is an integrating factor of $M(x, y)dx + N(x, y)dy = 0$ if and only if $\left(M \frac{\partial f}{\partial y} - N \frac{\partial f}{\partial x} \right) + \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) f = 0$

2.8. First Order First Degree Linear Differential Equation

The most general form of First order first degree Linear differential equation is $\frac{dy}{dx} + Py = Q$, ... (i), where P and Q are constants or functions of x alone (and not of y) is called a linear differential equation of the first order.

In general $\frac{dy}{dx} + Py = Q$ differential equation is non-exact differential equation.

$e^{\int P dx}$ is an integrating factor hence $\frac{dy}{dx} + Py = Q$ always reducible into exact differential equation by multiplying both sides by $e^{\int P dx}$

To solve such an equation $\left(\frac{dy}{dx} + Py\right)e^{\int P dx} = Qe^{\int P dx}$... (ii)

Written as follow $\frac{d}{dx} \left[ye^{\int P dx} - \int Q(x)e^{\int P dx} . dx \right] = 0$

Integrating both sides with respect to x , we get $ye^{\int P dx} = \int Q . e^{\int P dx} dx + c$, where c is an arbitrary constant as the general solution of (i).

Note: The factor $e^{\int P dx}$ on multiplying by which the L.H.S of the given equation (i) becomes exact differential coefficient of some function of x and y is called an integrating factor of the differential equation.

Example: Solve $\frac{dy}{dx} + \frac{y}{x} = x^2$ if $y = 1$ when $x = 1$

Solution: Here " P " = $1/x$ and " Q " = x^2 . Then integrating factor = $e^{\int P dx} = e^{\int (1/x) dx} = x$

Multiplying both sides of the given equation by the integrating factor we have $\left(x \frac{dy}{dx} + y\right) = x^3$

or $\frac{d}{dx}(xy) = x^3$. Integrating, $xy = \int x^3 dx + C$, where C is an arbitrary constant and $xy = \frac{1}{4}x^4 + C$ is the solution

of the given equation, Given $y(1) = 1$, So $1 = \frac{1}{4} + C$ or $C = \frac{3}{4}$

$\therefore$ The required solution is $xy = \frac{1}{4}x^4 + \frac{3}{4}$ or $4xy = x^4 + 3$.

Example: Solve $\frac{dy}{dx} + \frac{2xy}{1+x^2} = \frac{1}{(1+x^2)^2}$. Given that $y = 0$ when $x = 1$.

Solution: Here $P = \frac{2x}{1+x^2}$ and $Q = \frac{1}{(1+x^2)^2}$.

$\therefore$ Integrating factor = $e^{\int P dx} = e^{\int \frac{2x}{1+x^2} dx} = e^{\log(1+x^2)} = 1+x^2$. Multiplying both sides of the given equation by $(1+x^2)$ and integrating both sides with

respect to x we get $y(1+x^2) = c + \int \frac{1}{1+x^2} dx \Rightarrow y(1+x^2) = c + \int \frac{1}{1+x^2} dx$,

where c is constant of integration then $y(1+x^2) = c + \tan^{-1}(x)$. Now

$$y(1) = 0 \Rightarrow c = -\frac{\pi}{4}$$

$\therefore$ The corresponding solution is $y(1+x^2) = \tan^{-1}(x) - \frac{\pi}{4}$

Put Your Own Notes

2.9. Equations Reducible to Linear Form

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An equation of the form $f'(y)\frac{dy}{dx} + Pf(y) = Q, \dots(1)$, Where P and Q constants or functions of x alone (and not of y) can be reduced to linear form as follows. Putting $f(y) = v$ so that $f'(y)\left(\frac{dy}{dx}\right) = \frac{dv}{dx}$, (1) becomes, $\frac{dv}{dx} + Pv = Q, \dots(2)$, which is linear in v and x and its solution can be obtained by using working rule of Art 2.8. Thus, we have integrating factor $= e^{\int P dx}$ and solution is $v \cdot e^{\int P dx} = \int Q e^{\int P dx} dx + c$. finally, replace v by $f(y)$ to get solution in terms of x and y alone.

Another Form: An equation of the form $f'(x)\frac{dx}{dy} + P_1f(x) = Q_1 \dots(1)$,

Where P_1 and Q_1 are constants or functions of y alone can be reduced to linear form again as follows: Putting $f(x) = v$ so that $f'(x)\left(\frac{dx}{dy}\right) = \frac{dv}{dy}$, (1) gives $\frac{dv}{dy} + P_1v = Q_1, \dots(2)$, which is linear in variables v and y . Integrating

factor of (2) is $e^{\int P_1 dy}$ and hence solution of (2) is $v e^{\int P_1 dy} = \int Q_1 e^{\int P_1 dy} dy + c$

Replacing v by $f(x)$, we obtain the required solution of (1).

2.10. Bernoulli's Equation and Their Solution

An equation of the form $\left(\frac{dy}{dx}\right) + Py = Qy^n, n \neq 0, 1 \dots(1)$,

Where P and Q are constants or functions of x alone (and not of y) and n is constant except 0 and 1, is called a Bernoulli's differential equation. We first multiply by y^{-n} , thereby expressing it in the form (1)

$$y^{-n}\left(\frac{dy}{dx}\right) + Py^{1-n} = Q \dots(2)$$

$$\text{Let } y^{1-n} = v \dots(3)$$

Differentiating (3) w. r. t. x , gives

$$(1-n)y^{-n}\frac{dy}{dx} = \frac{dv}{dx}, \text{ or } y^{-n}\frac{dy}{dx} = \frac{1}{1-n}\frac{dv}{dx} \dots(4)$$

Using (3) and (4), (2) reduces to

$$\frac{1}{1-n}\frac{dv}{dx} + Pv = Q \text{ or } \frac{dv}{dx} + P(1-n)v = Q(1-n),$$

Which is linear in v and x . Its I.F. $= e^{\int P(1-n)dx} = e^{(1-n)\int P dx}$ and hence the required solution is $v \cdot e^{(1-n)\int P dx} = \int Q \cdot e^{(1-n)\int P dx} dx + c$, c being an arbitrary constant

$$y^{1-n} e^{(1-n)\int P dx} = \int Q \cdot e^{(1-n)\int P dx} dx + c, \text{ using (3)}$$

Remark: Equation $\frac{dx}{dy} + P_1x = Q_1x^n$ is also in the Bernoulli's form. Here P_1 and Q_1 are functions of y alone. Method of solution is similar to that of form (1) above.

Example: Solve $y^2 (dy/dx) = x + y^3$

Solution: The given equation can be rewritten as $y^2 \left(\frac{dy}{dx} \right) - y^3 = x \dots (i)$

Putting $y^3 = v$ or $3y^2 \left(\frac{dy}{dx} \right) = \frac{dv}{dx}$, the equation (i) reduces to $\frac{dv}{dx} - 3v = 3x$, which is the standard form of linear equation in v . Its integrating factor $= e^{\int -3dx} = e^{-3x}$

$\therefore$ Its solution is $ve^{-3x} = c + \int 3xe^{-3x} dx$ or $ve^{-3x} = c + 3 \left[x \frac{e^{-3x}}{(-3)} - \int 1 \cdot \frac{e^{-3x}}{(-3)} dx \right]$, or integrating by parts taking x as first function.

or $y^3 e^{-3x} = c - xe^{-3x} - \frac{1}{3} e^{-3x}$, where c is an arbitrary constant.

Example: Solve $(x^3 y^2 + xy) dx = dy$.

Solution: The given equation can be written as $\frac{dy}{dx} = xy + x^3 y^2$ or

$$\left(\frac{dy}{dx} \right) - xy = x^3 y^2$$

or $\frac{1}{y^2} \frac{dy}{dx} - x \cdot \frac{1}{y} = x^3$, dividing both sides by y^2

$$\frac{dv}{dx} + xv = x^3 \dots \left(\text{putting } v = -\frac{1}{y} \right)$$

This is in the standard form of linear equation and its integrating factor $= e^{\int x dx} = e^{x^2/2}$

Multiplying both sides by the integrating factor and integrating we get

$$v \cdot e^{\frac{1}{2}x^2} = C + \int x^3 e^{\frac{1}{2}x^2} dx, \text{ where } C \text{ is an arbitrary constant.}$$

$$v \cdot e^{\frac{1}{2}x^2} = C + 2 \int \frac{1}{2} x^2 e^{\frac{1}{2}x^2} \cdot x dx$$

$$= C + 2 \int t e^t dt, \text{ putting } t = \frac{1}{2} x^2$$

$$= C + 2 \left(t e^t - \int e^t dt \right) = C + 2e^t (t - 1)$$

$$\text{or } -(1/y) e^{\frac{1}{2}x^2} = C + 2e^{\frac{1}{2}x^2} \left(\frac{1}{2} x^2 - 1 \right)$$

$$\therefore t = \frac{1}{2} x^2 \text{ and } v = -(1/y)$$

$$\text{Or } 1 - 2y \left(1 - \frac{1}{2} x^2 \right) = -C y e^{-\frac{x^2}{2}}$$

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CHAPTER 3

GENERAL THEORY OF LINEAR DIFFERENTIAL EQUATION OF HIGHER ORDER

3.1. Definitions

Linear differential equations are those in which the dependent variable and its derivatives occurs only in the first degree and are not multiplied together.

Thus, the general linear differential equation of the n^{th} order is of the form

$$a_0(x) \cdot \frac{d^n y}{dx^n} + a_1(x) \cdot \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1}(x) \cdot y' + a_n(x) y = Q(x), x \in I = [a, b] \text{ Where } a_0(x), a_1(x), \dots, a_n(x) \text{ and } Q(x) \text{ are continuous functions of } x \text{ and } a_0(x) \neq 0 \forall x. \dots\dots (1)$$

It can be written as $[a_0(x)D^n + a_1(x)D^{n-1} + \dots + a_{n-1}(x)D + a_n(x)]y = Q(x)$

, where $D = \frac{d}{dx}$, this can be written as

$$L(D)y = Q(x) \text{ Where } L(D) = a_0(x)D^n + a_1(x)D^{n-1} + \dots + a_{n-1}(x)D + a_n(x)$$

3.2. Classification of Linear Differential Equation

Homogenous differential/ Non-Homogenous differential equation: The linear differential equation (1) is called Homogenous if $Q(x) \equiv 0$ otherwise non-homogenous differential equation

Variable Coefficient/Constant Coefficient: If all the coefficient $a_0(x), a_1(x), \dots, a_n(x)$ of the differential equation are constant then differential equation is called with constant coefficients. Otherwise differential equation is called variable coefficients

Example: $x^2 \frac{d^2 y}{dx^2} - x(x+2) \frac{dy}{dx} + (x+2)y = 0$ is second order linear homogenous differential equation with variable coefficients

Example: $y'' + xy' + (1-x^2)y = \sin x$ is second order linear non-homogenous differential equation with variable coefficients

Example: $\frac{d^2 y}{dx^2} - y = e^{-x}, x \in \mathbb{R}, y(0) = \frac{dy}{dx}(0) = 0$ is second order linear non-homogenous differential equation with Constant coefficients

Linear Combination: Let $f_1, f_2, \dots, f_n$ are n function defined on common domain D , then the expression $c_1 f_1 + c_2 f_2 + \dots + c_n f_n$ (Where $c_1, c_2, \dots, c_n$ are constants) is called the linear combination of $f_1, f_2, \dots, f_n$.

Convex Combination: A linear combination $\sum_{i=1}^n c_i f_i(x)$ is called convex

combination if $\sum_{i=1}^n c_i = 1$ and $c_i \geq 0, \forall i$

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Linearly Independent Function: The n -functions $f_1, f_2, \dots, f_n$ are called linearly independent on common domain D . If zero function does not have any non trivial linear combination of $f_1, f_2, \dots, f_n$ that is

$$\text{If } c_1 f_1 + c_2 f_2 + \dots + c_n f_n = 0 \quad \forall x \in D \Rightarrow c_1 = c_2 = \dots = c_n = 0.$$

Linearly Dependent: The n -functions $f_1, f_2, \dots, f_n$ are called linearly dependent on common domain D . If $\exists c_1, c_2, \dots, c_n$ (not all zero) such that $c_1 f_1 + c_2 f_2 + \dots + c_n f_n = 0 \quad \forall x \in D$ that is zero function have a non trivial linear combination of $f_1, f_2, \dots, f_n$.

Example: $f_1(x) = x$, $f_2(x) = |x|$ defined on $\mathbb{R}$. Then

- (a) f_1, f_2 are Linearly dependent on $[0, \infty)$
- (b) f_1, f_2 are Linearly dependent on $(-\infty, 0]$
- (c) f_1, f_2 are Linearly independent on $(-\infty, \infty)$

Example: $f_1(x) = e^{mx}$, $f_2(x) = e^{nx}$ defined on $\mathbb{R}$. f_1 and f_2 are linearly independent iff $m \neq n$

Example: $f_1 = x|x|$, $f_2 = x^2$ Then

- (a) f_1, f_2 are Linearly dependent on $[0, \infty)$
- (b) f_1, f_2 are Linearly dependent on $(-\infty, 0]$
- (c) f_1, f_2 are Linearly independent on $(-\infty, \infty)$

Principle of Superposition: Consider the n^{th} order linear differential equation (1) if $y_1, y_2, \dots, y_n$ are any n solutions of the differential equation then $y(x) = \sum_{i=1}^n c_i y_i$ is solution of the differential equation if either $Q(x) \equiv 0$ or

$$\sum_{i=1}^n c_i = 1$$

Example: If y_1 and y_2 are linearly independent solution of $y'' + xy' + (1-x^2)y = \sin x$. Then which of the following are also solution

- (a) $y_1 + y_2$
- (b) $y_1 - y_2$
- (c) $2y_1 - y_2$
- (d) $y_1 - 2y_2$

Ans: c

Example: Maximum number of linearly independent solution of $\frac{d^4 y}{dx^4} = 0$,

$$y(0) = 1$$

- (a) 1
- (b) 2
- (c) 3
- (d) 4

Ans: d

Hint: Solution of the differential equation is $y = \frac{Ax^3}{6} + \frac{Cx^2}{2} + Dx + 1$

Proposition 3.1: Consider the n^{th} order linear differential equation (1) if $y_1, y_2, \dots, y_n$ are any n solutions then

- (i) Linear Combination of any n solutions $y_1, y_2, \dots, y_n$ of differential equation (1) is also solution if either $Q(x) \equiv 0$ or $\sum_{i=1}^n c_i = 1$
- (ii) Every Convex Combination of any n solutions $y_1, y_2, \dots, y_n$ of differential equation (1) is also solution
- (iii) Every Linear Combination of any n solutions $y_1, y_2, \dots, y_n$ of n^{th} order homogeneous linear differential equation is also solution

Fundamental Set of Solution: If $f_1, f_2, \dots, f_n$ are n linearly independent solutions of the n^{th} -order homogeneous linear differential equation is called fundamental set of solutions.

Proposition 3.2:

- (i) The solution set of the n^{th} order homogeneous linear differential equation forms a vector space.
- (ii) The solution set of the n^{th} order homogeneous linear differential equation is vector space of dimension of n .
- (iii) Every n^{th} order homogeneous linear differential equation has exactly n linearly independent solutions i.e., every set of $n+1$ solutions is linearly dependent set.
- (iv) The solution set of the n^{th} order non-homogeneous linear differential equation never a vector space.

3.3. Wronskian of the n^{th} order Homogeneous Linear Differential Equation

Consider the n^{th} order linear differential equation of the form

$$a_0(x) \cdot \frac{d^n y}{dx^n} + a_1(x) \cdot \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1}(x) \cdot y' + a_n(x) y = Q(x), x \in I = [a, b] \quad \text{Where}$$

$a_0(x), a_1(x), \dots, a_n(x)$ and $Q(x)$ are continuous functions of x and $a_0(x) \neq 0 \forall x$. And let $y_1, y_2, \dots, y_n$ are any n solutions of differential equation then Wronskian of $y_1, y_2, \dots, y_n$ is defined as follow

$$W(x) = \begin{vmatrix} y_1(x) & y_2(x) & \dots & y_n(x) \\ y_1'(x) & y_2'(x) & \dots & y_n'(x) \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)}(x) & y_2^{(n-1)}(x) & \dots & y_n^{(n-1)}(x) \end{vmatrix} \quad \text{called Wronskian of } n \text{ solutions}$$

Let $y_1, y_2, \dots, y_n$ are any n solutions of n^{th} order linear differential equation then $W(x)$ Wronskian of $y_1, y_2, \dots, y_n$ is always continuous and differentiable function but higher order derivative may or may not exists

Wronskian of Second Order Linear Homogeneous Differential Equation

Consider the second order homogeneous linear differential equation

$$a_0(x)y'' + a_1(x)y' + a_2(x)y = 0, \forall x \in I = [a, b], a_0(x) \neq 0, \forall x \in I = [a, b], \dots (2)$$

Where $a_0(x), a_1(x), a_2(x)$ are continuous functions of x , $a_0(x) \neq 0 \forall x$

Moreover, let y_1, y_2 are any two solutions of differential equation then $W(x)$

Wronskian of y_1, y_2 is given by
$$W(x) = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix},$$

$$\Rightarrow W(x) = y_1(x)y_2'(x) - y_1'(x)y_2(x)$$

Note: Wronskian is a differentiable function & hence continuous

As y_1 and y_2 are two solution of (2) $\Rightarrow$

$$a_0(x)y_1'' + a_1(x)y_1' + a_2(x)y_1 = 0 \dots (3)$$

$$a_0(x)y_2'' + a_1(x)y_2' + a_2(x)y_2 = 0 \dots (4)$$

Multiply (3) by y_2 and (4) by y_1 and subtract we will get

$$a_0(x)[y_1''y_2 - y_1y_2''] + a_1(x)[y_2y_1' - y_1y_2'] = 0$$

$$\Rightarrow -a_0(x)W'(x) + a_1(x)(-W(x)) = 0$$

$$\Rightarrow a_0(x)W'(x) + a_1(x)W(x) = 0$$

$\Rightarrow W(x)$ is solution of first order linear differential equation hence differentiable and continuous

And given by $W(x) = Ae^{-\int \frac{a_1(x)}{a_0(x)} dx}$

Proposition 3.3: $a_0(x)y'' + a_1(x)y' + a_2(x)y = 0, \forall x \in I = [a, b]$,

$$a_0(x) \neq 0, \forall x \in I = [a, b],$$

Where $a_0(x), a_1(x), a_2(x)$ are continuous functions of x , $a_0(x) \neq 0 \forall x$

Moreover, let y_1, y_2 are any two solutions of differential equation then $W(x)$

Wronskian of y_1, y_2 is given by $W(x) = Ae^{-\int \frac{a_1(x)}{a_0(x)} dx}$ called **Abel's Formula**

3.4. Important Results on Wronskian

Let y_1, y_2 are any two solutions of the second order homogeneous linear differential equation

$$a_0(x)y'' + a_1(x)y' + a_2(x)y = 0, \forall x \in I = [a, b], a_0(x) \neq 0, \forall x \in I = [a, b], \text{ where}$$

$a_0(x), a_1(x), a_2(x)$ are continuous functions of x , $a_0(x) \neq 0 \forall x$ and $W(x)$

Wronskian of y_1, y_2 then

Proposition 3.4: $W(x)$ Wronskian of y_1, y_2 , $W(x) = Ae^{-\int \frac{a_1(x)}{a_0(x)} dx}$ is either identically zero or never zero that is

(i) If $W(x_0) = 0$ for Some $x_0 \Rightarrow W(x) = 0 \forall x$

(ii) If $W(x_0) \neq 0$ for Some $x_0 \Rightarrow W(x) \neq 0 \forall x$

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(iii) $W(x_0) = \alpha$

(a) If $\alpha > 0 \Rightarrow W(x) > 0 \forall x$

(b) If $\alpha < 0 \Rightarrow W(x) < 0 \forall x$

(c) If $\alpha = 0 \Rightarrow W(x) = 0 \forall x$

Proposition 3.5: If $W(x)$ Wronskian of y_1, y_2

(i) $W(x)$ Wronskian of y_1, y_2 identically zero iff y_1 & y_2 are linearly dependent

(ii) $W(x)$ Wronskian of y_1, y_2 never zero iff y_1 & y_2 are linearly independent

(iii) y_1 and y_2 are linearly independent $\Leftrightarrow \exists x_0$ such that $W(x_0) \neq 0$

(iv) y_1 and y_2 are linearly dependent $\Leftrightarrow \exists x_0$ such that $W(x_0) = 0$

Proposition 3.6: If $W(x)$ Wronskian of y_1, y_2 then $W(y_1, y_2) = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix}$

(i) If $\exists x_0$ such that $y_1(x_0) = 0 = y_2(x_0)$ i.e., x_0 is common zero of y_1 and $y_2 \Rightarrow$

$$W(x_0) = \begin{vmatrix} y_1(x_0) & y_2(x_0) \\ y_1'(x_0) & y_2'(x_0) \end{vmatrix} = \begin{vmatrix} 0 & 0 \\ y_1'(x_0) & y_2'(x_0) \end{vmatrix} = 0 \Rightarrow W(x) \equiv 0 \Rightarrow y_1 \& y_2$$

are linearly dependent

(ii) If y_1 & y_2 are linearly independent then they never have common zero

(iii) If $\exists x_0$ such that $y_1'(x_0) = 0 = y_2'(x_0)$ or x_0 is common point of extrema of y_1 and $y_2 \Rightarrow$

$$W(x_0) = \begin{vmatrix} y_1(x_0) & y_2(x_0) \\ y_1'(x_0) & y_2'(x_0) \end{vmatrix} = \begin{vmatrix} y_1(x_0) & y_2(x_0) \\ 0 & 0 \end{vmatrix} = 0 \Rightarrow W(x) \equiv 0 \Rightarrow y_1 \& y_2$$

are linearly dependent

(iv) If y_1 & y_2 are linearly independent then they never have common point of extrema

(v) If $\exists x_0$ such that $y_1(x_0) = 0 = y_1'(x_0)$, i.e. x_0 is repeated zero of $y_1 \Rightarrow$

$$W(x_0) = \begin{vmatrix} y_1(x_0) & y_2(x_0) \\ y_1'(x_0) & y_2'(x_0) \end{vmatrix} = \begin{vmatrix} 0 & y_2(x_0) \\ 0 & y_2'(x_0) \end{vmatrix} = 0 \Rightarrow W(x) \equiv 0 \Rightarrow y_1 \& y_2$$

are linearly dependent

(vi) If $\exists x_0$ such that $y_2(x_0) = 0 = y_2'(x_0)$, i.e. x_0 is repeated zero of $y_2 \Rightarrow$

$$W(x_0) = \begin{vmatrix} y_1(x_0) & y_2(x_0) \\ y_1'(x_0) & y_2'(x_0) \end{vmatrix} = \begin{vmatrix} y_1(x_0) & 0 \\ y_1'(x_0) & 0 \end{vmatrix} = 0 \Rightarrow W(x) \equiv 0 \Rightarrow y_1 \& y_2$$

are linearly dependent

(vii) let y_1 & y_2 are two solution of the second order linear differential equation. If either y_1 or y_2 has repeated root $\Rightarrow y_1$ and y_2 are linearly dependent

Put Your Own Notes

- (viii) If y_1 & y_2 are linearly independent then neither y_1 nor y_2 have repeated zero
- (ix) let y_1 be solution of the second order linear differential equation such that y_1 has repeated root $\Rightarrow y_1$ is linearly dependent to every solution $y_2 \Rightarrow y_1$ is trivial solution that is $y_1(x) \equiv 0$
- (x) Every non trivial solution y_1 of the second order linear differential equation never have repeated zero
- (xi) Every zero of non trivial solution y_1 of the second order linear differential equation is simple zero

Example: Let consider the second order linear homogeneous differential equation $a_0(x)y'' + a_1(x)y' + a_2(x)y = 0, \forall x \in I = [a, b]$, Where $a_0(x) \neq 0, \forall x \in I$ and $a_0(x), a_1(x), a_2(x)$ are continuous functions of x . If y_1 and y_2 are two solution such that $y_1(0) = 0, y_2(x) = 1, y_1'(0) = 1, y_2'(x) = -1$ then

- (a) y_1 and y_2 has common zero
- (b) y_1 & y_2 has common point of extrema
- (c) Neither y_1 nor y_2 has repeated root
- (d) y_1 and y_2 has never have common zero

Ans: d

Example: Let consider the second order linear homogeneous differential equation $a_0(x)y'' + a_1(x)y' + a_2(x)y = 0, \forall x \in I = [a, b]$, Where $a_0(x) \neq 0, \forall x \in I$ and $a_0(x), a_1(x), a_2(x)$ are continuous functions of x . If y_1 and y_2 are two solution of such that $y_1(0) = 0, y_2(0) = 1, y_1'(0) = 1, y_2'(0) = -1$

If $y_2(x) \neq 0 \forall x$ and $g(x) = \frac{y_1(x)}{y_2(x)}$ then

- (a) $g(x)$ is strictly increasing.
- (b) $g(x)$ is strictly decreasing.
- (c) $g(x)$ is constant function.
- (d) $g(x)$ has a point of extrema.

Ans: a

Example: Let y_1 and y_2 form a complete set (fundamental set) of solution of $y'' - 2xy' + \sin(e^{2x^2})y = 0 \forall x \in [0, 1]$ such that $y_1(0) = 0, y_2(0) = 1, y_1'(0) = 1, y_2'(0) = 0$ then  of y_1 and y_2 at $x = 1$

- (a) e^2
- (b) $-e$
- (c) $-e^2$
- (d) e

Ans: b

Hint: Apply Abel's Formula

Example: Let y_1 and y_2 are two solution defined on $[0,1]$,

$y''(x) + y'(x) + y(x) = 0$ and $W(x)$ of y_1 & y_2 such that $W\left(\frac{1}{2}\right) = 0$ then

(a) $W(x) = 0 \quad \forall x \in [0,1]$

(b) $W(x) \neq 0 \quad \forall x \in \left[0, \frac{1}{2}\right]$

(c) $W(x) > 0 \quad \forall x \in \left[\frac{1}{2}, 1\right]$

(d) $W(x) < 0 \quad \forall x \in \left[0, \frac{1}{2}\right]$

Ans: a

Hint: $W(x)$ either never zero or identically zero

Example: Let y_1 and y_2 are two linearly independent solution of $xy'' - 2x^2y' + e^xy = 0$ such that $y_1(0) = 1$, $y_1'(0) = 1$, $y_2(0) = -1$, $y_2'(0) = 1$ then find $W(2)$

(a) $2e^{-1}$

(b) $2e^{-2}$

(c) $2e^4$

(d) $2e^{-4}$

Ans: c

Hint: Use Able's Formula

Example: Let u and v are two solution of $y'' + P(x)y' + Q(x)y = 0$ defined on $[a,b]$

(a) $W(u,v)_{x=x_0} = 0 \Rightarrow u$ & v are linearly dependent, $x_0 \in [a,b]$

(b) $W(u,v) \equiv 0 \Rightarrow u$ & v are linearly dependent

(c) $W(u,v) = 0$ for some $x_0 \not\Rightarrow W(u,v) \equiv 0$

(d) $W(u,v) \neq 0, \forall x$ but u & v are linearly dependent.

Ans: b, c

3.5. Analysis of the Set of Zeros of Non-trivial Solutions of Linear Differential Equation

Consider the second order homogeneous linear differential equation $a_0(x)y'' + a_1(x)y' + a_2(x)y = 0, \forall x \in I = [a,b]$, $a_0(x) \neq 0, \forall x \in \mathbb{R}$, where $a_0(x), a_1(x), a_2(x)$ are continuous functions of x .

Put Your Own Notes

Proposition 3.7: If $y(x)$ be any solution of homogenous linear differential equation and let $A = \{a \in \mathbb{R} | y(a) = 0\}$ be the set of zeros of $y(x)$ then

- (i) If $\exists x_0 \in A$ s.t. $y'(x_0) = 0$ then $y(x)$ is trivial solution
- (ii) If $y(x)$ is non-trivial solution then every element of A is Isolated point
- (iii) If $y(x)$ is non-trivial solution then A is countable set.
- (iv) If $y(x)$ is non-trivial solution then A cannot be a bounded infinite set
- (v) A is a bounded and infinite then $y(x)$ cannot be non-trivial solution
- (vi) If $y(x)$ is non-trivial solution and A is an infinite set $\Leftrightarrow A$ is an unbounded set
- (vii) If $y(x)$ is non-trivial solution and A is an finite set $\Leftrightarrow A$ is an bounded set
- (viii) If differential equation defined on bounded domain then the set of zero of every non-trivial solution is bounded

***The above all the statements also hold for $y(x)$ solution of n^{th} order homogeneous differential equation.

Proposition 3.8: Let y_1 and y_2 are two linearly independent solutions of $a_0(x)y'' + a_1(x)y' + a_2(x)y = 0, \forall x \in I = [a, b]$, $a_0(x) \neq 0, \forall x \in \mathbb{R}$, where $a_0(x), a_1(x), a_2(x)$ are continuous functions of x , $a_0(x) \neq 0 \forall x$ and $A = \{a \in \mathbb{R} | y_1(a) = 0\}$ be a set of zeros of y_1 , $B = \{b \in \mathbb{R} | y_2(b) = 0\}$ be a set of zeros of y_2 then

- (i) $A \cap B = \emptyset$ i.e. y_1 & y_2 does not have common zero
- (ii) $\nexists x_0 \in A \cap B$ s.t. $y_1'(x_0) = 0$ and $y_2'(x_0) = 0$ i.e. y_1 & y_2 does not have common point of extrema
- (iii) Every point of A & B is isolated point
- (iv) A & B are countable set
- (v) Between any two successive zeros of $y_1(x)$ there must exists exactly one zero of $y_2(x)$ and vice versa that is $x_0, x_1 \in A$ successive elements of A there must exists $\beta \in B \cap (x_0, x_1)$ and vice versa
- (vi) If $|A| \geq 2 \Rightarrow |B| \geq 1$.
- (vii) If $|A| \geq 2 \Rightarrow \exists x_0 \in A^c$ (complement of A) such that $y_1'(x_0) = 0$.
- (viii) If $|A| \geq 3 \Rightarrow \exists x_0 \in B^c$ (complement of B) such that $y_2'(x_0) = 0$.
- (ix) If A is finite $\Leftrightarrow B$ is finite.
- (x) If A is bounded $\Leftrightarrow B$ is bounded
- (xi) If A is finite $\Leftrightarrow B$ is bounded
- (xii) If $\exists a_0, a_1 \in A$ such that $a_0 < b < a_1, \forall b \in B$ and A finite (i.e., If A is finite and smallest zero, largest zero of $y_1(x)$) $\Rightarrow |A| \neq |B|$.

- (xiii) If $\exists b_0, b_1 \in B$ such that $b_0 < a < b_1$, $\forall a \in A$ and B finite (i.e., If B is finite and smallest zero, largest zero of $y_2(x) \Rightarrow |A| \neq |B|$).
- (xiv) If A is finite and smallest zero of $y_1(x)$ and largest zero of $y_2(x) \Rightarrow |A| = |B|$
- (xv) If B is finite and smallest zero of $y_2(x)$ and largest zero of $y_1(x) \Rightarrow |A| = |B|$
- (xvi) If $\Rightarrow |A| = |B| \geq 1$ (finite) and if smallest zero of $y_1(x) \Rightarrow$ Largest zero of $y_2(x)$
- (xvii) If $\Rightarrow |A| = |B| \geq 1$ (finite) and if smallest zero of $y_2(x) \Rightarrow$ Largest zero of $y_1(x)$
- (xviii) If A is infinite then A and B are equivalent
- (xix) If $|A| \geq 2$ then $y_1(x)$ Non-monotonic solution
- (xx) If $|B| \geq 2$ then $y_2(x)$ Non-monotonic solution

3.6. Second Order Linear Differential Equation in Which First Order Term is Missing

- (a) Consider the differential equation $y'' + q(x)y = 0$, $q(x) < 0 \forall x \in \mathbb{R}$ and let $y(x)$ be a non-trivial solution then

Proposition 3.9:

- (i) $y(x)$ has at most one zero
 - (ii) $y(x)$ either strictly Increasing or strictly decreasing
 - (iii) $y'(x)$ either strictly Increasing or strictly decreasing
 - (iv) If $y'(x_0) > 0$ for some $x_0 \Rightarrow y(x)$ & $y'(x)$ are strictly increasing
 - (v) If $y'(x_0) < 0$ for some $x_0 \Rightarrow y(x)$ & $y'(x)$ both are strictly decreasing
 - (vi) $\nexists x_0$ such that $y'(x_0) = 0$ i.e. $y(x)$ does not have point of extrema
 - (vii) $\nexists x_0$ such that $y''(x_0) = 0$ i.e. $y'(x)$ does not have point of extrema
 - (viii) If $\exists x_0$ s.t. $y'(x_0) > 0 \Rightarrow \lim_{x \rightarrow \infty} y(x) = \infty, \lim_{x \rightarrow \infty} y'(x) = \infty$ and $\lim_{x \rightarrow -\infty} y(x) = -\infty, \lim_{x \rightarrow -\infty} y'(x) = -\infty$
 - (ix) If $\exists x_0$ s.t. $y'(x_0) < 0 \Rightarrow \lim_{x \rightarrow \infty} y(x) = -\infty, \lim_{x \rightarrow \infty} y'(x) = -\infty$ and $\lim_{x \rightarrow -\infty} y(x) = \infty, \lim_{x \rightarrow -\infty} y'(x) = \infty$
- (b) Let $y(x)$ be a non-trivial solution of $y'' + q(x)y = 0$ where $q(x) > 0 \forall x > 0$ and $\int_1^{\infty} q(x) dx = \infty$

Proposition 3.10:

- (i) $y(x)$ has infinite number of zero's on positive x-axis
 - (ii) $y(x)$ is oscillatory solution (non-monotonic)
 - (iii) $y(x)$ has infinite points of local extrema
- $\nexists x_0 \in \mathbb{R}^+$ such that $y(x_0) = 0$ & $y'(x_0) = 0$ i.e., Every zero of $y(x)$ simple zero

CHAPTER 4

SOLUTION OF LINEAR DIFFERENTIAL EQUATION WITH CONSTANT AND VARIABLE COEFFICIENTS

4.1 Linear Differential Equation with Constant Coefficient

$$\frac{d^n y}{dx^n} + k_1 \frac{d^{n-1} y}{dx^{n-1}} + k_2 \frac{d^{n-2} y}{dx^{n-2}} + \dots + k_n y = Q(x) \quad (1)$$

Operator D

Denoting $\frac{d}{dx}, \frac{d^2}{dx^2}, \frac{d^3}{dx^3}$ etc. by D, D^2, D^3 etc., that

$\frac{dy}{dx} = Dy, \frac{d^2 y}{dx^2} = D^2 y, \frac{d^3 y}{dx^3} = D^3 y$ etc., the equation (1) above can be written in

the symbolic form $(D^n + k_1 D^{n-1} + \dots + k_n) y = Q(x)$ i.e., $f(D) y = Q(x)$,

Where $f(D) = D^n + k_1 D^{n-1} + \dots + k_n$ i.e. a polynomial in D .

Thus the symbol D stands for the operations of differentiation and can be treated much the same as an algebraic quantity i.e. $f(D)$ can be factorized by ordinary rules of algebra and the factors may be taken in any order. For instance

Example: $\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} - 3y = (D^2 + 2D - 3)y = (D+3)(D-1)y$ or
 $(D-1)(D+3)y$.

4.1.1 Complementary Function of Homogeneous Linear Differential Equation

To solve the equation $\frac{d^n y}{dx^n} + k_1 \frac{d^{n-1} y}{dx^{n-1}} + k_2 \frac{d^{n-2} y}{dx^{n-2}} + \dots + k_n y = 0 \quad (1)$

Where k_i 's are constants.

The equation (1) in symbolic form is

$$(D^n + k_1 D^{n-1} + k_2 D^{n-2} + \dots + k_n) y = 0 \quad (2)$$

Its symbolic co-efficient equated to zero i.e.,

$$D^n + k_1 D^{n-1} + k_2 D^{n-2} + \dots + k_n = 0$$

Is called the auxiliary equation (A.E.) .Let $m_1, m_2, \dots, m_n$ be its roots.

Case I: If all the roots be real and different, then (2) is equivalent to

$$(D - m_1)(D - m_2) \dots (D - m_n) y = 0 \quad (3)$$

Now (3) will be satisfied by the solution of $(D - m_n) y = 0$ i.e., by

$$\frac{dy}{dx} - m_n y = 0$$

This is a Leibnitz's linear and I.F. = $e^{-m_n x}$

∴ Its solution is $ye^{-m_n x} = c_n$, i.e., $y = c_n e^{m_n x}$

Similarly, since the factors in (3) can be taken in any order, it will be satisfied by the solutions of $(D - m_1)y = 0, (D - m_2)y = 0$ etc. i.e., by $y = c_1 e^{m_1 x}, y = c_2 e^{m_2 x}$ etc.

Thus the complete solution of (1) is $y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \dots + c_n e^{m_n x}$ (4)

Case II: If two roots are equal (i.e. $m_1 = m_2$), then (4) becomes

$$y = (c_1 + c_2) e^{m_1 x} + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}$$

$$y = C e^{m_1 x} + c_3 e^{m_3 x} + \dots + c_n e^{m_n x} \quad [\because c_1 + c_2 = \text{one arbitrary constant } C]$$

It has only $n-1$ arbitrary constants and is, therefore, not the complete solution of (1). In this case, we proceed as follows:

The part of the complete solution corresponding to the repeated root is the complete solution of $(D - m_1)(D - m_1)y = 0$

Putting $(D - m_1)y = z$, it becomes $(D - m_1)z = 0$ or $\frac{dz}{dx} - m_1 z = 0$

This is a Leibnitz's linear in z and I.F. = $e^{-m_1 x}$

∴ Its solution is $ze^{-m_1 x} = c_1$ or $z = c_1 e^{m_1 x}$

Thus $(D - m_1)y = z = c_1 e^{m_1 x}$ or $\frac{dy}{dx} - m_1 y = c_1 e^{m_1 x}$ (5)

Its I.F. being $e^{-m_1 x}$, the solution of (5) is

$$ye^{-m_1 x} = \int c_1 e^{m_1 x} \cdot e^{-m_1 x} + c_2 \text{ or } y = (c_1 x + c_2) e^{m_1 x}$$

Thus the complete solution of (1) is $y = (c_1 x + c_2) e^{m_1 x} + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}$

If, however, the A.E. has three equal roots (i.e., $m_1 = m_2 = m_3$), then the complete solution is

$$y = (c_1 x^2 + c_2 x + c_3) e^{m_1 x} + c_4 e^{m_4 x} + \dots + c_n e^{m_n x}$$

Case III: If one pair of roots be imaginary, i.e., $m_1 = \alpha + i\beta, m_2 = \alpha - i\beta$, then the complete solution is

$$y = c_1 e^{(\alpha + i\beta)x} + c_2 e^{(\alpha - i\beta)x} + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}$$

$$= e^{\alpha x} (c_1 e^{i\beta x} + c_2 e^{-i\beta x}) + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}$$

$$= e^{\alpha x} [c_1 (\cos \beta x + i \sin \beta x) + c_2 (\cos \beta x - i \sin \beta x)] + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}$$

$$[\because \text{by Euler's Theorem, } e^{i\theta} = \cos \theta + i \sin \theta]$$

$$= e^{\alpha x} (C_1 \cos \beta x + C_2 \sin \beta x) + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}$$

Where $C_1 = c_1 + c_2$ and $C_2 = i(c_1 - c_2)$

Case IV: If two pairs of imaginary roots be equal i.e. $m_1 = m_2 = \alpha + i\beta$, $m_3 = m_4 = \alpha - i\beta$, the by case II, the complete solution is

$$y = e^{\alpha x} [(c_1 x + c_2) \cos \beta x + (c_3 x + c_4) \sin \beta x] + \dots + c_n e^{m_n x}$$

Example: Solve $(D^3 + D^2 + 4D + 4)y = 0$.

Solution: Here the A.E. is $D^3 + D^2 + 4D + 4 = 0$ i.e. $(D^2 + 4)(D + 1) = 0 \therefore$
 $D = -1, \pm 2i$.

Hence the C.S is $y = c_1 e^{-x} + e^{0x} (c_2 \cos 2x + c_3 \sin 2x)$

i.e. $y = c_1 e^{-x} + c_2 \cos 2x + c_3 \sin 2x$

Example: Solve $\frac{d^4 x}{dt^4} + 4x = 0$.

Solution: Given equation in symbolic form is $(D^4 + 4)x = 0$.

$\therefore$ A.E. is $D^4 + 4 = 0$ or $(D^4 + 4D^2 + 4) - 4D^2 = 0$ or $(D^2 + 2)^2 - (2D)^2 = 0$

or $(D^2 + 2D + 2)(D^2 - 2D + 2) = 0$

$\therefore$ either $D^2 + 2D + 2 = 0$ or $D^2 - 2D + 2 = 0$

Whence $D = \frac{-2 \pm \sqrt{(-4)}}{2}$ and $D = \frac{2 \pm \sqrt{(-4)}}{2}$ i.e. $D = -1 \pm i$ and $1 \pm i$,

Hence the required solution is $x = e^{-t} (c_1 \cos t + c_2 \sin t) + e^t (c_3 \cos t + c_4 \sin t)$.

Inverse Operator

- (i) **Definition:** $\frac{1}{f(D)}Q(x)$ is that function of x , not containing arbitrary constants which when operated upon by $f(D)$ gives $Q(x)$.

i.e. $f(D) \left\{ \frac{1}{f(D)}Q(x) \right\} = Q(x)$

Thus $\frac{1}{f(D)}Q(x)$ satisfies the equation $f(D)y = Q(x)$ and is, therefore its particular integral.

Obviously, $f(D)$ and $\frac{1}{f(D)}$ are inverse operators.

(ii) $\frac{1}{D}Q(x) = \int Q(x) dx$

Let $\frac{1}{D}Q(x) = y$ (i)

Operating by D , $D \frac{1}{D}Q(x) = Dy$ i.e. $Q(x) = \frac{dy}{dx}$

Integrating both sides w. r. t $x, y = \int Q(x)dx$, no constant being added as (i) does not contain any constant.

$$\text{Thus } \frac{1}{D} Q(x) = \int Q(x)dx$$

$$(iii) \quad \frac{1}{D-a} Q(x) = e^{ax} \int Q(x)e^{-ax} dx.$$

$$\text{Let } \frac{1}{D-a} Q(x) = y \quad (ii)$$

$$\text{Operating by } D-a, (D-a) \cdot \frac{1}{D-a} Q(x) = (D-a)y$$

or $Q(x) = \frac{dy}{dx} - ay$, i.e. $\frac{dy}{dx} - ay = Q(x)$ which is a Leibnitz's linear equation.

$\therefore$ I.F. being e^{-ax} , its solution is

$ye^{-ax} = \int Q(x)e^{-ax} dx$, no constant being added as (ii) doesn't contain any constant.

$$\text{Thus } \frac{1}{D-a} Q(x) \Rightarrow y = e^{ax} \int Q(x)e^{-ax} dx$$

4.1.2 Rules for Finding The Particular Integral of Non-Homogeneous Linear Differential Equation

$$\text{Consider the equation } \frac{d^n y}{dx^n} + k_1 \frac{d^{n-1} y}{dx^{n-1}} + k_2 \frac{d^{n-2} y}{dx^{n-2}} + \dots + k_n y = Q(x)$$

$$\text{Which in symbolic form is } (D^n + k_1 D^{n-1} + k_2 D^{n-2} + \dots + k_n) y = Q(x)$$

$$\therefore \text{P.I.} = \frac{1}{D^n + k_1 D^{n-1} + k_2 D^{n-2} + \dots + k_n} Q(x)$$

Case I: When $Q(x) = e^{ax}$

$$\text{Since } De^{ax} = ae^{ax}$$

$$D^2 e^{ax} = a^2 e^{ax}$$

.....

.....

$$D^n e^{ax} = a^n e^{ax}$$

$\therefore$

$$(D^n + k_1 D^{n-1} + \dots + k_n) e^{ax} = (a^n + k_1 a^{n-1} + \dots + k_n) e^{ax} \text{ i.e. } f(D) e^{ax} = f(a) e^{ax}$$

$$\text{Operating on both sides by } \frac{1}{f(D)}, \frac{1}{f(D)} f(D) e^{ax} = \frac{1}{f(D)} f(a) e^{ax} \text{ or}$$

$$e^{ax} = f(a) \frac{1}{f(D)} e^{ax}.$$

∴ Dividing by $f(a)$.

$$\frac{1}{f(D)} e^{ax} = \frac{1}{f(a)} e^{ax} \text{ Provided } f(a) \neq 0 \quad (1)$$

If $f(a) = 0$ the above rule fails and we proceed further

Since a is a root of A.E. $f(D) = D^n + k_1 D^{n-1} + \dots + k_n = 0$

∴ $D - a$ is a factor of $f(D)$. Suppose $f(D) = (D - a)\phi(D)$, where $\phi(a) \neq 0$. Then

$$\frac{1}{f(D)} e^{ax} = \frac{1}{D - a} \cdot \frac{1}{\phi(D)} e^{ax} = \frac{1}{D - a} \cdot \frac{1}{\phi(a)} e^{ax} \quad (\text{by (1)})$$

$$= \frac{1}{\phi(a)} \cdot \frac{1}{D - a} e^{ax} = \frac{1}{\phi(a)} \cdot e^{ax} \int e^{ax} \cdot e^{-ax} dx$$

$$= \frac{1}{\phi(a)} e^{ax} \int dx = x \frac{1}{\phi(a)} e^{ax}$$

$$\text{i.e. } \frac{1}{f(D)} e^{ax} = x \frac{1}{f'(a)} e^{ax} \text{ provided } f'(a) \neq 0 \quad (2)$$

If $f'(a) = 0$, then applying (2) again, we get $\frac{1}{f(D)} e^{ax} = x^2 \frac{1}{f''(a)} e^{ax}$, provided $f''(a) \neq 0$ (3) and so on.

Case II: When $Q(x) = \sin(ax + b)$ or $\cos(ax + b)$

Since $D \sin(ax + b) = a \cos(ax + b)$

$$D^2 \sin(ax + b) = -a^2 \sin(ax + b)$$

$$D^3 \sin(ax + b) = -a^3 \cos(ax + b)$$

$$D^4 \sin(ax + b) = a^4 \sin(ax + b)$$

$$\text{i.e. } D^2 \sin(ax + b) = (-a^2) \sin(ax + b)$$

$$(D^2)^2 \sin(ax + b) = (-a^2)^2 \sin(ax + b)$$

$$\text{In general } (D^2)^r \sin(ax + b) = (-a^2)^r \sin(ax + b)$$

$$\therefore f(D^2) \sin(ax + b) = f(-a^2) \sin(ax + b)$$

Operating on both sides $\frac{1}{f(D^2)}$,

$$\frac{1}{f(D^2)} \cdot f(D^2) \sin(ax + b) = \frac{1}{f(D^2)} f(-a^2) \sin(ax + b)$$

$$\text{Or } \sin(ax + b) = f(-a^2) \frac{1}{f(D^2)} \sin(ax + b)$$

$$\therefore \text{Dividing by } f(-a^2) \cdot \frac{1}{f(D^2)} \sin(ax + b) = \frac{1}{f(-a^2)} \sin(ax + b) \text{ provided}$$

$$f(-a^2) \neq 0 \quad (4)$$

If $f(-a^2) = 0$, the above rule fails and we proceed further.

Since $\cos(ax+b) + i \sin(ax+b) = e^{i(ax+b)}$ [Euler's theorem]

$$\therefore \frac{1}{f(D^2)} \sin(ax+b) = \text{P. I. of } \frac{1}{f(D^2)} e^{i(ax+b)} \left[\text{Since } f(-a^2) = 0 \therefore \text{by (2)} \right]$$

$$= \text{P.I. of } x \frac{1}{f'(D^2)} e^{i(ax+b)} \text{ Where } D^2 = -a^2$$

$$\therefore \frac{1}{f(D^2)} \sin(ax+b) = x \frac{1}{f'(-a^2)} \sin(ax+b), \text{ provided } f'(-a^2) \neq 0 \quad (5)$$

If $f'(-a^2) = 0$, $\frac{1}{f(D^2)} \sin(ax+b) = x^2 \frac{1}{f''(-a^2)} \sin(ax+b)$, provided $f''(-a^2) \neq 0$, and so on.

Similarly, $\frac{1}{f(D^2)} \cos(ax+b) = \frac{1}{f'(-a^2)} \cos(ax+b)$, provided $f'(-a^2) \neq 0$,

If $f'(-a^2) = 0$, $\frac{1}{f(D^2)} \cos(ax+b) = x \cdot \frac{1}{f''(-a^2)} \cos(ax+b)$, provided $f''(-a^2) \neq 0$.

If $f''(-a^2) = 0$, $\frac{1}{f(D^2)} \cos(ax+b) = x^2 \frac{1}{f'''(-a^2)} \cos(ax+b)$, provided $f'''(-a^2) \neq 0$ and so on.

Case III: When $Q(x) = x^m$

Here P.I. $\frac{1}{f(D)} x^m = [f(D)]^{-1} x^m$

Expand $[f(D)]^{-1}$ in ascending powers of D as far as the term in D^m and operate on x^m term by term. Since the $(m+1)^{\text{th}}$ and higher derivatives of x^m are zero, we need not consider terms beyond D^m .

Case IV: $Q(x) = e^{ax}V$, V being a function of x .

If u is a function of x , then

$$D(e^{ax}u) = e^{ax}Du + ae^{ax}u = e^{ax}(D+a)u$$

$$D^2(e^{ax}u) = e^{ax}D^2u + 2ae^{ax}Du + a^2e^{ax}u = e^{ax}(D+a)^2u$$

and in general, $(D^n e^{ax}u) = e^{ax}(D+a)^n u$

$$\therefore f(D)(e^{ax}u) = e^{ax}f(D+a)^n u$$

Operating both sides by $\frac{1}{f(D)}$,

$$\frac{1}{f(D)} \cdot f(D)(e^{ax}u) = \frac{1}{f(D)}[e^{ax}f(D+a)u]$$

$$e^{ax}u = \frac{1}{f(D)}[e^{ax}f(D+a)u]$$

Now put $f(D+a)u = V$, i.e. $u = \frac{1}{f(D+a)}V$, so that

$$e^{ax} \frac{1}{f(D+a)}V = \frac{1}{f(D)}(e^{ax}V)$$

$$\text{i.e. } \frac{1}{f(D)}(e^{ax}V) = e^{ax} \frac{1}{f(D+a)}V.$$

Case V: When X is any other function of x

$$\text{Here P.I.} = \frac{1}{f(D)}X$$

If $f(D) = (D-m_1)(D-m_2)\dots(D-m_n)$, resolving into partial fractions,

$$\frac{1}{f(D)} = \frac{A_1}{D-m_1} + \frac{A_2}{D-m_2} + \dots + \frac{A_n}{D-m_n}$$

$$\therefore \text{P.I.} = \left[\frac{A_1}{D-m_1} + \frac{A_2}{D-m_2} + \dots + \frac{A_n}{D-m_n} \right] X$$

$$= A_1 \frac{1}{D-m_1} X + A_2 \frac{1}{D-m_2} X + \dots + A_n \frac{1}{D-m_n} X$$

$$= A_1 \cdot e^{m_1 x} \int X e^{-m_1 x} dx + A_2 \cdot e^{m_2 x} \int X e^{-m_2 x} dx + \dots + A_n \cdot e^{m_n x} \int X e^{-m_n x} dx$$

Working Procedure To Solve The Equation

$$\frac{d^n y}{dx^n} + k_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + k_{n-1} \frac{dy}{dx} + k_n y = Q(x) \text{ of which the symbolic form is}$$

$$(D^n + k_1 D^{n-1} + \dots + k_{n-1} D + k_n) y = Q(x)$$

Step I. To find the complementary function

(i) Write the A.E. i.e. $D^n + k_1 D^{n-1} + \dots + k_{n-1} D + k_n = 0$ and solve it for D .

S.No.	Roots of A.E.	C.F.
1.	$m_1, m_2, m_3, \dots$ (real and different roots)	$c_1 e^{m_1 x} + c_2 e^{m_2 x} + c_3 e^{m_3 x} + \dots$
2.	$m_1, m_1, m_3, \dots$ (real and two equal roots)	$(c_1 + c_2 x) e^{m_1 x} + c_3 e^{m_3 x} + \dots$
3.	$m_1, m_1, m_1, m_4, \dots$ (real and three equal roots)	$(c_1 + c_2 x + c_3 x^2) e^{m_1 x} + c_4 e^{m_4 x} + \dots$

4.	$\alpha + i\beta, \alpha - i\beta, m_3, \dots$ (one pair of imaginary roots)	$e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x) + c_3 e^{m_3 x} + \dots$
5.	$\alpha \pm i\beta, \alpha \pm i\beta, m_5, \dots$ (two pair of imaginary roots)	$e^{\alpha x} [(c_1 + c_2 x) \cos \beta x + (c_3 + c_4 x) \sin \beta x] + c_5 e^{m_5 x} + \dots$

Step II. Find the Particular integral by using any of the above form.

Sum of the complementary function and Particular integral represents the general solution

Two Other Methods of Finding P.I.

(i) **Method of Variation of Parameters:** This method is quite general and applies to equations of the form $y'' + py' + qy = Q(x)$ (1)

where p, q and $Q(x)$ are functions of x . It gives P.I.

$$= -y_1 \int \frac{y_2 Q(x)}{W} dx + y_2 \int \frac{y_1 Q(x)}{W} dx \quad (2)$$

where y_1 and y_2 are the solutions of $y'' + py' + qy = 0$ (3)

and $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$ is Wronskian of y_1, y_2 .

Proof: Let the C.F. of (1) be $y = c_1 y_1 + c_2 y_2$.

Replacing c_1, c_2 (regarded as parameters) by unknown functions $u(x)$ and $v(x)$ let the P.I. be

$$y = uy_1 + vy_2 \quad (4)$$

Differentiating (4) w.r.t. x we get $y' = uy_1' + vy_2' + u'y_1 + v'y_2$

$$= uy_1' + vy_2' \quad (5)$$

$$\text{on assuming that } u'y_1 + v'y_2 = 0 \quad (6)$$

Differentiate (4) and substitute in (1). Then noting that y_1 and y_2 , satisfy (3), we obtain

$$u'y_1' + v'y_2' = Q(x) \quad (7)$$

Solving (6) and (7), we get

$$u' = -\frac{y_2 Q(x)}{W}, v' = \frac{y_1 Q(x)}{W} \text{ Where } W = y_1 y_2' - y_2 y_1'$$

Integrating $u = -\int \frac{y_2 Q(x)}{W} dx, v = \int \frac{y_1 Q(x)}{W} dx$. Substituting these in (4), we get (2).

(ii) **Method of Undetermined Coefficients:** To find the P.I. of $f(D)y = Q(x)$, we assume a trial solution containing unknown constants which are determined by substitution in the given equation.

The trial solution to be assumed in each case, depends on the form of X . Thus when

- (a) $Q(x) = 2e^{3x}$, trial solution $= ae^{3x}$.
- (b) $Q(x) = 3 \sin 2x$, trial solution $= a_1 \sin 2x + a_2 \cos 2x$
- (c) $Q(x) = 2x^3$, trial solution $= a_1 x^3 + a_2 x^2 + a_3 x + a_4$

However when $X = \tan x$ or $\sec x$, this method fails, since the number of terms obtained by differentiating $X = \tan x$ or $\sec x$ is infinite.

The above method holds as long as no term in the trial solution appears in the C.F. If any term of the trial solution appears in the C.F., we multiply this trial solution by the lowest positive integral power of x which is large enough so that none of the terms which are then present, appear in the C.F.

4.2 Linear Differential Equations with Variable Coefficients

4.2.1 Cauchy's Homogeneous Linear Differential Equation

An equation of the form $x^n \frac{d^n y}{dx^n} + k_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + k_{n-1} x \frac{dy}{dx} + k_n y = X$ (1)

where X is a function of x , is called Cauchy's homogeneous linear equations.

Such equations can be reduced to linear differential equations with constant coefficients by putting $x = e^t$ or $t = \log x$. Then if $D = \frac{d}{dt}$

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{dy}{dt} \cdot \frac{1}{x} \text{ i.e. } x \frac{dy}{dx} = Dy.$$

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{1}{x} \frac{dy}{dt} \right) = -\frac{1}{x^2} \frac{dy}{dt} + \frac{1}{x} \frac{d}{dx} \left(\frac{dy}{dt} \right) \frac{dy}{dx} = -\frac{1}{x^2} \frac{dy}{dt} + \frac{1}{x} \frac{d^2 y}{dt^2} \frac{dy}{dx} = \frac{1}{x^2} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right)$$

$$\text{i.e. } x^2 \frac{d^2 y}{dt^2} = D(D-1)y. \text{ Similarly, } x^3 \frac{d^3 y}{dt^3} = D(D-1)(D-2)y \text{ and so on.}$$

After making these substitutions in (1), this results a linear equation with constant coefficients, which can be solved as before.

4.2.2 Legendre's Linear Differential Equation

An equation of the form $(ax+b)^n \frac{d^n y}{dx^n} + k_1 (ax+b)^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + k_n y = X$ (2)

Where k_i 's are constants and X is a function of x , is called Legendre's linear equations. Such equations can be reduced to linear equations with constant coefficients by the substitution $ax+b = e^t$, i.e., $t = \log(ax+b)$.

Then, if $D = \frac{d}{dt}, \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{a}{ax+b} \cdot \frac{dy}{dt}$ i.e. $(ax+b) \frac{dy}{dx} = aDy$

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{a}{ax+b} \frac{dy}{dt} \right) = \frac{-a^2}{(ax+b)^2} \frac{dy}{dt}$$

$$+ \frac{a}{ax+b} \frac{d}{dt} \left(\frac{dy}{dt} \right) \frac{dt}{dx} = \frac{a^2}{(ax+b)^2} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right)$$

$$\text{i.e. } (ax+b)^2 \frac{d^2 y}{dx^2} = a^2 D(D-1)y.$$

$$\text{Similarly } (ax+b)^3 \frac{d^3 y}{dx^3} = a^3 D(D-1)(D-2)y \text{ and so on.}$$

After making these replacements in (2), there results a linear equation with constant coefficients.

4.3 Linear Ordinary Differential Equations of II Order With Variable Coefficients

An equation of the form $\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = R$ (1)

Where P, Q, R are functions of x only and called LODE of II order

The Complete Solution When One Integral is Known

Let $y = u$ be a known integral in the C.F. of (1)

$$\text{i.e. } y = u \text{ is a solution of } \frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = 0 \quad (2)$$

$$\Rightarrow \frac{d^2 u}{dx^2} + P \frac{du}{dx} + Qu = 0 \quad (3)$$

Let $y = uv$ be the solution of (1)

$$y = uv \Rightarrow \frac{dy}{dx} = v \frac{du}{dx} + u \frac{dv}{dx}$$

$$\Rightarrow \frac{d^2 y}{dx^2} = v \frac{d^2 u}{dx^2} + 2 \frac{du}{dx} \frac{dv}{dx} + u \frac{d^2 v}{dx^2}$$

Hence (1) becomes:

$$v \frac{d^2 u}{dx^2} + 2 \frac{du}{dx} \frac{dv}{dx} + u \frac{d^2 v}{dx^2} + P \left(v \frac{du}{dx} + u \frac{dv}{dx} \right) + Quv = R$$

$$\Rightarrow u \frac{d^2 v}{dx^2} + \frac{dv}{dx} \left(2 \frac{du}{dx} + Pu \right) + v \left(\frac{d^2 u}{dx^2} + P \frac{du}{dx} + Qu \right) = R$$

$$\Rightarrow u \frac{d^2 v}{dx^2} + \frac{dv}{dx} \left(2 \frac{du}{dx} + Pu \right) = R \text{ from } \quad (3)$$

$$\Rightarrow \frac{d^2 v}{dx^2} + \left(P + \frac{2}{u} \frac{du}{dx} \right) \frac{dv}{dx} = \frac{R}{u} \quad (4)$$

Now putting $\frac{dv}{dx} = p \Rightarrow \frac{d^2v}{dx^2} = \frac{dp}{dx}$

$$\Rightarrow \frac{dp}{dx} + \left(P + \frac{2}{u} \frac{du}{dx} \right) p = \frac{R}{u} \quad (5)$$

Which is linear in p as dependent variable

$$\begin{aligned} \text{I.F.} &= e^{\int \left(P + \frac{2}{u} \frac{du}{dx} \right) dx} = e^{\int P dx} \cdot e^{\int \frac{2}{u} du} \\ &= u^2 e^{\int P dx} \end{aligned}$$

Hence solution of (5) is

$$pu^2 e^{\int P dx} = \int \left[\frac{R}{u} u^2 e^{\int P dx} \right] dx + c_1$$

$$\Rightarrow p = \frac{c_1}{u^2} e^{-\int P dx} + \frac{e^{-\int P dx}}{u^2} \left[\int R u e^{\int P dx} dx \right]$$

$\Rightarrow$ Integrating then,

$$v = c_2 + c_1 \int \frac{e^{-\int P dx}}{u^2} dx + \int \frac{e^{-\int P dx}}{u^2} \left[\int R u e^{\int P dx} dx \right] dx$$

$y = uv$ is a solution of (1)

Second part of the complementary function is $u \int \frac{e^{-\int P dx}}{u^2} dx$ and particular integral is

$$= u \int \frac{e^{-\int P dx}}{u^2} \left[\int R u e^{-\int P dx} dx \right] dx$$

To Find one Integral in C.F. by Inspection i.e. to Find a Solution of
 $D^2y + PDy + Qy = 0$

- (i) $y = x$ is a solution of (1) if $P + Qx = 0$
- (ii) $y = x^2$ is a solution of (1) if $2 + 2Px + Qx^2 = 0$
- (iii) $y = e^x$ is a solution of (1) if $1 + P + Q = 0$
- (iv) $y = e^{-x}$ is a solution of (1) if $1 - P + Q = 0$
- (v) $y = e^{mx}$ is a solution of (1) if $m^2 + mP + Q = 0$
- (vi) $y = x^m$ is a part of C.F. if $m(m-1) + Pmx + Qx^2 = 0$

* If in a linear equation $A \frac{d^2y}{dx^2} + B \frac{dy}{dx} + Cy = x$

$A + B + C = 0$ then $y = e^x$ is a part of C.F.

Example: $x \frac{d^2 y}{dx^2} - 2(x+1) \frac{dy}{dx} + (x+2)y = (x-2)e^x$

Solution: $\frac{d^2 y}{dx^2} - \frac{2(x+1)}{x} \frac{dy}{dx} + \frac{x+2}{x} y = \left(\frac{x-2}{x}\right)e^x$ (1)

Here $1 + P + Q = 1 - \frac{2(x+1)}{x} + \frac{x+2}{x} = 0$

$\Rightarrow y = e^x$ in a part of the C.F.

Putting $y = ve^x \Rightarrow \frac{dy}{dx} = e^x \left(\frac{dv}{dx} + v \right)$

$\Rightarrow \frac{d^2 y}{dx^2} = e^x \left[\frac{d^2 v}{dx^2} + 2 \frac{dv}{dx} + v \right]$

(1) $\Rightarrow e^x \left[\frac{d^2 v}{dx^2} + 2 \frac{dv}{dx} + v \right] - \frac{2(x+1)}{x} e^x \left[\frac{dv}{dx} + v \right] + \frac{(x+2)}{x} ve^x = \frac{(x-2)}{x} e^x$

$\Rightarrow \frac{d^2 v}{dx^2} + \frac{dv}{dx} \left(2 - \frac{2(x+1)}{x} \right) + v \left(1 - \frac{2(x+1)}{x} + \frac{x+2}{x} \right) = \frac{x-2}{x}$

$\Rightarrow \frac{dp}{dx} + \left(-\frac{2}{x} \right) p = \frac{x-2}{x} \quad \left[\because \frac{dv}{dx} = p \right]$

I.F. $= e^{\int -\frac{2}{x} dx} = \frac{1}{x^2}$

$\therefore \frac{p}{x^2} = \int \frac{x-2}{x^3} dx = \int \frac{dx}{x^2} - \int \frac{2dx}{x^3} = \frac{1}{x} + \frac{1}{x^2} + c_1$

$\Rightarrow p = c_1 x^2 - x + 1$

$\Rightarrow v = c_2 + \frac{c_1 x^3}{3} - \frac{x^2}{2} + x$

Hence complete solution is

$y = ve^x = e^x \left(x - \frac{x^2}{2} + \frac{c_1}{3} x^3 + c_2 \right)$

Example: Solve $x \frac{d^2 y}{dx^2} + (x-2) \frac{dy}{dx} - 2y = x^3$

$\Rightarrow \frac{d^2 y}{dx^2} + \frac{x-2}{x} \frac{dy}{dx} - \frac{2}{x} y = x^2$ (1)

We have $1 - P + Q = 1 - \frac{x-2}{x} - \frac{2}{x} = 0$

$\Rightarrow y = e^{-x}$ is a part of C.F.

$\Rightarrow y = ve^{-x}$ is complete solution of (1)

$\frac{dy}{dx} = e^{-x} \left(\frac{dv}{dx} - v \right)$

Put Your Own Notes

$$\frac{d^2y}{dx^2} = e^{-x} \left(\frac{d^2v}{dx^2} - 2 \frac{dv}{dx} + v \right)$$

$$(1) \Rightarrow e^{-x} \left[\frac{d^2v}{dx^2} - 2 \frac{dv}{dx} + v \right] + \frac{x-2}{x} e^{-x} \left(\frac{dv}{dx} - v \right) - \frac{2}{x} v e^{-x} = x^2$$

$$\Rightarrow \frac{d^2v}{dx^2} + \frac{dv}{dx} \left(\frac{x-2}{x} - 2 \right) + v \left(1 - \frac{x-2}{x} - \frac{2}{x} \right) = x^2 e^x$$

$$\Rightarrow \frac{dp}{dx} - \frac{(x+2)}{x} p = x^2 e^x$$

$$\text{I. F.} = e^{\int \left(-1 - \frac{2}{x} \right) dx} = \frac{e^{-x}}{x^2}$$

$$p \frac{e^{-x}}{x^2} = \int x^2 e^x \cdot \frac{e^{-x}}{x^2} dx + c_1$$

$$\Rightarrow p = c_1 x^2 e^x + x^3 e^x$$

$$\Rightarrow v = c_2 + \int x^3 e^x dx + c_1 \int x^2 e^x dx$$

$$= c_2 + x^3 e^x - \int 3x^2 e^x dx + c_1 \int x^2 e^x dx$$

$$= c_2 + x^3 e^x + (c_1 - 3) \left[x^2 e^x - \int 2x e^x dx \right]$$

$$= c_2 + x^3 e^x + (c_1 - 3) \left[x^2 e^x - 2x e^x + 2e^x \right]$$

$$= x^3 e^x + (c_1 - 3) x^2 e^x - 2(c_1 - 3) x e^x + 2(c_1 - 3) e^x + c_2$$

$$\therefore y = v e^{-x} = x^3 + (c_1 - 3) x^2 - 2(c_1 - 3) x + 2(c_1 - 3) + c_2 e^{-x}$$

Removal of The First Derivative (Reduction to Normal Form)

When unable to obtain a part of C.F. of solution of $\frac{d^2y}{dx^2} + P \frac{dy}{dx} + Qy = R$ (1)

Putting $y = uv$ where u is some for of x .

$$\frac{dy}{dx} = u \frac{dv}{dx} + \frac{du}{dx} \quad \& \quad \frac{d^2y}{dx^2} = u \frac{d^2v}{dx^2} + 2 \frac{du}{dx} \frac{dv}{dx} + v \frac{d^2u}{dx^2}$$

$$(1) \Rightarrow \left[v \frac{d^2u}{dx^2} + 2 \frac{du}{dx} \frac{dv}{dx} + u \frac{d^2v}{dx^2} \right] + P \left[v \frac{du}{dx} + u \frac{dv}{dx} \right] + Quv = R$$

$$\Rightarrow u \frac{d^2v}{dx^2} + \left[P + \frac{2}{u} \frac{du}{dx} \right] u \frac{dv}{dx} + v \left[\frac{d^2u}{dx^2} + P \frac{du}{dx} + Qu \right] = R \quad (2)$$

Now to remove the term of the first derivative from to (2) we choose u s.t.

$$P + \frac{2}{u} \frac{du}{dx} = 0 \Rightarrow \frac{du}{u} + \frac{P}{2} dx = 0$$

$$u = e^{-\frac{1}{2} \int P dx} \quad (3)$$

$$(2) \Rightarrow \frac{d^2 v}{dx^2} + \frac{v}{u} \left[\frac{d^2 u}{dx^2} + P \frac{du}{dx} + Qu \right] = \frac{R}{u} \quad (4)$$

$$\therefore \frac{du}{dx} = -\frac{1}{2} Pu \Rightarrow \frac{d^2 u}{dx^2} = -\frac{1}{2} \left[P \frac{du}{dx} + u \frac{dP}{dx} \right]$$

$$\Rightarrow \frac{d^2 u}{dx^2} = -\frac{1}{2} \left[-\frac{1}{2} P^2 u + u \frac{dP}{dx} \right] = \frac{1}{4} P^2 u - \frac{u}{2} \frac{dP}{dx}$$

$$(4) \Rightarrow \frac{d^2 v}{dx^2} + v \left[\frac{1}{4} P^2 - \frac{1}{2} \frac{dP}{dx} - \frac{1}{2} P^2 + Q \right] = \frac{R}{e^{\frac{1}{2} \int P dx}}$$

$$\Rightarrow \frac{d^2 v}{dx^2} + v \left[Q - \frac{P^2}{4} - \frac{1}{2} \frac{dP}{dx} \right] = R e^{\frac{1}{2} \int P dx}$$

$$\Rightarrow \frac{d^2 v}{dx^2} + Xv = z \text{ Normal form} \quad (5)$$

$$\text{where } X = Q - \frac{1}{2} \frac{dP}{dx} - \frac{P^2}{4}$$

$$z = R e^{\frac{1}{2} \int P dx}$$

Method of The Variation of Parameters: Linear ordinary differential

$$\text{equation } \frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = R \quad (1)$$

Let $y = Au + Bv$ is the C.F. of (1)

Where A, B are constants & u, v are function of x .

Then $y = Au + Bv$ is the solution of

$$\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = 0$$

$$\Rightarrow Au'' + Bv'' + P(Au' + Bv') + Q(Au + Bv) = 0$$

$$\Rightarrow A(u'' + Pu' + Qu) + B(v'' + Pv' + Qv) = 0$$

$$\Rightarrow (u'' + Pu' + Qu) = 0 \quad (2)$$

$$\text{and } (v'' + Pv' + Qv) = 0 \quad (3)$$

Now suppose that $y = Au + Bv$ in the complete primitive of (1) where A & B are function of x , so (1) should be satisfied.

$$\therefore y = Au + Bv$$

$$\Rightarrow \frac{dy}{dx} = A'u + B'v + Au' + Bv'$$

$$\text{Let us choose } A \text{ \& } B \text{ s. t. } A'u + B'v = 0 \quad (4)$$

Then $\frac{dy}{dx} = Au' + Bv'$

$$\Rightarrow \frac{dy}{dx} = A'u' + B'v' + Au'' + Bv''$$

Now (1) $\Rightarrow (Au'' + Bv'' + A'u' + B'v') + P(Au' + Bv') + Q(Au + Bv) = R$

$$\Rightarrow A(u'' + Pu' + Qu) + B(v'' + Pv' + Qv) + A'u' + B'v' = R$$

$$\Rightarrow A'u' + B'v' = R \quad (5)$$

$$\& A'u + B'v = 0 \quad (4)$$

Solving (4) & (5) we get A' & B' where on integration give A & B

Substituting these values of A & B in $y = Au + Bv$, we get complete primitive of (1).

Example: Solve $\frac{d^2y}{dx^2} + 4y = 4 \tan 2x$ (*)

Solution : C.F. of given equation in solution of $\frac{d^2y}{dx^2} + 4y = 0$ is (*)

i.e. $y = c_1 \cos 2x + c_2 \sin 2x$ (1)

Let $y = A \cos 2x + B \sin 2x$ be complete primitive of (1) where A, B are function of x .

Now $\frac{dy}{dx} = \frac{dA}{dx} \cos 2x + \frac{dB}{dx} \sin 2x - 2A \sin 2x + 2B \cos 2x$

Choose A & B s. t. $\frac{dA}{dx} \cos 2x + \frac{dB}{dx} \sin 2x = 0$ (2)

$$\Rightarrow \frac{dy}{dx} = -2A \sin 2x + 2B \cos 2x$$

$$\Rightarrow \frac{d^2y}{dx^2} = -2 \sin 2x \frac{dA}{dx} + 2 \cos 2x \frac{dB}{dx} - 4A \cos 2x - 4B \sin 2x$$

$$\Rightarrow -2 \sin 2x \frac{dA}{dx} + 2 \cos 2x \frac{dB}{dx} - 4A \cos 2x - 4B \sin 2x + 4A \cos 2x$$

$$\Rightarrow -2 \sin 2x \frac{dA}{dx} + 2 \cos 2x \frac{dB}{dx} = 4 \tan 2x \quad (3)$$

$$\text{and } \cos 2x \frac{dA}{dx} + \sin 2x \frac{dB}{dx} = 0 \quad (2)$$

$$(3) \Rightarrow -\sin 2x \frac{dA}{dx} + \cos 2x \frac{dB}{dx} = 2 \tan 2x \quad (4)$$

Solving these two equations, we get

$$\frac{dB}{dx} = 2 \sin 2x, \frac{dA}{dx} = -2 \frac{\sin^2 2x}{\cos 2x}$$

$$\Rightarrow B = -\cos 2x + c_2,$$

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$$\begin{aligned} A &= -2 \int \left(\frac{1 - \cos^2 2x}{\cos 2x} \right) dx + c_1 \\ &= -2 \int (\sec 2x - \cos 2x) dx + c_1 \\ &= -\log(\sec 2x + \tan 2x) + \sin 2x + c_1 \\ y &= c_1 \cos 2x + c_2 \sin 2x - \log[\sec 2x + \tan 2x] \cdot \cos 2x \end{aligned}$$

Example: Solve $\frac{d^3 y}{dx^3} - x \frac{d^2 y}{dx^2} - \frac{dy}{dx} + xy = 0$ (1)

Solution: Since sum of the coefficient in zero

$\Rightarrow$ One of the solutions of ODE in $y = e^x$

Let v be another solution of

Then putting $y = ve^x$ in (1) we have

$$\left. \begin{aligned} \frac{dy}{dx} &= e^x \left[\frac{dv}{dx} + v \right] \\ \frac{d^2 y}{dx^2} &= e^x \left[\frac{d^2 v}{dx^2} + 2 \frac{dv}{dx} + v \right] \end{aligned} \right\} \Rightarrow \frac{d^3 y}{dx^3} = e^x \left[\frac{d^3 v}{dx^3} + 3 \frac{d^2 v}{dx^2} + 3 \frac{dv}{dx} + v \right]$$

$$(1) \Rightarrow e^x \left[\frac{d^3 v}{dx^3} + 3 \frac{d^2 v}{dx^2} + 3 \frac{dv}{dx} + v \right] - x \left[\frac{d^2 v}{dx^2} + 2 \frac{dv}{dx} + v \right] e^x - e^x \left[v + \frac{dv}{dx} \right] + xve^x = 0$$

$$\Rightarrow \frac{d^3 v}{dx^3} + (3-x) \frac{d^2 v}{dx^2} + (3-2x-1) \frac{dv}{dx} = 0$$

$$\Rightarrow \frac{d^2 p}{dx^2} + (3-x) \frac{dp}{dx} + (2-2x)p = 0 \quad (2) \quad \left(\text{where } p \equiv \frac{dv}{dx} \right)$$

Hence $P = 3-x, Q = 2-2x$

$$m^2 + mP + Q = (-2)^2 + (-2)(3-x) + (2-2x) = 0 \quad \text{where } m = -2$$

$$\Rightarrow p = e^{-2x} \text{ in a solution of} \quad (2)$$

Putting $p = qe^{-2x}$ in (2)

$$\begin{aligned} \Rightarrow \frac{dp}{dx} &= e^{-2x} \left[\frac{dq}{dx} - 2q \right] \\ \& \frac{d^2 p}{dx^2} &= e^{-2x} \left[\frac{d^2 q}{dx^2} - 4 \frac{dq}{dx} + 4q \right] \\ \Rightarrow e^{-2x} \left[\frac{d^2 q}{dx^2} - 4 \frac{dq}{dx} + 4q \right] &+ (3-x)e^{-2x} \left[\frac{dq}{dx} - 2q \right] + (2-2x)qe^{-2x} = 0 \\ \Rightarrow \frac{d^2 q}{dx^2} - (1+x) \frac{dq}{dx} &= 0 \end{aligned}$$

$$\text{Let } u \equiv \frac{dq}{dx}$$

Then $\frac{du}{dx} = (1+x)u$

$$\Rightarrow \log u = x + \frac{x^2}{2} + \log c_1$$

$$\Rightarrow u = c_1 e^{x + \frac{x^2}{2}}$$

$$\Rightarrow q = \int u dx = \int c_1 e^{x + \frac{x^2}{2}} dx + c_2$$

$$\therefore p = e^{-2x} \int c_1 e^{x + \frac{x^2}{2}} dx + c_2 e^{-2x}$$

$$\Rightarrow v = \int p dx = \int \left(e^{-2x} \int c_1 e^{x + \frac{x^2}{2}} dx \right) dx + \int c_2 e^{-2x} dx + c_3$$

$$= c_1 \left[-\frac{1}{2} e^{-2x} \int e^{x + \frac{x^2}{2}} dx - \int -\frac{1}{2} e^{-2x} \cdot e^{x + \frac{x^2}{2}} dx \right] + c_2 \int e^{-2x} dx + c_3$$

$$= c_1 \left[-\frac{1}{2} e^{-2x} \int e^{x + \frac{x^2}{2}} dx + \frac{1}{2} \int e^{-2x} e^{x + \frac{x^2}{2}} dx \right] + c_2 \int e^{-2x} dx + c_3$$

Hence $y = ve^x$

$$\Rightarrow y = c_1 \left[-\frac{1}{2} e^{-x} \int e^{x + \frac{x^2}{2}} dx + \frac{1}{2} e^x \int e^{-2x} e^{x + \frac{x^2}{2}} dx \right] - \frac{1}{2} c_2 e^{-x} + c_3 e^x$$

Example: Solve by method of variation of parameters

$$\frac{d^2 y}{dx^2} + n^2 y = \sec nx \quad (1)$$

C.F. = $c_1 \cos nx + c_2 \sin nx$

Let $y = A \cos nx + B \sin nx$ be complete primitive of (1)

$$\Rightarrow \frac{dy}{dx} = \frac{dA}{dx} \cos nx + \frac{dB}{dx} \sin nx - nA \sin nx + nB \cos nx$$

We choose A & B such that

$$\frac{dA}{dx} \cos nx + \frac{dB}{dx} \sin nx = 0 \quad (2)$$

$$\Rightarrow \frac{dy}{dx} = -nA \sin nx + nB \cos nx$$

$$\Rightarrow \frac{d^2 y}{dx^2} = -n \frac{dA}{dx} \sin nx + n \frac{dB}{dx} \cos nx - n^2 A \cos nx - n^2 B \sin nx$$

Substituting these values in (1), we get

$$-n \frac{dA}{dx} \sin nx + n \frac{dB}{dx} \cos nx - n^2 A \cos nx - n^2 B \sin nx + n^2 A \cos nx + n^2 B \sin nx = \sec nx$$

$$\Rightarrow -\frac{dA}{dx} \sin nx + \frac{dB}{dx} \cos nx = \frac{1}{n} \sec nx \quad (3)$$

$$\& \frac{dA}{dx} \cos nx + \frac{dB}{dx} \sin nx = 0 \quad (2)$$

Solving these two equations, we have

$$\frac{dB}{dx} = \frac{1}{n} \& \frac{dA}{dx} = -\frac{1}{n} \tan nx$$

$$\Rightarrow B = \frac{x}{n} + c_3 \& A = -\frac{1}{n^2} \log \cos nx + c_4$$

Hence solution in

$$y = \frac{1}{n^2} \cos nx \log \cos nx + c_4 \cos nx + c_3 \sin nx + \frac{x}{n} \sin nx$$

Example : Solve by method of variation of parameter $\frac{d^2y}{dx^2} + a^2y = \operatorname{cosec} ax$ (1)

Solution: C. F. = $c_1 \cos ax + c_2 \sin ax$

Let $y = A \cos ax + B \sin ax$ be complete primitive of (1)

$$\Rightarrow \frac{dy}{dx} = \frac{dA}{dx} \cos ax + \frac{dB}{dx} \sin ax - aA \sin ax + aB \cos ax.$$

$$\text{Choose } A \& B \text{ s. t. } \frac{dA}{dx} \cos ax + \frac{dB}{dx} \sin ax = 0 \quad (2)$$

$$\text{Then } \frac{dy}{dx} = -aA \sin ax + aB \cos ax$$

$$\frac{d^2y}{dx^2} = -a \frac{dA}{dx} \sin ax + a \frac{dB}{dx} \cos ax - a^2 A \cos ax - a^2 B \sin ax \quad (1)$$

$$\Rightarrow -a \frac{dA}{dx} \sin ax + a \frac{dB}{dx} \cos ax - a^2 A \cos ax - a^2 B \sin ax + a^2 A \cos ax + a^2 B \sin ax = \operatorname{cosec} ax$$

$$\Rightarrow -\frac{dA}{dx} \sin ax + \frac{dB}{dx} \cos ax = \frac{1}{a} \operatorname{cosec} ax \quad (3)$$

$$\& \frac{dA}{dx} \cos ax + \frac{dB}{dx} \sin ax = 0 \quad (2)$$

Solving these,

$$\frac{dA}{dx} = -\frac{1}{a}, \quad \frac{dB}{dx} = \frac{1}{a} \cot ax$$

$$\Rightarrow A = -\frac{x}{a} + c_3 \& B = \frac{1}{a^2} \log \sin ax + c_4.$$

$$\text{Hence } y = c_3 \cos ax + c_4 \sin ax - \frac{x}{a} \cos ax + \frac{1}{a^2} \sin ax \log \sin ax$$

Example : Solve the method of the variation of parameters

$$\frac{d^2y}{dx^2} + (1 - \cot x) \frac{dy}{dx} - y \cot x = \sin^2 x \quad (1)$$

$$\text{Solution: To find C.F. i.e. solution of } \frac{d^2y}{dx^2} + (1 - \cot x) \frac{dy}{dx} - y \cot x = 0 \quad (2)$$

$$\text{Clearly, } 1 - P + Q = 1 - 1 + \cot x - \cot x = 0$$

$$\Rightarrow y = e^{-x} \text{ is a part of C.F.}$$

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⇒ Now putting $y = ve^{-x}$ in equation (2)

$$e^{-x} \left[\frac{d^2 v}{dx^2} - 2 \frac{dv}{dx} + v \right] + (1 - \cot x) e^{-x} \left[\frac{dv}{dx} - v \right] - ve^{-x} \cot x = 0$$

$$\Rightarrow \frac{d^2 v}{dx^2} - (1 + \cot x) \frac{dv}{dx} = 0 \quad (3)$$

Let $p \equiv \frac{dv}{dx}$

$$(3) \Rightarrow \frac{dp}{dx} - (1 + \cot x) p = 0$$

$$\Rightarrow \log p = x + \log \sin x + \log c_1$$

$$\Rightarrow p / \sin x = c_1 e^x$$

$$\Rightarrow p = c_1 e^x \sin x$$

$$\Rightarrow v = \int p dx = c_1 \int e^x \sin x + c_2 = c_1 \cdot \frac{1}{2} e^x (\sin x - \cos x) + c_2$$

$$\text{Hence C.F.} = \frac{c_1}{2} (\sin x - \cos x) + c_2 e^{-x}$$

Let $y = A(\sin x - \cos x) + Be^{-x}$ be complete primitive of $\quad (1)$

$$\Rightarrow \frac{dy}{dx} = \frac{dA}{dx} (\sin x - \cos x) + \frac{dB}{dx} e^{-x} + A(\cos x + \sin x) - Be^{-x}$$

Choose A & B such that $\frac{dA}{dx} (\sin x - \cos x) + \frac{dB}{dx} e^{-x} = 0 \quad (4)$

$$\Rightarrow \frac{dy}{dx} = A(\cos x + \sin x) - Be^{-x}$$

$$\Rightarrow \frac{d^2 y}{dx^2} = \frac{dA}{dx} (\cos x + \sin x) - \frac{dB}{dx} e^{-x} + A(\cos x - \sin x) + Be^{-x}$$

Hence (1) $\Rightarrow$

$$\frac{dA}{dx} (\cos x + \sin x) - \frac{dB}{dx} e^{-x} + A(\cos x - \sin x) + Be^{-x} + A(1 - \cot x)(\cos x + \sin x)$$

$$- B(1 - \cot x) e^{-x} - A \cot x (\sin x - \cos x) - B \cot x e^{-x} = \sin^2 x$$

$$\Rightarrow \frac{dA}{dx} (\cos x + \sin x) - \frac{dB}{dx} e^{-x} = \sin^2 x \quad (5)$$

$$\& \frac{dA}{dx} (\sin x - \cos x) + \frac{dB}{dx} e^{-x} = 0$$

$$\Rightarrow \frac{dA}{dx} = \frac{1}{2} \sin x \& \frac{dB}{dx} = \frac{e^x}{2} \left[(\sin x \cos x - \sin^2 x) \right]$$

$$\Rightarrow A = -\frac{1}{2} \cos x + c_3 \& B = \frac{1}{4} \int e^x [\sin 2x + \cos 2x - 1] dx + c_4$$

$$\Rightarrow B = -\frac{e^x}{4} + C_4 + \frac{e^x}{20} (3 \sin 2x - \cos 2x)$$

Hence

$$y = -\frac{1}{2} \cos x (\sin x - \cos x) + C_3 (\sin x - \cos x) + C_4 e^{-x} - \frac{1}{4} + \frac{1}{20} (3 \sin 2x - \cos 2x)$$

$$\begin{aligned}
 &= C_3 (\sin x - \cos x) + C_4 e^{-x} + \frac{1}{20} (3 \sin 2x - \cos 2x) - \frac{1}{4} (\sin 2x - 2 \cos^2 x + 1) \\
 &= C_3 (\sin x - \cos x) + C_4 e^{-x} + \frac{1}{20} (3 \sin 2x - \cos 2x) - \frac{1}{4} (\sin 2x - \cos 2x) \\
 &= C_3 (\sin x - \cos x) + C_4 e^{-x} + \frac{1}{20} (-2 \sin 2x + 4 \cos 2x)
 \end{aligned}$$

Put Your Own Notes

CHAPTER 5

UNIQUENESS AND EXISTENCE (FIRST ORDER FIRST DEGREE ORDINARY DIFFERENTIAL EQUATION)

5.1 Picard's Method of Successive Approximations (or Picard's iteration method)

consider an initial value problem of the form $\frac{dy}{dx} = f(x, y), y(x_0) = y_0$ (1)

By integrating over the interval (x_0, x) , (1) gives

$$\int_{x_0}^x dy = \int_{x_0}^x f(x, y) dx, y(x_0) = y_0 \text{ or } y(x) - y_0 = \int_{x_0}^x f(x, y) dx \text{ or}$$

$$y(x) = y_0 + \int_{x_0}^x f(x, y) dx \quad (2)$$

Thus, the solving of initial value problem (1) is equivalent to finding a function $y(x)$ which satisfies the equation (2), since by differentiating (2)

we get $\frac{dy}{dx} = f(x, y)$ and putting $x = x_0$ in (2) yields $y(x_0) = y_0 + 0$ i.e.

$y(x_0) = y_0$. Conversely, (2) has been obtained from (1) by integration over the interval (x_0, x) and employing the initial condition $y(x_0) = y_0$. Since the information concerning the expression of y in terms of x is absent in the integral on the R.H.S. of (2) cannot be evaluated. Hence the exact value of y cannot be obtained. Therefore we determine a sequence of approximations to the solution (2) as follows. As a crude approximation, we put $y = y_0$ in the integral on the right of (2) and obtain

$$y_1(x) = y_0 + \int_{x_0}^x f(x, y_0) dx \quad (3)$$

Where $y_1(x)$ is the corresponding value of $y(x)$ and is called first approximation and is a better approximation of $y(x)$ at any x . To determine still better approximation we replace y by y_1 in the integral on R.H.S in (2) and obtain the second approximation y_2 as

$$y_2(x) = y_0 + \int_{x_0}^x f(x, y_1) dx \quad (4)$$

Proceeding in this way, the n th approximation y_n is given by

$$y_n(x) = y_0 + \int_{x_0}^x f(x, y_{n-1}) dx \quad (5)$$

Thus, we arrive at a sequence of approximate solutions

$$y_1(x), y_2(x), y_3(x), \dots, y_n(x), \dots$$

Example: Using the Picard's method of successive approximations, find the third approximation of the solution of the equation: $\frac{dy}{dx} = x + y^2$, where $y = 0$ when $x = 0$.

Solution: Given problem is $\frac{dy}{dx} = x + y^2$, where $y = 0$ when $x = 0$ (1)

We know that the n th approximation y_n of the initial problem

$$\frac{dy}{dx} = f(x, y) \text{ where } y = y_0 \text{ when } x = x_0. \quad (2)$$

$$\text{Given by } y_n = y_0 + \int_{x_0}^x f(x, y_{n-1}) dx \quad (3)$$

Comparing (1) and (2) $f(x, y) = x + y^2$, $x_0 = 0$ and $y_0 = 0$ (4)

$$\therefore \text{From (3) } y_n = \int_0^x [x + y_{n-1}^2] dx \quad (5)$$

First approximation, putting $n = 1$ in (5) and using (4) we have

$$y_1 = \int_0^x (x + y_0^2) dx = \int_0^x x dx = \frac{1}{2} x^2 \quad (6)$$

Second approximation, putting $n = 2$ in (5) and using (6), we have

$$y_2 = \int_0^x (x + y_1^2) dx = \int_0^x \left[x + \frac{x^4}{4} \right] dx = \frac{1}{2} x^2 + \frac{1}{20} x^5 \quad (7)$$

Third approximation, putting $n = 3$ in (5), we get

$$y_3 = \int_0^x (x + y_2^2) dx = \int_0^x \left[x + \left(\frac{1}{2} x^2 + \frac{1}{20} x^5 \right)^2 \right] dx, \text{ using (7)}$$

$$= \int_0^x \left[x + \frac{1}{4} x^4 + \frac{1}{400} x^{10} + \frac{1}{20} x^7 \right] dx = \frac{1}{2} x^2 + \frac{1}{20} x^5 + \frac{1}{4400} x^{11} + \frac{1}{160} x^8$$

Example 2: Use Picard's method to obtain a solution of differential equation: $\frac{dy}{dx} = x^2 - y$, $y(0) = 0$. Find at least the fourth approximation to each solution.

Solution: Given $\frac{dy}{dx} = x^2 - y$, where $y = 0$ when $x = 0$ (1)

We know that the n th approximation y_n of the initial value problem

$$\frac{dy}{dx} = f(x, y), \text{ where } y = y_0 \text{ when } x = x_0 \quad (2)$$

is given by $y_n = y_0 + \int_{x_0}^x f(x, y_{n-1}) dx$ (3)

Comparing (1) and (2), $f(x, y) = x^2 - y, x_0 = 0$ and $y_0 = 0$ (4)

$\therefore$ From (3) $y_n = \int_{x_0}^x (x^2 - y_{n-1}) dx$ (5)

First approximation, putting $n = 1$ in (5) and using (2), we get

$$y_1 = \int_{x_0}^x (x^2 - y_0) dx = \int_{x_0}^x x^2 dx = \left[\frac{1}{3} x^3 \right]_0^x = \frac{1}{3} x^3 \quad (6)$$

Second approximation, putting $n = 2$ in (5) and using (6), we get

$$y_2 = \int_{x_0}^x (x^2 - y_1) dx = \int_0^x \left(x^2 - \frac{1}{3} x^3 \right) dx = \left[\frac{1}{3} x^3 - \frac{1}{12} x^4 \right]_0^x = \frac{1}{3} x^3 - \frac{1}{12} x^4 \quad (7)$$

Third approximation, putting $n = 2$ in (5) and using (7), we get

$$y_3 = \int_{x_0}^x (x^2 - y_2) dx = \int_0^x \left[x^2 - \left(\frac{1}{3} x^3 - \frac{1}{12} x^4 \right) \right] dx = \left[\frac{1}{3} x^3 - \frac{1}{12} x^4 + \frac{1}{60} x^5 \right]_0^x$$

$$\text{or } y_3 = \frac{1}{3} x^3 - \frac{1}{12} x^4 + \frac{1}{60} x^5$$

Fourth approximation, putting $n = 4$ in (5) and using (7), we get

$$y_4 = \int_{x_0}^x (x^2 - y_3) dx = \int_0^x \left[x^2 - \left(\frac{1}{3} x^3 - \frac{1}{12} x^4 + \frac{1}{60} x^5 \right) \right] dx$$

$$= \left[\frac{1}{3} x^3 - \frac{1}{12} x^4 + \frac{1}{60} x^5 - \frac{1}{360} x^6 \right]_0^x = \frac{1}{3} x^3 - \frac{1}{12} x^4 + \frac{1}{60} x^5 - \frac{1}{360} x^6$$

Example: Find the third approximation of the solution of equation

$$\frac{dy}{dx} = z, \frac{dz}{dx} = x^2 z + x^4 y \text{ by Picard's method, } y = 5, z = 1 \text{ when } x = 0.$$

Solution: Given $\frac{dy}{dx} = z, \frac{dz}{dx} = x^2 z + x^4 y, y = 5, z = 1$ when $x = 0$ (1)

We know that the n th approximation (y_n, z_n) to initial value problem

$$\frac{dy}{dx} = f(x, y, z), \frac{dz}{dx} = g(x, y, z) \text{ where } y = y_0, z = z_0, \text{ when } x = x_0 \quad (2)$$

is given by $y_n = y_0 + \int_{x_0}^x f(x, y_{n-1}, z_{n-1}) dx$ (3)

and $z_n = z_0 + \int_{x_0}^x g(x, y_{n-1}, z_{n-1}) dx$ (4)

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Comparing (1) and (2) we have

$$f(x, y, z) = z, g(x, y, z) = x^2 z + x^4 y, y_0 = 5, z_0 = 1, x_0 = 0 \quad (5)$$

$$\therefore \text{From (3)} \quad y_n = 5 + \int_0^x z_{n-1} dx \quad (6)$$

$$\text{and from (4), } z_n = 1 + \int_0^x (x^2 z_{n-1} + x^4 y_{n-1}) dx \quad (7)$$

First approximation, putting $n=1$ in (6) and using (5), we get

$$y_1 = 5 + \int_0^x z_0 dx = 5 + \int_0^x dx = 5 + x \quad (8)$$

Next putting $n=1$ in (7) and using (5), we get

$$z_1 = 1 + \int_0^x (x^2 z_0 + x^4 y_0) dx = 1 + \int_0^x (x^2 + 5x^4) dx = 1 + \frac{x^3}{3} + x^5 \quad (9)$$

Second approximation, putting $n=2$ in (6) and using (9), we get

$$y_2 = 5 + \int_0^x z_1 dx = 5 + \int_0^x \left(1 + \frac{x^3}{3} + x^5\right) dx = 5 + x + \frac{x^4}{12} + \frac{x^6}{6} \quad (10)$$

Next, putting $n=2$ in (7) and using (8) and (9), we get

$$\begin{aligned} z_2 &= 1 + \int_0^x (x^2 z_1 + x^4 y_1) dx = 1 + \int_0^x \left[x^2 \left(1 + \frac{x^3}{3} + x^5\right) + x^4 (5 + x) \right] dx \\ &= 1 + \int_0^x \left(x^2 + 5x^4 + \frac{4}{3}x^5 + x^7 \right) dx = 1 + \frac{1}{3}x^3 + x^5 + \frac{2}{9}x^6 + \frac{1}{8}x^8 \end{aligned} \quad (11)$$

Third approximation, putting $n=3$ in (6) and using (11), we get

$$\begin{aligned} y_3 &= 5 + \int_0^x z_2 dx = 5 + \int_0^x \left(1 + \frac{1}{3}x^3 + x^5 + \frac{2}{9}x^6 + \frac{1}{8}x^8\right) dx \\ &= 5 + x + \frac{1}{12}x^4 + \frac{1}{6}x^6 + \frac{2}{63}x^7 + \frac{1}{72}x^8 \end{aligned}$$

Next, putting $n=3$ in (7) and using (10) and (11), we have

$$\begin{aligned} z_3 &= 1 + \int_0^x (x^2 z_2 + x^4 y_2) dx \\ &= 1 + \int_0^x \left[x^4 \left(5 + x + \frac{x^4}{12} + \frac{x^6}{6}\right) + x^2 \left(1 + \frac{x^3}{3} + x^5 + \frac{2x^6}{9} + \frac{x^8}{8}\right) \right] dx \\ &= 1 + \frac{1}{3}x^3 + x^5 + \frac{2}{9}x^6 + \frac{1}{8}x^8 + \frac{11}{324}x^9 + \frac{7}{264}x^{11}. \end{aligned}$$

5.2 Lipschitz Condition

A function $f(x, y)$ is said to satisfy a Lipschitz condition in a region D in xy -plane if there exists a positive constant k such that

$$|f(x, y_2) - f(x, y_1)| \leq K |y_2 - y_1|$$

Whenever the points (x, y_1) and (x, y_2) both lie in D . The constant K is called a Lipschitz constant for the function $f(x, y)$.

Example: Let $f(t, y) = 1 + ty^2$ and $D = \{(t, y) | 0 \leq t \leq 2, -1 \leq y \leq 1\}$. Does f satisfy a Lipschitz condition on D ? If so, find its Lipschitz constant.

Solution: Let (t, y_1) and (t, y_2) be in D , i.e., t is in $[0, 2]$ and y is in $[-1, 1]$. Observe that

$$\begin{aligned} |f(t, y_1) - f(t, y_2)| &= |1 + ty_1^2 - (1 + ty_2^2)| = |t| |y_1^2 - y_2^2| \\ &= |t| (y_1 + y_2)(y_1 - y_2) = |t| |y_1 + y_2| |y_1 - y_2|. \end{aligned}$$

Because $|t| \leq 2$ and $|y_1 + y_2| \leq 2$, we have

$$|f(t, y_1) - f(t, y_2)| \leq (2)(2) |y_1 - y_2| = 4 |y_1 - y_2|.$$

So, f satisfies a Lipschitz condition and its Lipschitz constant is 4.

Note that the Lipschitz constant L is not unique, that is, for any $\bar{L} > 4$, the inequality

$$|f(t, y_1) - f(t, y_2)| \leq \bar{L} |y_1 - y_2|$$

also holds. So, in practice, we want to find a L as small as possible.

A Sufficient Condition for Lipschitz Condition

Theorem: Suppose $f(t, y)$ is defined on a convex set D in R^2 . If a constant $L > 0$ exists with

$$\left| \frac{\partial f}{\partial y}(t, y) \right| \leq L, \text{ for } (t, y) \text{ in } D,$$

then f satisfies a Lipschitz condition on D in the variable y with Lipschitz constant L .

Example: Let $f(t, y) = 1 + ty^2$ and $D = \{(t, y) | 0 \leq t \leq 2, -1 \leq y \leq 1\}$. Does f satisfy a Lipschitz condition on D ? If so, find its Lipschitz constant.

The partial derivative of f with respect to y is $\frac{\partial f}{\partial y}(t, y) = 2ty$. Because

$$\left| \frac{\partial f}{\partial y}(t, y) \right| = |2ty| = 2|t||y| \leq 2(2)(1) = 4,$$

f satisfies a Lipschitz condition with Lipschitz constant 4.

Example: Let $f(t, y) = 1 + t \sin(ty)$. Determine if f satisfies a Lipschitz condition in

$$D = \{(t, y) | 0 \leq t \leq 2, -\infty < y < \infty\}.$$

Put Your Own Notes

Compute the partial derivative of f with respect to

$$y: \frac{\partial f}{\partial y}(t, y) = t \cos(ty) (t) = t^2 \cos(ty). \text{ Because } |\cos(ty)| \leq 1 \text{ and } |t| \leq 2,$$

$$\left| \frac{\partial f}{\partial y}(t, y) \right| = |t^2 \cos(ty)| = |t^2| |\cos(ty)| \leq |t^2| \leq 4.$$

So, f satisfies a Lipschitz condition with a constant 4.

5.3 Existence and Uniqueness Theorem

Statement: Let $f(x, y)$ be continuous in a domain D of the (x, y) plane and let M be a constant such that $|f(x, y)| \leq M$ (1) in D . Let $f(x, y)$ satisfy in D the Lipschitz condition in y namely

$$|f(x, y_1) - f(x, y_2)| \leq K |y_1 - y_2|, \quad (2)$$

where the constant K is independent of x, y_1, y_2 .

* Let the rectangle R defined by

$$|x - x_0| \leq h, |y - y_0| \leq k, \quad (3)$$

lie in D , where $Mh < k$. Then for $|x - x_0| \leq h$, the differential equation

$$\frac{dy}{dx} = f(x, y) \text{ has a unique solution } y = y(x) \text{ for which } y(x_0) = y_0.$$

Proof: We shall prove this theorem by the method of successive approximations. Let x be such that $|x - x_0| \leq h$. we now define a sequence of functions $y_1(x), y_2(x), \dots, y_n(x) \dots$, called the successive approximations (or Picard iterants) as follows:

$$y_1(x) = y_0 + \int_{x_0}^x f(x, y_0) dx$$

$$y_2(x) = y_0 + \int_{x_0}^x f(x, y_1) dx$$

$$y_{n-1}(x) = y_0 + \int_{x_0}^x f(x, y_{n-2}) dx$$

$$y_n(x) = y_0 + \int_{x_0}^x f(x, y_{n-1}) dx \quad (4)$$

We shall divide the proof into five main steps.

First Step: We prove that, for $x_0 - h \leq x \leq x_0 + h$ the curve $y = y_n(x)$ lies in the rectangle R , that is to say $y_0 - k < y < y_0 + k$.

$$\text{Now, } |y_1 - y_0| = \left| \int_{x_0}^x f(x, y_0) dx \right| \leq \int_{x_0}^x |f(x, y_0)| \cdot |dx|, \text{ by (4)}$$

$$\text{or } |y_1 - y_0| \leq M |x - x_0| \leq Mh < k,$$

by (1), (3) and the given result viz $Mh < k$.

This proves the desired result for $n=1$. Assume that $y = y_{n-1}(x)$ lies in R and so $f(x, y_{n-1})$ is defined continuous and satisfies $|f(x, y_{n-1})| \leq M$ on $[x_0 - h, x_0 + h]$

$$\text{From (4) we have } |y_n - y_0| = \left| \int_{x_0}^x f(x, y_{n-1}) dx \right| \leq \int_{x_0}^x |f(x, y_{n-1})| \cdot |dx|$$

$$\leq M|x - x_0| \leq Mh < k$$

As before which shows that $y_n(x)$ lies in R and hence $f(x, y_n)$ is defined and continuous on $[x_0 - h, x_0 + h]$. The above arguments show that the desired results holds for all n by induction.

Second Step: We prove again by induction, that $|y_n - y_{n-1}| \leq \frac{MK^{n-1}}{n!} |x - x_0|^n$

(5)

we have already verified (5) for $n=1$, in first step we have shown that $|y_1 - y_0| \leq M|x - x_0|$. Assume that this inequality (5) holds for $n-1$ in place of n , that is let

$$|y_{n-1} - y_{n-2}| \leq \frac{MK^{n-2}}{(n-1)!} |x - x_0|^{n-1}$$

(6)

Then we have

$$|y_n - y_{n-1}| = \left| \int_{x_0}^x \{f(x, y_{n-1}) - f(x, y_{n-2})\} dx \right|, \text{ by (4)}$$

$$\text{or } |y_n - y_{n-1}| \leq \int_{x_0}^x |f(x, y_{n-1}) - f(x, y_{n-2})| \cdot |dx|$$

(7)

Lipstchitz conditions (2) gives

$$|f(x, y_{n-1}) - f(x, y_{n-2})| \leq K|y_{n-1} - y_{n-2}|$$

(8)

From (7) and (8), we get

$$|y_n - y_{n-1}| \leq \int_{x_0}^x K|y_{n-1} - y_{n-2}| \cdot |dx|$$

$$\leq K \cdot \frac{MK^{n-1}}{(n-1)!} \cdot \frac{|x - x_0|^n}{n}, \text{ by (6)}$$

Hence, by mathematical induction, we conclude that (5) is true for each natural number n .

Third Step: We shall now prove that the sequence y_n converges uniformly to a limit for $x_0 - h \leq x \leq x_0 + h$.

For the interval under consideration, $|x - x_0| \leq h$.

Hence, from second step we have

$$|y_n - y_{n-1}| \leq \frac{MK^{n-1}h^n}{n!} \text{ is true for all } n.$$

Using this, the infinite series

$$\begin{aligned} & y_0 + (y_1 - y_0) + (y_2 - y_1) - \dots + (y_n - y_{n-1}) + \dots \\ & \leq y_0 + Mh + \frac{1}{2!}MKh^2 + \dots + \frac{1}{n!}MK^{n-1}h^n + \dots \\ & \leq y_0 + \frac{M}{K}[e^{Kh} - 1], \end{aligned} \quad (9)$$

Which is known to be convergent for all values of K, h and M . Consequently, the series (9) is surely convergent. Thus, by the Weirstrass M -test, the series (9) converges uniformly on $[x_0 - h, x_0 + h]$. Now since the terms of (9) are continuous function of x , its sum

$$= \lim_{n \rightarrow \infty} y_n(x) = y(x), \text{ say } \left[\because y_n = y_0 + \sum_{n=1}^n (y_n - y_{n-1}) \right] \quad (10)$$

must be continuous.

Fourth Step: We now show that $y = y(x)$ satisfies the differential equation

$$\frac{dy}{dx} = f(x, y).$$

Since $y_n(x)$ tends uniformly to $y(x)$ in $[x_0 - h, x_0 + h]$ and by Lipschitz condition,

$$|f(x, y) - f(x, y_n)| \leq K|y - y_n|,$$

it shows that $f[x, y_n(x)]$ tends uniformly to $f[x, y(x)]$. Again from (4) we have

$$y_n(x) = y_0 + \int_{x_0}^x [f(x, y_{n-1}(x))] dx$$

$$\text{or } \lim_{n \rightarrow \infty} y_n(x) = y_0 + \lim_{n \rightarrow \infty} \int_{x_0}^x f[x, y_{n-1}(x)] dx, \text{ letting } n \rightarrow \infty.$$

Since the sequence $f[x, y_n(x)]$, consisting of continuous functions on the given interval, converge uniformly to $f[x, y(x)]$ on the same interval, the interchanges of limiting operations given below are valid. Thus using (10),

$$y(x) = y_0 + \int_{x_0}^x \lim_{n \rightarrow \infty} f[x, y_{n-1}(x)] dx$$

$$\text{or } y(x) = y_0 + \int_{x_0}^x f[x, y(x)] dx \quad (11)$$

The integrand on the right hand side of (11) being a continuous function of x , we conclude that the integral has the derivative. Thus, the limit function $y(x)$ satisfies the differential equation on $[x_0 - h, x_0 + h]$ and is such that $y(x_0) = y_0$. [In the above four steps we have thus proved the existence of a solution of the given initial value problem. The next step will show that the solution $y(x)$ is unique.]

Fifth Step: Uniqueness of the Solution: We now prove that the solution $y = y(x)$ just found is the only solution for which $y(x_0) = y_0$.

Assume if possible $y = Y(x)$, say is another solution of the given initial value problem.

Let $|Y(x) - y(x)| \leq B$ where $x_0 - h \leq x \leq x_0 + h$ (12)

It may be noted here that we can surely take $B = 2K$.

From (11), we get

$$|Y(x) - y(x)| = \left| \int_{x_0}^x [f\{x, Y(x)\} - f\{x, y(x)\}] dx \right|$$

$$\text{or } |Y(x) - y(x)| \leq \int_{x_0}^x |f\{x, Y(x)\} - f\{x, y(x)\}| \cdot |dx|$$

$$\text{or } |Y(x) - y(x)| \leq K \int_{x_0}^x |Y(x) - y(x)| \cdot |dx| \quad (13)$$

$$\therefore |f\{x, Y(x)\} - f\{x, y(x)\}| \leq K |Y(x) - y(x)| \text{ by Lipschitz condition}$$

$$\text{or } |Y(x) - y(x)| \leq K \cdot B \cdot |x - x_0|, \text{ using (12)} \quad (14)$$

Now substituting (14) for integrand in (13), we get

$$|Y(x) - y(x)| \leq K^2 B \int_{x_0}^x |x - x_0| dx \leq \frac{K^2 B |x - x_0|^2}{2!} \quad (15)$$

Again, substituting (15) for the integrand in (13), we get

$$|Y(x) - y(x)| \leq \frac{K^3 B}{2!} \int_{x_0}^x |x - x_0|^2 dx \leq \frac{K^3 B |x - x_0|^3}{3!}$$

Continuing in this way, we shall surely get

$$|Y(x) - y(x)| \leq \frac{K^n B |x - x_0|^n}{n!} \leq B \frac{(Kh)^n}{n!} [\because |x - x_0| \leq h] \quad (16)$$

Now the series $\sum_{n=0}^{\infty} B \frac{(Kh)^n}{n!}$ converges and so $\lim_{n \rightarrow \infty} B \frac{(Kh)^n}{n!} = 0$

Thus $|Y(x) - y(x)|$ can be made less than any number however small and consequently we conclude that $Y(x) - y(x) = 0$ i.e. $Y(x) = y(x)$.

This shows that the solution $y = y(x)$ is always unique, and the proof of the theorem is complete.

Remark 1: If the existence theorem is asked then you need not mention uniqueness in the statement of theorem and finish up the proof just after the fourth step. Again if you are asked to state and prove the uniqueness theorem, give the complete proof by mentioning the necessary matter of first four steps in short and fifth step in details.

Remark 2: If $f(x, y)$ satisfies the condition

$$\left| \frac{\partial f}{\partial y} \right| \leq K \quad (i)$$

For all values of x, y in the given range then for the same constant K the Lipschitz's condition is also satisfied. By the mean value theorem of differential calculus, we get

$$f(x, y_2) - f(x, y_1) = (y_2 - y_1) \left(\frac{\partial f}{\partial y} \right)_{y=\bar{y}}, \text{ where } y_1 < \bar{y} < y_2, \quad (ii)$$

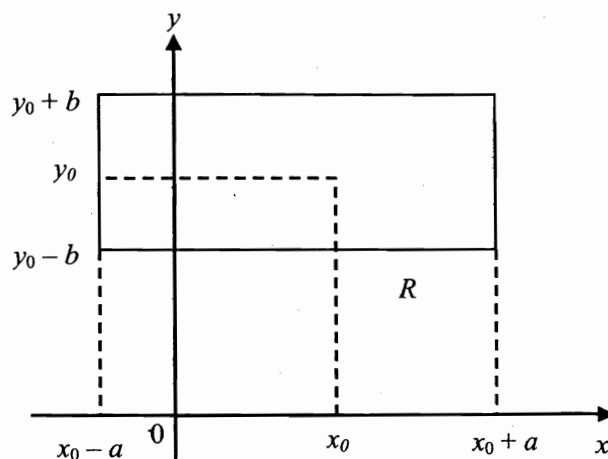
where (x, y_1) and (x, y_2) are assumed in the given range.

$$\text{Now (i) and (ii)} \Rightarrow |f(x, y_2) - f(x, y_1)| \leq K |y_2 - y_1|, \quad (iii)$$

which is Lipschitz condition. It follows that the Lipschitz condition (iii) can be replaced by the stronger condition (i)

Existence Theorem (Sufficient Conditions)

Let $f(x, y)$ be continuous at all points in the closed rectangular region $R: |x - x_0| \leq a, |y - y_0| \leq b$ and $|f(x, y)| \leq M$. Then the initial value problem $y' = f(x, y), y(x_0) = y_0$ has at least one solution. This solution is defined at least for all x in the interval. $I = |x - x_0| \leq h$ let where $h = \min\{a, b/M\}$



The Proof is Omitted

Since $y' = f(x, y)$ and $|f(x, y)| \leq M$ implies $|y'| \leq M$. The slope of every solution curve lies in the interval $[-M, M]$. If $h = \min\{a, b/M\} = a$, then the solution exist at least in the interval $[x_0 - a, x_0 + a]$. If $h = \min\{a, b/M\} = b/M$. Then the solution exists at least in the small

interval $[x_0 - b/m, x_0 + b/M]$. For values of x out side these ranges, the curve leaves the region R and hence nothing can be said about the existence of solution at these values.

It may be pointed out that the conditions given in the theorem are sufficient and not necessary, that is even if the conditions are violated, the problem may have a solution.

Example: Test the existence and uniqueness of the solutions of the initial value problem $y' = \sqrt{y}, y(1) = 0$ in the suitable rectangle R . If more than one solution exists, then find all solutions.

Solution: Here $f(x, y) = \sqrt{y}$, $x_0 = 1, y_0 = 0$. Now $f(x, y)$ is continuous and bounded in R . Hence at least one solution exists in some rectangle containing $(1, 0)$. Let us now test the Lipschitz condition. We have

$$|f(x, y_2) - f(x, y_1)| = |\sqrt{y_2} - \sqrt{y_1}|$$

$$= \left| \frac{(\sqrt{y_2} + \sqrt{y_1})(\sqrt{y_2} - \sqrt{y_1})}{(\sqrt{y_2} + \sqrt{y_1})} \right| = \left| \frac{y_2 - y_1}{\sqrt{y_2} + \sqrt{y_1}} \right|$$

or $\frac{|f(x, y_2) - f(x, y_1)|}{|y_2 - y_1|} = \frac{1}{\sqrt{y_2} + \sqrt{y_1}}$

This equation can be made as large as possible by choosing y_1 and y_2 sufficiently small, that is, a finite value for the Lipschitz constant M can not be determined.

We have $\sqrt{y_1} + \sqrt{y_2} < 2\sqrt{y}$, if $y = \max\{y_1, y_2\}$

And $\left| \frac{1}{\sqrt{y_1} + \sqrt{y_2}} \right| > \frac{1}{2\sqrt{y}} > M$, if $\sqrt{y} < \frac{1}{2M}$

In the neighbourhood of $y = 0$, this criterion is satisfied for every $M > 0$

Therefore, the initial value problem does not passess a unique solution. Infact, there are two solutions of the IVP. Which are

$$y = (x-1)^2/4 \text{ and } y^2 = 0 \quad (i)$$

it can be easily verified for $y \neq 0$, from the differential equation $\frac{dy}{\sqrt{y}} = dx$

Integrating, We have

$$2\sqrt{y} = x + c \text{ or } y = \left(\frac{x+c}{2} \right)^2 \quad (ii)$$

The conditions $y(1) = 0$ is satisfied both by (i) and (ii) $\left[(c+1)^2/4 \right] = 0$ or $c = -1$

Hence, the solutions are $y \equiv 0$ and $y = \frac{(x-1)^2}{4}$

Note: If c can be determine uniquely then we can not say the IVP has a unique solution.

Example: Show that $f(x, y) = xy^2$ satisfies Lipschitz condition on the rectangle $|x| \leq 1, |y| \leq 1$ but does not satisfy a Lipschitz condition on the strip $|x| \leq 1, |y| < \infty$.

Solution: $|f(x, y_2) - f(x, y_1)| = |xy_2^2 - xy_1^2|$, as $f(x, y) = xy^2$

$$\text{or } |f(x, y_2) - f(x, y_1)| = |x||y_2 + y_1||y_2 - y_1| \quad (1)$$

Hence in the rectangle $|x| \leq 1, |y| \leq 1$ (1) reduces to

$$|f(x, y_2) - f(x, y_1)| \leq (1)(2)|y_2 - y_1|,$$

Showing that Lipschitz condition is satisfied.

Next, $\left| \frac{f(x, y_2) - f(x, y_1)}{y_2 - y_1} \right| = |x||y_2 + y_1| \rightarrow \infty$ when $|y_2| \rightarrow \infty$ if $|x| \neq 0$, showing that the Lipschitz condition is not satisfied on the strip $|x| \leq 1, |y| < \infty$.

Example: Show that the solution of the initial value problem $\frac{dy}{dx} = f(x, y), y(x_0) = y_0$ may not be unique although $f(x, y)$ is continuous.

Solution: Consider the initial value problem

$$\frac{dy}{dx} = \sqrt{|y|}, y(0) = 0. \quad (1)$$

Clearly $f(x, y) = |y|$ is continuous for all y and (1) has the following two solutions

$$y = 0 \text{ and } y = \begin{cases} \frac{x^2}{4}, & \text{when } x \geq 0 \\ -\frac{x^2}{4}, & \text{when } x < 0 \end{cases}$$

We now show that Lipschitz condition $\frac{|f(x, y_2) - f(x, y_1)|}{|y_2 - y_1|} \leq K \quad (2)$

does not hold good in any region which includes the line $y = 0$. For example when $y_1 = 0$ and y_2 is positive, then

$$\frac{|f(x, y_2) - f(x, y_1)|}{|y_2 - y_1|} = \frac{\sqrt{y_2}}{y_2} = \frac{1}{\sqrt{y_2}}, \quad (\because \sqrt{y_2} > 0)$$

showing that quantity on L.H.S of (2) can be made as large as we please by choosing y_2 sufficiently small. This violates (2) because the quantity on L.H.S. of (2) should not exceed a fixed constant.

Example : Illustrate by an example that a continuous function may not satisfy a Lipschitz condition on a rectangle.

Solution: As example, consider the function

$$f(x, y) = y^{2/3} \text{ on the rectangle } S : |x| \leq 1, |y| \leq 1 \quad (1)$$

Clearly, $f(x, y)$ is continuous in the rectangle S .

Here $\left| \frac{\partial}{\partial y} f(x, y) \right| = \left| \frac{2}{3y^{1/3}} \right| \rightarrow \infty$ when $y \rightarrow 0$.

Since $y=0$ is a point of the rectangle S , (2) shows that the Lipschitz condition is not satisfied by the function $f(x, y) = y^{2/3}$ on the rectangle S .

Example: For the initial value problem $\frac{dy}{dx} = e^y, y(0) = 0$. Find the largest interval $|x| \leq a$ in which the Picard's theorem guarantees existence of a unique solution.

Solution: Here the condition of boundedness of $f(x, y)$, namely,

$$|f(x, y)| \leq M \text{ for } |y - y_0| \leq Ma,$$

reduces to $e^y \leq M$ for $|y - 0| \leq Ma$ ($\because f(x, y) = e^y$)

Let y_1, y_2 lie in the range $|y| \leq Ma$ and $y_1 < y_2$. Then using the mean value theorem, we have

$$e^{y_2} - e^{y_1} = (y_2 - y_1) \left(\frac{\partial e^y}{\partial y} \right)_{y=\bar{y}}, \text{ where } y_1 < \bar{y} < y_2$$

$$\text{or } e^{y_2} - e^{y_1} \leq (y_2 - y_1)M, \text{ since } e^{\bar{y}} \leq M.$$

showing that the function e^y satisfies the Lipschitz condition.

Again, the inequality $e^y \leq M$ will be satisfied for all values of y such that $|y| \leq Ma$ provided it is satisfied by $y = Ma$.

$$\therefore e^{Ma} \leq M \text{ or } a \leq (\log M) / M$$

Using the well known methods of finding maximum and minimum values of a function, we can easily show that the function $(\log M) / M$ is maximum when $M = e = 2.718$. Hence uniqueness theorem of the given initial value problem gives $|x| \leq a$ where $a = 1/e = 0.308$. Thus, the required largest interval is $|x| \leq 0.308$.

Example: Show that for the problem $\frac{dy}{dx} = y, y(0) = 1$, the constant a in Picard's theorem must be smaller than unity.

Solution: The condition of boundedness of $f(x, y)$ namely

$$|f(x, y)| \leq M \text{ for } |y - y_0| \leq Ma \text{ reduces to } |y| \leq M \text{ for } |y - 1| \leq Ma$$

Choose $M \geq 1$. Then the Lipschitz condition is also satisfied because in this situation, the Lipschitz condition takes the form $|f(x, y_2) - f(x, y_1)| = |y_2 - y_1| \leq M |y_2 - y_1|$ as $M \geq 1$.

$$\text{Again, } |y - 1| \leq Ma \Rightarrow |y| - 1 \leq Ma.$$

It follows that the condition $|y| \leq M$ will be satisfied for all values of $|y - 1| \leq Ma$ provided it is satisfied

$$\text{for } |y| \leq 1 + Ma.$$

$$1 + Ma \leq M \text{ or } a < (M - 1) / M = 1 - (1 / M) < 1 \text{ as } 1 \leq M < \infty.$$

Example: Determine if the initial - value problem

$$y'(t) = \frac{2}{t}y + t^2e^t, \quad 1 \leq t \leq 2, \quad y(1) = 0$$

has a unique solution for $1 \leq t \leq 2$. If so, find the solution exactly or numerically.

Let $D = \{(t, y) | 1 \leq t \leq 2, -\infty < y < \infty\}$. Check if $f(t, y) = \frac{2}{t}y + t^2e^t$ satisfies a Lipschitz condition in D . Beause

$$\left| \frac{\partial f}{\partial y} \right| = \left| \frac{2}{t} \right| \leq \frac{2}{1} = 2,$$

$f(t, y)$ satisfies a Lipschitz condition in D and therefore by Theorem 2 the initial-value problem has a unique solution.

Example: Find approximations $y_k(t)$ of the solution to the initial-value problem:

$$\frac{dy}{dt} = y + t, \quad y(0) = -1, \text{ by Picard's method}$$

For this problem, $f(t, y) = y + t$ and $y_0(t) = -1$. Then

$$y_1(t) = -1 + \int_0^t (-1 + x) dx = \frac{1}{2}t^2 - t - 1$$

$$y_2(t) = -1 + \int_0^t \left(\frac{1}{2}x^2 - x - 1 + x \right) dx = -1 + \int_0^t \left(\frac{1}{2}x^2 - 1 \right) dx = \frac{1}{6}t^3 - t - 1$$

$$y_k(t) = \frac{1}{(k+1)!}t^{k+1} - t - 1$$

Review the sufficient condition for a function f satisfying a Lipschitz condition.

Show that each of the following functions satisfies a Lipschitz condition in y on the indicated set D .

- (i) $t(t, y) = \frac{t\sqrt{y}}{1+t^2}, D = \{(t, y); t \text{ is in } R, y \geq 1\}$
- (ii) $f(t, y) = 1 - y + e^{2t}y^2, D = \{(t, y); 0 \leq t \leq 1, -5 \leq y \leq 5\}$
- (iii) $f(t, y) = y^2 - \frac{t}{y}, D = \{(t, y); -2 \leq t \leq 2, 1 \leq y \leq 10\}$

Let $M > 0$. Show that the function $f(t, y) = t^4 - \frac{4y}{t}$ satisfies a Lipschitz condition in y on the set.

$D = \{(t, y); t \geq M, y \text{ is real}\}$. Does f satisfy a Lipschitz condition in y on the set

$$D = \{(t, y); t \geq 0, y \text{ is real}\}?$$

Example: Consider the ODE

$$y' = xy - \sin y, \quad y(0) = 2.$$

Here f and $\partial f / \partial y$ are continuous in a closed rectangle about $x_0 = 0$ and $y_0 = 2$. Hence, there exists unique solution in the neighbourhood of $(0, 2)$.

ASSIGNMENT SHEET -01

1. The initial value problem $\dot{x}(t) = 3x^{2/3}, x(0) = 0$, in an interval around $t = 0$, has
 - (a.) No solution
 - (b.) A unique solution
 - (c.) Finitely many linearly independent solutions
 - (d.) Infinitely many linearly independent solutions
2. The initial value problem $\frac{dy}{dx} = 2y^{1/3}, y(0) = 0$ has
 - (a.) no solution
 - (b.) infinitely many solutions
 - (c.) exactly one solution
 - (d.) finitely many solutions
3. Let $y(t)$ be a solution to the differential equation $y' = y^2 + t$, then $y(t)$ is differentiable
 - (a.) once but not twice
 - (b.) twice but not 3 times
 - (c.) 3 times but not 4 times
 - (d.) infinitely many times
4. The differential equation $\frac{dy}{dx} = y^{1/3}, y(0) = 0$ has
 - (a.) A unique solution
 - (b.) No non-trivial solution
 - (c.) Finite number of solutions
 - (d.) Infinite number of solutions.
5. The order of the differential equation whose general solution is $y = c_1 \cos 2x + c_2 \cos^2 x + c_3 \sin^2 x + c_4$ is
 - (a.) Two
 - (b.) Three
 - (c.) Four
 - (d.) One
6. Order and degree of the differential equation $e^{y''} - y'' + xy = 0$
 - (a.) 3, 1
 - (b.) 3, not defined
 - (c.) Both degree and order are not defined
 - (d.) not defined, 3
7. The differential equation of the system of circles touching the y -axis at the origin is
 - (a.) $x^2 + y^2 - 2xy \frac{dy}{dx} = 0$
 - (b.) $x^2 + y^2 + 2xy \frac{dy}{dx} = 0$
 - (c.) $x^2 - y^2 + 2xy \frac{dy}{dx} = 0$
 - (d.) $x^2 - y^2 - 2xy \frac{dy}{dx} = 0$
8. The degree of the equation $\left(\frac{d^3 y}{dx^3}\right)^{2/3} + \left(\frac{d^3 y}{dx^3}\right)^{3/2} = 0$ is
 - (a.) $\frac{3}{2}$
 - (b.) 5
 - (c.) 4
 - (d.) 9
9. The order and degree of the differential equation $\frac{d^2}{dx^2} \left(\frac{d^2 y}{dx^2} \right)^{-3/2} = 0$ is
 - (a.) 1, 4
 - (b.) 4, 1
 - (c.) 4, 4
 - (d.) 1, 1
10. The differential equation $\left(\frac{dx}{dy}\right)^2 + 5y^{1/3} = 2$, where y is dependent variable is
 - (a.) Linear differential equation of order 2 and degree 1
 - (b.) Non linear differential equation of order 1 and degree 2
 - (c.) Non linear differential equation of order 2 and degree 1
 - (d.) Linear differential equation of order 1 and degree 2
11. If $y = y(x)$ then the differential equation, $\frac{dy}{dx} + Py = Q$, is a linear equation of first order if,
 - (a.) P is a constant but Q is a function of y
 - (b.) P and Q are functions of y
 - (c.) P is a function of y but Q is a constant
 - (d.) P and Q are functions of x or constants

12. The order and degree of the differential equation

$$\left(\frac{d^2y}{dx^2}\right)^{1/4} + \left(\frac{d^4y}{dx^4}\right)y^{10} + 4e^x \cos(2x+3) = 0$$

are

respectively

- (a.) 4, 10
(b.) 4, 4
(c.) $\frac{1}{4}, 10$
(d.) $4, \frac{1}{4}$

13. Differential equation of the curve

$$y = cx + \frac{1}{c}$$

- (a.) $xy'^2 - yy' + 1 = 0$
(b.) $xy'^2 - yy' + 2 = 0$
(c.) $xy'^2 + 4 = 0$
(d.) $xy'^2 - 4 = 0$

14. The differential equation

$$\frac{dy}{dx} = 60(y^2)^{1/5}; x > 0, y(0) = 0 \text{ has}$$

- (a.) A unique solution
(b.) Two solutions
(c.) No solution
(d.) Infinite number of solutions.

15. A curve γ in the xy -plane is such that the line joining the origin to any point $P(x, y)$ on the curve and the line parallel to the y -axis through P are equally inclined to the tangent to the curve at P . Then, the differential equation of the curve γ is

- (a.) $x\left(\frac{dy}{dx}\right)^2 + 2y\left(\frac{dy}{dx}\right) = x$
(b.) $x\left(\frac{dy}{dx}\right)^2 + 2y\left(\frac{dy}{dx}\right) = 0$
(c.) $x\left(\frac{dx}{dy}\right)^2 + 2y\left(\frac{dx}{dy}\right) = 0$

(d.) $x\left(\frac{dx}{dy}\right)^2 + 2y\left(\frac{dx}{dy}\right) = x$

16. If f is a differentiable function, define $(Df)(x) = \frac{1}{2x} \frac{df}{dx}$. Given $n \in \mathbb{N}$ and n -times differentiable function f , define $(D^n f)(x)$ by successively applying D for n times to the function f . Then $D^8(e^{-x^2})$ at $x=0$ is.

- (a.) 1
(b.) 0
(c.) e
(d.) e^{-1}

17. The differential equation corresponding to the family of circles of radius ' r ' whose centre lies on the x -axis is

- (a.) $y\left(\frac{dy}{dx}\right) + y^2 = r$
(b.) $y\left\{\frac{dy}{dx} + 1\right\} = r^2$
(c.) $y^2\left\{\frac{dy}{dx} + 1\right\} = r^2$
(d.) $y^2\left\{\left(\frac{dy}{dx}\right)^2 + 1\right\} = r^2$

18. Consider the Curve $y(x) = ax + \frac{1}{a}$, where $a \neq 0$ is an arbitrary constant. Then the corresponding differential equation is
(a.) Second order, first degree, linear differential equation.
(b.) First order, Second degree, non-linear differential equation.
(c.) First order, first degree, non-linear differential equation.
(d.) First order, first degree, linear differential equation.

19. The differential equation corresponding to $y = k \cos^{-1} x$ (Where k is a parameter) is

- (a.) $y - \sqrt{1-x^2} \cos^{-1} x \frac{dy}{dx} = 0$
(b.) $y - \sqrt{1-x^2} \sin^{-1} x \frac{dy}{dx} = 0$
(c.) $y + \sqrt{1-x^2} \cos^{-1} x \frac{dy}{dx} = 0$
(d.) $y + \sqrt{1-x^2} \sin^{-1} x \frac{dy}{dx} = 0$

20. If $y = y(x)$ Then the differential equation $y' = \sqrt{x} + \sqrt{y}$ is
(a.) Linear differential equation of order 1.
(b.) Non-linear differential equation of order 1 and degree 1.

- (c.) Linear differential equation of order 1 and degree 1.
(d.) Non- linear differential equation of order 1 and degree 2.

21. The differential equation. $\left| \frac{dy}{dx} \right| + |y| + a = 0$

- (a.) Has infinite many real solution if $a < 0$
(b.) Has unique real solution if $a < 0$
(c.) Has no real solution if $a > 0$
(d.) Has unique real solution if $a = 0$

The differential equation

$\frac{dy}{dx} = 60(y^2)^{1/5}; x > 0, y(0) = 0$ has

- (a.) A unique solution
(b.) Two solutions
(c.) No solution
(d.) Infinite number of solutions.

23. The degree of the differential equation $(y'')^2 + (y')^2 = x \sin\left(\frac{d^2y}{dx^2}\right)$ is

- (a.) 1
(b.) 2
(c.) 3
(d.) None of these

24. Which of the following differential equation is linear if $y = y(x)$?

- (a.) $\frac{dy}{dx} + x^2y = \sqrt{y}$
(b.) $\frac{dy}{dx} - \sin x = x^n y$
(c.) $(1+y)\frac{dy}{dx} + P(x)y = Q(x)$
(d.) $\sqrt{y + \frac{dy}{dx}} = x + y$

ASSIGNMENT SHEET-02

- Consider the ODE on $\mathbb{R}$ $y'(x) = f(y(x))$. If f is an even function and y is an odd function, then
 - $-y(-x)$ is also a solution.
 - $y(-x)$ is also a solution.
 - $-y(x)$ is also a solution.
 - $y(x)y(-x)$ is also a solution.
- Let $u(t)$ be a continuously differentiable function taking nonnegative values for $t > 0$ and satisfying $u'(t) = 4u^{3/4}(t)$; $u(0) = 0$. Then
 - $u(t) = 0$.
 - $u(t) = t^4$.
 - $u(t) = \begin{cases} 0 & \text{for } 0 < t < 1 \\ (t-1)^4 & \text{for } t \geq 1 \end{cases}$
 - $u(t) = \begin{cases} 0 & \text{for } 0 < t < 10 \\ (t-10)^4 & \text{for } t \geq 10 \end{cases}$
- Let $y(x)$ be a continuous solution of the initial value problem $y' + 2y = f(x)$, $y(0) = 0$, where $f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ 0, & x > 1 \end{cases}$. Then $y\left(\frac{3}{2}\right)$ is equal to
 - $\frac{\sinh(1)}{e^3}$
 - $\frac{\cosh(1)}{e^3}$
 - $\frac{\sinh(1)}{e^2}$
 - $\frac{\cosh(1)}{e^2}$
- Let $u(t)$ be a continuously differentiable function taking non-negative values for $t > 0$ satisfying $u'(t) = 3u(t)^{2/3}$ and $u(0) = 0$. Which of the following are possible solutions of the above equation?
 - $u(t) = 0$
 - $u(t) = t^3$
 - $u(t) = \begin{cases} 0 & \text{for } 0 < t < 1 \\ (t-1)^3 & \text{for } t \geq 1 \end{cases}$
 - $u(t) = \begin{cases} 0 & \text{for } 0 < t < 3 \\ (t-3)^3 & \text{for } t \geq 3 \end{cases}$
- Let $y: \mathbb{R} \rightarrow \mathbb{R}$ satisfy the initial value problem $y'(t) = 1 - y^2(t)$, $t \in \mathbb{R}$, $y(0) = 0$. Then
 - $y(t_1) = 1$ for some $t_1 \in \mathbb{R}$.
 - $y(t) > -1$ for all $t \in \mathbb{R}$.
 - y is strictly increasing in $\mathbb{R}$.
 - y is increasing in $(0, 1)$ and decreasing in $(1, \infty)$.
- Let $y_1(x)$ and $y_2(x)$ be the solutions of the differential equation $\frac{dy}{dx} = y + 17$ with initial conditions $y_1(0) = 0$, $y_2(0) = 1$. Then
 - y_1 and y_2 will never intersect.
 - y_1 and y_2 will intersect at $x = 17$
 - y_1 and y_2 will intersect at $x = e$
 - y_1 and y_2 will intersect at $x = 1$
- Consider the initial value problem $y'(t) = f(t)y(t)$, $y(0) = 1$ Where $f: \mathbb{R} \rightarrow \mathbb{R}$ is continuous. Then this initial value problem has
 - infinitely many solutions for some f
 - a unique solution in $\mathbb{R}$
 - no solution in $\mathbb{R}$ for some f
 - a solution in an interval containing 0, but not on $\mathbb{R}$ for some f
- Consider the equation $\frac{dy}{dt} = (1 + f^2(t))y(t)$, $y(0) = 1$; $t \geq 0$, Where f is a bounded continuous function on $[0, \infty)$. Then
 - The equation admits a unique solution $y(t)$ and further $\lim_{t \rightarrow \infty} y(t)$ exists and is finite
 - This equation admits two linearly independent solutions
 - This equation admits a bounded solution for which $\lim_{t \rightarrow \infty} y(t)$ does not exist
 - This equation admits a unique solution $y(t)$ and further $\lim_{t \rightarrow \infty} y(t) = \infty$

9. The equation $(\alpha xy^3 + y \cos x)dx + (x^2 y^2 + \beta \sin x)dy = 0$ is exact for

(a.) $\alpha = \frac{3}{2}, \beta = 1$

(b.) $\alpha = 1, \beta = \frac{3}{2}$

(c.) $\alpha = \frac{2}{3}, \beta = 1$

(d.) $\alpha = 1, \beta = \frac{2}{3}$

10. An integrating factor for $(\cos y \sin 2x)dx + (\cos^2 y - \cos^2 x)dy = 0$ is

(a.) $\sec^2 y + \sec y \tan y$

(b.) $\tan^2 y + \sec y \tan y$

(c.) $1/(\sec^2 y + \sec y \tan y)$

(d.) $1/(\tan^2 y + \sec y \tan y)$

11. If the integrating factor of $(x^7 y^2 + 3y)dx + (3x^8 y - x)dy = 0$ is $x^m y^n$, then

(a.) $m = -7, n = 1$

(b.) $m = 1, n = -7$

(c.) $m = n = 0$

(d.) $m = n = 1$

12. The initial value problem $(x^2 - x)\frac{dy}{dx} = (2x - 1)y, y(x_0) = y_0$ has a unique solution if (x_0, y_0) equals

(a.) (2,1)

(b.) (1,1)

(c.) (0,0)

(d.) None of these

13. Which of the following is not an integrating factor of $xdy - ydx = 0$?

(a.) $\frac{1}{x^2}$

(b.) $\frac{1}{x^2 + y^2}$

(c.) $\frac{1}{xy}$

(d.) $\frac{x}{y}$

14. The general solution of the differential equation $\frac{dy}{dx} + \tan y \tan x = \cos x \sec y$ is

(a.) $2 \sin y = (x + c - \sin x \cos x) \sec x$

(b.) $\sin y = (x + c) \cos x$

(c.) $\cos y = (x + c) \sin x$

(d.) $\sec y = (x + c) \cos x$

15. The particular solution of the equation $y' \sin x = y$ satisfying the initial condition $y\left(\frac{\pi}{2}\right) = e$, is

(a.) $e \tan(x/2)$

(b.) $e \cot(x/2)$

(c.) $\ln \tan\left(\frac{x}{2}\right)$

(d.) $\ln \cot\left(\frac{x}{2}\right)$

16. The differential equation $\frac{dy}{dx} = h(a - y)(b - y)$, when solved with the condition $y(0) = 0$, yields the result

(a.) $\frac{b(a - y)}{a(b - y)} = e^{(a-b)hx}$

(b.) $\frac{b(a - x)}{a(b - x)} = e^{(b-a)hy}$

(c.) $\frac{a(b - y)}{b(a - y)} = e^{(a-b)hx}$

(d.) $xy = he$

17. The differential equation $(3a^2 x^2 + b y \cos x)dx + (2 \sin x - 4ay^3)dy = 0$ is exact for

(a.) $a = 3, b = 2$

(b.) $a = 2, b = 3$

(c.) $a = 3, b = 4$

(d.) $a = 2, b = 5$

18. For the differential equation $xy' - y = 0$ which of the following function is not an integrating factor?

(a.) $\frac{1}{x^2}$

(b.) $\frac{1}{y^2}$

(c.) $\frac{1}{xy}$

(d.) $\frac{1}{x+y}$

19. If $y' - x \neq 0$, a solution of the differential equation $y'(y' + y) = x(x + y)$ is given by $y =$

(a.) $1 - x - e^{-x}$

(b.) $1 - x + e^x$

(c.) $1 + x + e^{-x}$

(d.) $1 + x + e^x$

20. For $a, b, c \in \mathbb{R}$, if the differential equation $(ax^2 + bxy + y^2)dx + (2x^2 + cxy + y^2)dy = 0$ is exact, then

(a.) $b = 2, c = 2a$

(b.) $b = 4, c = 2$

(c.) $b = 2, c = 4$

(d.) $b = 2, a = 2c$

21. The differential equation $(1 + x^2y^3 + \alpha x^2y^2)dx + (2 + x^3y^2 + x^3y)dy = 0$ is exact if α equals

(a.) $\frac{1}{2}$

(b.) $\frac{3}{2}$

(c.) 2

(d.) 3

22. An integrating factor for the differential equation

$(2xy + 3x^2y + 6y^3)dx + (x^2 + 6y^2)dy = 0$ is

(a.) x^3

(b.) y^3

(c.) e^{3x}

(d.) e^{3y}

23. If y^a is an integrating factor of the differential equation $2xy dx - (3x^2 - y^2)dy = 0$, then the value of a is

(a.) -4

(b.) 4

(c.) -1

(d.) 1

24. Consider the differential equation $\frac{dy}{dx} = ay - by^2$, where $a, b > 0$ and $y(0) = y_0$.

As $x \rightarrow +\infty$, the solution $y(x)$ tends to

(a.) 0

(b.) $\frac{a}{b}$

(c.) $\frac{b}{a}$

(d.) y_0

25. Consider the differential equation $(x + y + 1)dx + (2x + 2y + 1)dy = 0$. Which of the following statements is true?

(a.) The differential equation is linear

(b.) The differential equation is exact

(c.) e^{x+y} is an integrating factor of the differential equation

(d.) A suitable substitution transforms the differentiable equation to the variables separable form.

26. Consider the differential equation $2\cos(y^2)dx - xy\sin(y^2)dy = 0$. Then

- (a.) e^x is an integrating factor
- (b.) e^{-x} is an integrating factor
- (c.) $3x$ is an integrating factor
- (d.) x^3 is an integrating factor.

27. If k is a constant such that $xy + k = e^{(x-1)^2/2}$ satisfies the differential equation $x \frac{dy}{dx} = (x^2 - x - 1)y + (x - 1)$, then k is equal to

- (a.) 1
- (b.) 0
- (c.) -1
- (d.) -2

28. One of the integrating factors of the differential equation

$$(y^2 - 3xy)dx + (x^2 - xy)dy = 0 \text{ is}$$

- (a.) $\frac{1}{(x^2y^2)}$
- (b.) $\frac{1}{(x^2y)}$
- (c.) $\frac{1}{(xy^2)}$
- (d.) $\frac{1}{(xy)}$

29. An integrating factor of

$$x \frac{dy}{dx} + (3x + 1)y = xe^{-2x} \text{ is}$$

- (a.) xe^{3x}
- (b.) $3xe^x$
- (c.) xe^x
- (d.) x^3e^x

30. The solution of differential equation

$$\frac{dy}{dx} = \frac{x+y+1}{x+y-1}, y\left(\frac{2}{3}\right) = \frac{1}{3} \text{ is}$$

- (a.) $y = x + \log(x+y) - \frac{1}{3}$
- (b.) $y = x - \log(x+y) + \frac{2}{3}$
- (c.) $y = x + \log(x+y) + \frac{2}{3}$
- (d.) $2y = x - \log(x+y) + 1$

Assignment 0.

Find $\int \frac{dx}{x^2 + 1}$ using $\tan^{-1}x = \arctan x$

$$\frac{d}{dx} \tan^{-1}x = \frac{1}{1+x^2}$$

ASSIGNMENT SHEET -03

1. Consider the functions $f(x) = x|x|$ and $g(x) = x^2$. Then
 - (a.) $\{f, g\}$ is a linearly independent pair of functions on $(-\infty, 0)$
 - (b.) $\{f, g\}$ is a linearly independent pair of functions on $(0, \infty)$
 - (c.) $\{f, g\}$ is a linearly dependent pair of functions on $\mathbb{R}$
 - (d.) $\{f, g\}$ is a linearly independent pair of functions on $\mathbb{R}$
2. Consider the two functions $f(x) = |x| \sin x$ and $g(x) = x \sin x$. Then, $\{f, g\}$ is
 - (a.) linearly independent on $(-\infty, 0)$
 - (b.) linearly independent on $(0, \infty)$
 - (c.) linearly dependent on $\mathbb{R}$
 - (d.) linearly independent on $\mathbb{R}$
3. Consider $f(x) = |x|e^{ax}$ and $g(x) = xe^{ax}$. Then the pair $\{f, g\}$ is
 - (a.) Linearly independent on $\mathbb{R}$
 - (b.) Linearly dependent on $(-\varepsilon, \varepsilon)$ for some $0 < \varepsilon < 1$
 - (c.) Linearly independent on $(0, \infty)$
 - (d.) Linearly independent on $(-\infty, 0)$.
4. Let P be a continuous function on $\mathbb{R}$ and W be the Wronskian of two linearly independent solutions y_1 and y_2 of the ODE: $\frac{d^2 y}{dx^2} + (1+x^2) \frac{dy}{dx} + P(x)y = 0, x \in \mathbb{R}$. Let $W(1) = a, W(2) = b$ and $W(3) = c$, then
 - (a.) $a < 0$ and $b > 0$
 - (b.) $a < b < c$ or $a > b > c$
 - (c.) $\frac{a}{|a|} = \frac{b}{|b|} = \frac{c}{|c|}$
- (d.) $0 < a < b$ and $b > c > 0$
5. Let P, Q be continuous real valued function defined on $[-1, 1]$ and $u_i : [-1, 1] \rightarrow \mathbb{R}, i = 1, 2$ be solutions of the ODE: $\frac{d^2 u}{dx^2} + P(x) \frac{du}{dx} + Q(x)u = 0, x \in [-1, 1]$ satisfying $u_1(0) = 0, u_2(0) = 0$. Let w denote the Wronskian of u_1 and u_2 , then
 - (a.) u_1 and u_2 are linearly independent
 - (b.) u_1 and u_2 are linearly dependent
 - (c.) $w(x) = 0$ for all $x \in [-1, 1]$
 - (d.) $w(x) \neq 0$ for some $x \in [-1, 1]$
6. Let $Y_1(x)$ and $Y_2(x)$ defined on $[0, 1]$ be twice continuously differentiable functions satisfying $Y''(x) + Y'(x) + Y(x) = 0$. Let $W(x)$ be the Wronskian of Y_1 and Y_2 and satisfy $W\left(\frac{1}{2}\right) = 0$. Then
 - (a.) $W(x) = 0$ for $x \in [0, 1]$
 - (b.) $W(x) \neq 0$ for $x \in \left[0, \frac{1}{2}\right] \cup \left[\frac{1}{2}, 1\right]$
 - (c.) $W(x) > 0$ for $x \in \left[\frac{1}{2}, 1\right]$
 - (d.) $W(x) < 0$ for $x \in \left[0, \frac{1}{2}\right]$
7. Let $y_1(x)$ and $y_2(x)$ form a complete set of solutions to the differential equation $y'' - 2xy' + \sin(e^{2x^2})y = 0, x \in [0, 1]$ with $y_1(0) = 0, y_1'(0) = 1, y_2(0) = 1, y_2'(0) = 1$. Then the Wronskian $W(x)$ of $y_1(x)$ and $y_2(x)$ at $x = 1$ is
 - (a.) e^2
 - (b.) $-e$
 - (c.) $-e^2$

- (d.) e
8. Let W be the Wronskian of two linearly independent solutions of ODE $2y'' + y' + t^2y = 0; t \in \mathbb{R}$. Then, for all $t \in \mathbb{R}$, there exists a constant $C \in \mathbb{R}$ such that $W(t)$ is
- (a.) Ce^{-1}
(b.) $Ce^{-t/2}$
(c.) Ce^{2t}
(d.) Ce^{-2t}
9. Let y_1 and y_2 be two linearly independent solution of $y'' + (\sin x)y = 0, 0 \leq x \leq 1$. Let $g(x) = W(y_1, y_2)(x)$ be the wronskian of y_1 and y_2 . Then
- (a.) $g' > 0$ on $[0, 1]$
(b.) $g' < 0$ on $[0, 1]$
(c.) g' vanishes at only one point of $[0, 1]$
(d.) g' vanishes at all point of $[0, 1]$
10. Let y be a nontrivial solution of the boundary value problem, $y'' + xy = 0, x \in [a, b], y(a) = y(b) = 0$ then
- (a.) $b > 0$.
(b.) y is monotonic in $(a, 0)$ if $a < 0 < b$.
(c.) $y'(a) = 0$.
(d.) y has infinitely many zeros in $[a, b]$.
11. Let $\frac{d^2y}{dx^2} - q(x)y = 0, 0 \leq x < \infty$, $y(0) = 1, \frac{dy}{dx}(0) = 1$ where $q(x)$ is a positive monotonically increasing continuous function. Then
- (a.) $y(x) \rightarrow \infty$ as $x \rightarrow \infty$
(b.) $\frac{dy}{dx} \rightarrow \infty$ as $x \rightarrow \infty$
(c.) $y(x)$ has finitely many zeros in $[0, \infty)$
(d.) $y(x)$ has infinitely many zeros in $[0, \infty)$
12. Let $y_1(x)$ and $y_2(x)$ form a fundamental set of solution of $\frac{d^2y}{dx^2} + p(x)\frac{dy}{dx} + q(x)y = 0, a \leq x \leq b$, Where $p(x)$ and $q(x)$ are real valued continuous functions on $[a, b]$. If x_0 and x_1 , with $x_0 < x_1$, are consecutive zeros of $y_1(x)$ in (a, b) , then
- (a.) $y_1(x) = (x - x_0)q_0(x)$, where $q_0(x)$ is continuous on $[a, b]$ with $q_0(x_0) \neq 0$
(b.) $y_1(x) = (x - x_0)^2 p_0(x)$, where $p_0(x)$ is continuous on $[a, b]$ with $p_0(x_0) \neq 0$
(c.) $y_2(x)$ has no zeros in (x_0, x_1)
(d.) $y_2(x_0) = 0$ but $y_2'(x_0) \neq 0$
13. Consider the ODE $u''(t) + P(t)u'(t) + Q(t)u(t) = R(t), t \in [0, 1]$ There exist continuous function P, Q and R defined on $[0, 1]$ and two solutions u_1 and u_2 of the ODE such that the Wronskian $W(t)$ of u_1 and u_2 is
- (a.) $W(t) = 2t - 1, 0 \leq t \leq 1$
(b.) $W(t) = \sin 2\pi t, 0 \leq t \leq 1$
(c.) $W(t) = \cos 2\pi t, 0 \leq t \leq 1$
(d.) $W(t) = 1, 0 \leq t \leq 1$
14. Let $y_1(x)$ and $y_2(x)$ form a fundamental set of solutions to the differential equation $\frac{d^2y}{dx^2} + p(x)\frac{dy}{dx} + q(x)y = 0, a \leq x \leq b$, Where $p(x)$ and $q(x)$ are continuous in $[a, b]$ and x_0 is a point in (a, b) . Then
- (a.) Both $y_1(x)$ and $y_2(x)$ cannot have a local maximum at x_0 .
(b.) Both $y_1(x)$ and $y_2(x)$ cannot have a local minimum at x_0 .

- (c.) $y_1(x)$ cannot have a local maximum at x_0 and $y_2(x)$ cannot have local minimum at x_0 simultaneously.
- (d.) Both $y_1(x)$ and $y_2(x)$ cannot vanish at x_0 simultaneously.
15. Consider the linear homogenous differential equation
 $y^n + P_1(x)y^{(n-1)} + \dots + P_n(x)y = 0$ on $[0, 1]$,
 where $P_1, \dots, P_n$ are continuous real valued functions on $[0, 1]$. Then the set of solutions of the above equation
- (a.) Is a linear space of infinite dimension
- (b.) Is a linear space of dimension n
- (c.) Is a linear space of dimension less than n
- (d.) Is not a linear space.
16. Let u and v be two solutions of $y'' + P(x)y' + Q(x)y = 0$ on $[a, b]$. Let $W(u, v)$ denote the Wronskian of u and v . Then
- (a.) $W(u, v)$ vanishes at a point $x_0 \in [a, b] \Rightarrow u$ and v are linearly dependent.
- (b.) $W(u, v)$ vanishes identically on $[a, b] \Rightarrow u$ and v are linearly dependent.
- (c.) $W(u, v)$ does not vanish at any point on $[a, b]$ but u and v are linearly dependent.
- (d.) None of these
17. Consider the linear differential equation $y''' + 3xy'' + 4y' + 2x^2y = \sin x, x \in [0, 1]$. Then the set of solutions of the above equation
- (a.) Is a linear space of infinite dimension
- (b.) Is a linear space of dimension 3
- (c.) Is not a linear space
- (d.) Is a linear space of dimension less than 3
18. The differential equation $y'' + 2x(y') = 0$, satisfying the condition $y(1) = 0, y'(1) = 1$
- (a.) Has no solution
- (b.) Has a unique solution
- (c.) Has two distinct solutions
- (d.) Has infinite solutions
19. If $y_1(x) = x$ and $y_2(x) = xe^x$ are two linearly independent solution of $x^2 \frac{d^2y}{dx^2} - x(x+2) \frac{dy}{dx} + (x+2)y = 0$ then the interval on which they form a fundamental set of solutions is
- (a.) $x > 0$ or $x < 0$
- (b.) $-1 < x < \infty$
- (c.) $-1 < x < 2$
- (d.) $-\infty < x < \infty$
20. Let $y_1(x)$ and $y_2(x)$ be two linearly independent solutions of the differential equation $x \frac{d^2y}{dx^2} - 2x^2 \frac{dy}{dx} + e^x y = 0$ satisfying $y_1(0) = 1, y_2(0) = -1, y_1'(0) = 1$ and $y_2'(0) = 1$. Then the Wronskian of $y_1(x)$ and $y_2(x)$ at $x = 2$, i.e. $W(y_1, y_2)|_{x=2}$ is equal to
- (a.) $2e^{-1}$
- (b.) $2e^{-2}$
- (c.) $2e^4$
- (d.) $2e^{-4}$
- Linked Questions**
- Let $W(y_1, y_2)$ be the Wronskian of two linearly independent solutions y_1 and y_2 of the equation $y'' + P(x)y' + Q(x)y = 0$.
21. The product $W(y_1, y_2)P(x)$ equals
- (a.) $y_2y_1'' - y_1y_2''$
- (b.) $y_1y_2' - y_2y_1'$
- (c.) $y_1'y_2'' - y_2'y_1''$
- (d.) $y_2'y_1' - y_1'y_2''$

22. If $y_1 = e^{2x}$ and $y_2 = xe^{2x}$, then the value of $P(0)$ is

- (a.) 4
- (b.) -4
- (c.) 2
- (d.) -2

23. If a transformation $y = uv$ transforms the given differential equation $f(x)y'' - 4f'(x)y' + g(x)y = 0$ into the equation of the form $v'' + h(x)v = 0$, then u must be

- (a.) $\frac{1}{f^2}$
- (b.) xf
- (c.) $\frac{1}{2f}$
- (d.) f^2

Common data for questions

Let $y_1(x) = 1+x$ and $y_2(x) = e^x$ be two solutions of $y''(x) + P(x)y'(x) + Q(x)y(x) = 0$

24. $P(x) =$

- (a.) $1+x$
- (b.) $-1-x$
- (c.) $\frac{1+x}{x}$
- (d.) $\frac{-1-x}{x}$

25. The set of initial conditions for which the above differential equation has No solution is

- (a.) $y(0) = 2, y'(0) = 1$
- (b.) $y(1) = 0, y'(1) = 1$
- (c.) $y(1) = 1, y'(1) = 0$
- (d.) $y(2) = 1, y'(2) = 2$

26. Let $y_1(x)$ and $y_2(x)$ be two solutions of $(1-x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + (\sec x)y = 0$ With

Wronskian $W(x)$. If $y_1(0) = 1, \left(\frac{dy_1}{dx}\right)_{x=0} =$

0 and $W\left(\frac{1}{2}\right) = \frac{1}{3}$, then $\left(\frac{dy_2}{dx}\right)_{x=0}$ equals

- (a.) $1/4$
- (b.) 1
- (c.) $3/4$
- (d.) $4/3$

27. Let $y = \phi(x)$ and $y = \psi(x)$ be solutions of $y'' - 2xy' + (\sin x^2)y = 0$ such that $\phi(0) = 1, \phi'(0) = 1$ and $\psi(0) = 1, \psi'(0) = 2$. Then the value of the Wronskian $W(\phi, \psi)$ at $x = 1$ is

- (a.) 0
- (b.) 1
- (c.) e
- (d.) e^2

28. Let $y_1(x)$ and $y_2(x)$ be solutions of $y''x^2 + y' + (\sin x)y = 0$, which satisfy the boundary conditions $y_1(0) = 0, y_1'(1) = 1$ and $y_2(0) = 1, y_2'(1) = 0$ respectively. Then

- (a.) y_1 and y_2 do not have common zeros
- (b.) y_1 and y_2 have common zeros
- (c.) Either y_1 or y_2 has a zero of order 2
- (d.) Both y_1 and y_2 have zeros of order 2

29. Linear combinations of solutions of an ordinary differential equation are also solutions if the differential equation is

- (a.) Linear non homogeneous
- (b.) Linear homogeneous
- (c.) Nonlinear homogeneous
- (d.) Nonlinear non homogeneous

30. Determine the type of the differential equation $\frac{d^2y}{dx^2} + \sin(x+y) = \sin x$

- (a.) Linear, homogeneous

- (b.) Nonlinear, homogeneous
(c.) Linear, non homogeneous
(d.) Nonlinear, non homogeneous
31. The value of the Wronskian of the functions $x^2, 3x+2$ and $2x+3$ is
(a.) 0
(b.) -10
(c.) -5
(d.) 8
32. If $y_1(x)$ and $y_2(x)$ are solutions of $y'' + x^2 y' + (1-x)y = 0$ such that $y_1(0) = 0$, $y_1'(0) = -1$ and $y_2(0) = 1$ then the Wronskian $W(y_1, y_2)$ on $\mathbb{R}$
(a.) Is never zero
(b.) Is identically zero
(c.) Is zero only at a finite number of points
(d.) Is zero at countably infinite number of points
33. If $y_1(x)$ and $y_2(x)$ are solutions of $y'' + xy' + (1-x^2)y = y \sin x$ Then which of the following is also its solution?
(a.) $y_1(x) + y_2(x)$
(b.) $y_1(x) - y_2(x)$
(c.) $2y_1(x) - y_2(x)$
(d.) $y_1(x) - 2y_2(x)$
34. Suppose that y_1 and y_2 form a fundamental set of solutions of a second order ordinary differential equation on the interval $-\infty < t < +\infty$, then
(a.) There is only one zero of y_1 between consecutive zeros of y_2
(b.) There are two zeros of y_1 between consecutive zeros of y_2
(c.) There is finite number of zeros of y_1 between consecutive zeros of y_2
(d.) None of the above
35. Let $y_1(x)$ and $y_2(x)$ be linearly independent solutions of the differential equation $y'' + P(x)y' + Q(x)y = 0$, where $P(x)$ and $Q(x)$ are continuous functions on an interval I . Then $y_3(x) = ay_1(x) + by_2(x)$ and $y_4(x) = cy_1(x) + dy_2(x)$ are linearly independent solutions of the given differential equation if
(a.) $ad = bc$
(b.) $ac = bd$
(c.) $ad \neq bc$
(d.) $ac \neq bd$

1 d 2 d 3 a 4 b 5 b c 6 a 7 c 8 b 9 d
10 a b c
11 a b c
12 a b c
13 a b c
14 a b c
15 a b c
16 a b c
17 a b c
18 a b c
19 a b c
20 a b c

ASSIGNMENT SHEET-04

- Consider the second-order ODE $x^2 y'' + Axy' + y = 0$. This equation
 - admits a linearly independent pair of solutions consisting of trigonometric functions for each $A > 0$
 - admits a linearly independent pair of solutions consisting of power of x for some values of $A > 0$,
 - does not admit any linearly independent pair of solutions consisting of power of x for any $A > 0$
 - has no solutions.
- The number of non-trivial polynomial solution of the differential equation $x^3 y'(x) = y(x^2)$ is
 - zero
 - one
 - three
 - infinity
- Let $y: \mathbb{R} \rightarrow \mathbb{R}$ be a solution of the ODE $\frac{d^2 y}{dx^2} - y = e^{-x}$, $x \in \mathbb{R}$, $y(0) = \frac{dy}{dx}(0) = 0$ then
 - y attains its minimum on $\mathbb{R}$
 - y is bounded on $\mathbb{R}$
 - $\lim_{x \rightarrow \infty} e^{-x} y(x) = \frac{1}{4}$
 - $\lim_{x \rightarrow \infty} e^{-x} y(x) = \frac{1}{2}$
- Let V be the set of all bounded solutions of the ODE $u''(t) - 4u'(t) + 3u(t) = 0$, $t \in \mathbb{R}$. Then V
 - Is a real vector space of dimension 2
 - Is a real vector space of dimension 1
 - Contains only the trivial function $u \equiv 0$
 - Contains exactly two functions.
- The particular solution of the boundary value problem $\frac{d^2 y}{dx^2} + y = \csc x$; $0 < x < \frac{\pi}{2}$, $y(0) = y\left(\frac{\pi}{2}\right) = 0$
 - Convex
 - Concave
 - Negative
 - Positive
- The particular integral $y_p(x)$ of the differential equation $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = \frac{1}{x+1}$, $x > 0$ is given by $y_p(x) = xv_1(x) + \frac{1}{x}v_2(x)$ where $v_1(x)$ and $v_2(x)$ are given by
 - $xv_1'(x) - \frac{1}{x^2}v_2'(x) = 0$, $v_1'(x) - \frac{1}{x^2}v_2' = \frac{1}{x+1}$
 - $xv_1'(x) + \frac{1}{x^2}v_2'(x) = 0$, $v_1'(x) - \frac{1}{x^2}v_2' = \frac{1}{x+1}$
 - $xv_1'(x) - \frac{1}{x^2}v_2'(x) = 0$, $v_1'(x) + \frac{1}{x^2}v_2' = \frac{1}{x+1}$
 - None of these
- The solution to the initial value problem $\frac{d^2 y}{dt^2} + 2\frac{dy}{dt} + 5y = 3e^{-t} \sin t$, $y(0) = 0$ and $\frac{dy}{dt}(0) = 3$, is
 - $y(t) = e^t(\sin t + \sin 2t)$
 - $y(t) = e^{-t}(\sin t + \sin 2t)$
 - $y(t) = 3e^t \sin t$
 - $y(t) = 3e^{-t} \sin t$
- The ordinary differential equation $x \frac{dy}{dx} - y = 2x^2$ with initial conditions $y(0) = 0$ has
 - No solution
 - A unique solution
 - Two distinct solutions
 - Has infinite solutions
- The initial value problem $2x \frac{dy}{dx} = 3(2y-1)$, $y(0) = \frac{1}{2}$ has
 - A unique solution
 - More than one, but only finitely many solutions
 - Infinitely many solutions
 - No solution

10. A homogenous linear differential equation with real constant coefficients, which has $y = xe^{-3x} \cos 2x + e^{-3x} \sin 2x$ as one of its solutions, is given by:
- $(D^2 + 6D + 13)y = 0$
 - $(D^2 - 6D + 13)y = 0$
 - $(D^2 - 6D + 13)^2 y = 0$
 - $(D^2 + 6D + 13)^2 y = 0$
11. For solving the differential equation $y'' - 3y' + 2y = \frac{e^{2x}}{1 + e^{2x}}$ by the method of variation of parameter we assume $y(x) = A(x)e^x + B(x)e^{2x}$ as its general solution. Then $A(x)$ and $B(x)$ satisfy one of the following
- $A'(x)e^x + B'(x)e^{2x} = 0$
 $A'(x)e^x + 2B'(x)e^{2x} = \frac{e^{2x}}{1 + e^{2x}}$
 - $A'(x)e^x + B'(x)e^{2x} = 0$
 $A(x)e^x + A'(x)e^x + 2B'(x)e^{2x} = \frac{e^{2x}}{1 + e^{2x}}$
 - $A'(x)e^x + B'(x)e^{2x} = 0$
 $2B(x)e^x + A'(x)e^x + 2B'(x)e^{2x} = \frac{e^{2x}}{1 + e^{2x}}$
 - $A'(x)e^{2x} + B'(x)e^x = 0$
 $A'(x)e^x + 2B'(x)e^{2x} = \frac{e^{2x}}{1 + e^{2x}}$
12. Consider the initial value problem $x^2 y'' - 6y = 0$, $y(1) = \alpha$, $y'(1) = 6$. If $y(x) \rightarrow 0$ as $x \rightarrow 0^+$, then α is equal to
- 2
 - 4
 - 6
 - 7
13. The minimum possible order of a homogeneous linear ordinary differential equation with real constant coefficients having $x^2 \sin(x)$ as a solution is equal to
- 2
 - 3
 - 6
 - 9
14. If $y_1(x) = x$ is a solution to the differential equation $(1 - x^2) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + 2y = 0$, then its general solution is
- $y(x) = c_1 x + c_2 (x \ln |1 + x^2| - 1)$
 - $y(x) = c_1 x + c_2 \left(\ln \left| \frac{1-x}{1+x} \right| + 1 \right)$
 - $y(x) = c_1 x + c_2 \left(\frac{x}{2} \ln |1 - x^2| + 1 \right)$
 - $y(x) = c_1 x + c_2 \left(\frac{x}{2} \ln \left| \frac{1+x}{1-x} \right| - 1 \right)$
15. Assume that all the zeros of the polynomial $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ have negative real parts. If $u(t)$ is any solution to the ordinary differential equation $a_n \frac{d^n u}{dt^n} + a_{n-1} \frac{d^{n-1} u}{dt^{n-1}} + \dots + a_1 \frac{du}{dt} + a_0 u = 0$, then $\lim_{t \rightarrow \infty} u(t)$ is equal to
- 0
 - 1
 - $n - 1$
 - ∞
16. Let $f(x)$ and $xf(x)$ be the particular solutions of a differential equation $y'' + R(x)y' + S(x)y = 0$. Then the solution of the differential equation $y'' + R(x)y' + S(x)y = f(x)$ is
- $y = \left(-\frac{x^2}{2} + \alpha x + \beta \right) f(x)$
 - $y = \left(\frac{x^2}{2} + \alpha x + \beta \right) f(x)$
 - $y = (-x^2 + \alpha x + \beta) f(x)$
 - $y = (x^3 + \alpha x + \beta) f(x)$
17. The maximum number of linearly independent solutions of the differential equation $\frac{d^4 y}{dx^4} = 0$, with the condition $y(0) = 1$, is
- 4
 - 3
 - 2
 - 1

18. Let $y(x)$ be the solution of the initial value problem $y''' - y'' + 4y' - 4y = 0$, $y(0) = y'(0) = 2$, $y''(0) = 0$ Then the value of $y\left(\frac{\pi}{2}\right)$ is

- (a.) $\frac{1}{5}\left(4e^{\frac{\pi}{2}} - 6\right)$
(b.) $\frac{1}{5}\left(6e^{\frac{\pi}{2}} - 4\right)$
(c.) $\frac{1}{5}\left(8e^{\frac{\pi}{2}} - 2\right)$
(d.) $\frac{1}{5}\left(8e^{\frac{\pi}{2}} + 2\right)$

19. Let $y(x)$ be the solution of the initial value problem $x^2 y'' + xy' + y = x$,

$y(1) = y'(1) = 1$ Then the value of $y\left(e^{\frac{\pi}{2}}\right)$ is

- (a.) $\frac{1}{2}\left(1 - e^{\frac{\pi}{2}}\right)$
(b.) $\frac{1}{2}\left(1 + e^{\frac{\pi}{2}}\right)$
(c.) $\frac{1}{2} + \frac{\pi}{4}$
(d.) $\frac{1}{2} - \frac{\pi}{4}$

20. If $D \equiv \frac{d}{dx}$ then the value of $\frac{1}{(xD+1)}(x^{-1})$ is

- (a.) $\log x$
(b.) $\frac{\log x}{x}$
(c.) $\frac{\log x}{x^2}$
(d.) $\frac{\log x}{x^3}$

21. If $y(x) = x$ is a solution of the differential equation $y'' - \left(\frac{2}{x^2} + \frac{1}{x}\right)(xy' - y) = 0$, $0 < x < \infty$, then its general solution is

- (a.) $(\alpha + \beta e^{-2x})x$
(b.) $(\alpha + \beta e^{2x})x$
(c.) $\alpha x + \beta e^x$
(d.) $(\alpha e^x + \beta)x$

22. One particular solution of $y''' - y'' - y' + y = -e^x$ is

- (a.) xe^{-x}
(b.) xe^x
(c.) $x^2 e^{-x}$
(d.) $x^2 e^x$

23. Suppose $y_p(x) = x \cos(2x)$ is a particular solution of $y'' + \alpha y = -4 \sin(2x)$. Then the constant α equals

- (a.) -4
(b.) -2
(c.) 2
(d.) 4

24. A particular solution of $x^2 \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} + \frac{y}{4} = \frac{1}{\sqrt{x}}$ is

- (a.) $\frac{1}{2\sqrt{x}}$
(b.) $\frac{\log x}{2\sqrt{x}}$
(c.) $\frac{(\log x)^2}{2\sqrt{x}}$
(d.) $\frac{(\log x)\sqrt{x}}{2}$

25. The set of linearly independent solutions of the differential equation $\frac{d^4 y}{dx^4} - \frac{d^2 y}{dx^2} = 0$ is

- (a.) $\{1, x, e^x, e^{-x}\}$
(b.) $\{1, x, e^{-x}, xe^{-x}\}$
(c.) $\{1, x, e^x, xe^x\}$
(d.) $\{1, x, e^x, xe^{-x}\}$

26. If $y = \varphi(x)$ is a particular solution of $y'' + (\sin x)y' + 2y = e^x$ and $y = \psi(x)$ is a particular solution of $y'' + (\sin x)y' + 2y = \cos 2x$, then a particular solution of $y'' + (\sin x)y' + 2y = e^x + 2\sin^2 x$, is given by
- (a.) $\varphi(x) - \psi(x) + \frac{1}{2}$
(b.) $\psi(x) - \varphi(x) + \frac{1}{2}$
(c.) $\varphi(x) - \psi(x) + 1$
(d.) $\psi(x) - \varphi(x) + 1$
27. All real solutions of the differential equation $y'' + 2ay' + by = \cos x$ (where a and b are real constants) are periodic if
- (a.) $a = 1$ and $b = 0$
(b.) $a = 0$ and $b = 1$
(c.) $a = 1$ and $b \neq 0$
(d.) $a = 0$ and $b \neq 1$
28. The differential equation whose linearly independent solutions are $\cos 2x, \sin 2x$ and e^{-x} is
- (a.) $(D^3 + D^2 + 4D + 4)y = 0$
(b.) $(D^3 - D^2 + 4D - 4)y = 0$
(c.) $(D^3 + D^2 - 4D - 4)y = 0$
(d.) $(D^3 - D^2 - 4D + 4)y = 0$
29. Every solution of $y'' + ay' + by = 0$, where a and b are constants, approximate to zero as $x \rightarrow \infty$ provided
- (a.) $a > 0, b > 0$
(b.) $a > 0, b < 0$
(c.) $a < 0, b < 0$
(d.) $a < 0, b > 0$
30. Which of the following pair of functions is not a linearly independent pair of solutions of $y'' + 9y = 0$?
- (a.) $\sin 3x, \sin 3x - \cos 3x$
(b.) $\sin 3x + \cos 3x, 3\sin x - 4\sin^3 x$
(c.) $\sin 3x, \sin 3x \cos 3x$
(d.) $\sin 3x + \cos 3x, 4\cos^3 x - 3\cos x$
31. A nontrivial solutions of $x^2 y'' + xy' + 4y = 0, x > 0$ are
- (a.) Bounded and non periodic
(b.) Unbounded and non-periodic
(c.) Bounded and periodic
(d.) Unbounded and periodic
32. The solution $y(x)$ of the differential equation $\frac{d^2 y}{dx^2} + 4\frac{dy}{dx} + 4y = 0$ satisfying the conditions $y(0) = 4, \frac{dy}{dx}(0) = 8$ is
- (a.) $4e^{2x}$
(b.) $(16x + 4)e^{-2x}$
(c.) $4e^{-2x} + 16x$
(d.) $4e^{-2x} + 16xe^{2x}$
33. If $\frac{(c_1 + c_2 \ln x)}{x}$ is the general solution of the differential equation $x^2 \frac{d^2 y}{dx^2} + kx \frac{dy}{dx} + y = 0, x > 0$ then k equals
- (a.) 3
(b.) -3
(c.) 2
(d.) -1
34. The general solution of $x^2 \frac{d^2 y}{dx^2} - 5x \frac{dy}{dx} + 9y = 0$ is
- (a.) $(c_1 + c_2 x)e^{3x}$
(b.) $(c_1 + c_2 \ln x)x^3$
(c.) $(c_1 + c_2 x)x^3$
(d.) $(c_1 + c_2 \ln x)e^{x^3}$
(Here c_1 and c_2 are arbitrary constants.)

35. The solution of the first order ODE $xy' = xy + x + y + 1$ is
(a.) $y = Cx(e^x - 1)$
(b.) $y = (Cxe^x) - 1$
(c.) $y = (Ce^x) - x$
(d.) $y = (Ce^x) - x - 1$
(C is a constant)
36. The general solution of the first-order ODE $xy' + x^2y - y = 0$ is
(a.) $y(x) = xe^{-x^2/2} + C$
(b.) $y(x) = e^{x^2/2}(x + C)$
(c.) $y(x) = e^{x^2/2} - Cx$
(d.) $y(x) = Cxe^{-x^2/2}$
(C is a constant)
37. The general solution of the first order ODE $xy' + 2x^2y - xe^{-x^2} = 0$ is
(a.) $y(x) = e^{x^2}(x + C)$
(b.) $y(x) = e^{-x^2}(x + C)$
(c.) $y(x) = xe^{x^2} + C$
(d.) $y(x) = x + C$
(C is a constant)
38. Consider the second ordinary differential equation $y'' + by' + cy = 0$, where b, c are real constants. Given that $y = \exp(2x)$ is a solution. Then,
(a.) $b^2 + 4c < 0$
(b.) $b^2 + 4c \geq 0$
(c.) $b^2 - 4c < 0$
(d.) $b^2 - 4c \geq 0$
39. Consider the second order ordinary differential equation $y'' + 3y' + 2y = 0$. The $\lim_{t \rightarrow \infty} y(t)$ is
(a.) a non zero finite number
(b.) 0
(c.) $-\infty$
(d.) ∞
40. Consider the ordinary differential equation $y'' + 4y = \sin 2t, y(0) = 0$. Then the solution $y(t)$
(a.) converges to 0 as $t \rightarrow \infty$ with no oscillations
(b.) converges to 0 as $t \rightarrow \infty$ and the solution is oscillating
(c.) is oscillating and bounded
(d.) is unbounded
41. The number of independent solution of the differential equation $y^{(4)} - 2y^{(2)} + y = 0$ (here $y^{(2)}$ and $y^{(4)}$ present the second and fourth derivatives of y respectively) is
(a.) 4
(b.) 3
(c.) 2
(d.) 1
42. Let ϕ be a solution of the ODE $x^2y' + 2xy = 1$ on $0 < x < \infty$. Then the limit of $\phi(x)$ as $x \rightarrow \infty$
(a.) Is zero
(b.) Is one
(c.) Is ∞
(d.) Does not exist.
43. Let ϕ be the solution of $y' + iy = x$ such that $\phi(0) = 2$. Then $\phi(\pi)$ equals
(a.) $i\pi$
(b.) $-i\pi$
(c.) π
(d.) $-\pi$
44. Let $y(x)$ be a solution to the ordinary differential equation $\frac{d^2y}{dx^2} + A\frac{dy}{dx} + By = 0$ with constants $A \neq 0, B \neq 0$ and $y(0) \neq 0$. Then
(a.) $y(x)$ is always bounded on $\mathbb{R}$
(b.) $y(x)$ is always unbounded on $\mathbb{R}$
(c.) $y(x)$ is bounded or unbounded on $\mathbb{R}$ depending on the value of $y'(0)$
(d.) Such a solution cannot exist.
45. Consider the second order ordinary differential equation $\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$, where b and c are real constants. If $y = xe^{-5x}$ is a solution, then
(a.) Both b and c are positive
(b.) b is positive but c is negative
(c.) b is negative but c is positive
(d.) Both b and c are negative.
46. The solution of the ordinary differential equation $\frac{dy}{dx} = y, y(0) = 0$
(a.) Is unbounded
(b.) Is positive
(c.) Is negative
(d.) Is zero

ASSIGNMENT SHEET -05

- The initial value problem $y' = 2\sqrt{y}$, $y(0) = a$, has
 - a unique solution if $a < 0$
 - no solution if $a > 0$
 - infinitely many solutions if $a = 0$
 - a unique solution if $a \geq 0$
- For the initial value problem $\frac{dy}{dx} = y^2 + \cos^2 x$, $x > 0$, $y(0) = 0$. The largest interval of existence of the solution predicted by Picard's theorem is
 - $[0, 1]$
 - $[0, 1/2]$
 - $[0, 1/3]$
 - $[0, 1/4]$
- Let $y: \mathbb{R} \rightarrow \mathbb{R}$ be differentiable and satisfy the ODE: $\frac{dy}{dx} = f(y)$, $x \in \mathbb{R}$ Where $f: \mathbb{R} \rightarrow \mathbb{R}$ is a Lipschitz continuous function. Then $y(0) = y(1) = 0$
 - $y(x) = 0$ if and only if $x \in \{0, 1\}$
 - y is bounded
 - y is strictly increasing
 - $\frac{dy}{dx}$ is unbounded
- Consider the initial value problem (I.V.P.) $\frac{dy}{dx} = y^2$, $y(0) = 1$, $(x, y) \in \mathbb{R} \times \mathbb{R}$. Then there exists a unique solution of the I.V.P. on
 - $(-\infty, \infty)$
 - $(-\infty, 1)$
 - $(-2, 2)$
 - $(-1, \infty)$
- Consider the initial value problem (I.V.P.) $\frac{dy}{dx} = xy^{1/3}$, $y(0) = 0$, $(x, y) \in \mathbb{R} \times \mathbb{R}$. Then which of the following is/are correct?
 - The function $f(x, y) = xy^{1/3}$ does not satisfy the Lipschitz's condition with respect to y in any neighbourhood of $y = 0$
 - There exists a unique solution for the I.V.P.
 - There exists no solution for the I.V.P.
 - There exists more than one solution for the I.V.P.
- The differential equation $\frac{dy}{dx} = 60(y^2)^{1/5}$; $x > 0$, $y(0) = 0$ has
 - A unique solution
 - Two solutions
 - No solution
 - Infinite number of solutions.
- The initial value problem $\frac{dy}{dx} = y^2$, $y(0) = 1$ has
 - A unique solution defined on $(-\infty, 1)$
 - A unique solution defined on $(-1, \infty)$
 - A unique solution defined on $(-\infty, \infty)$
 - A solution on $(-\infty, 1)$ but not unique.
- The largest interval on which the solution of the initial value problem $\frac{dx}{dt} = x^3$ is defined, is
 - $t > \frac{1}{2}$
 - $t < \frac{1}{2}$
 - $-\frac{1}{2} < t < \frac{1}{2}$
 - $-1 < t < 0$
- Consider the differential equation $\frac{dy}{dx} = y^2$, $(x, y) \in \mathbb{R} \times \mathbb{R}$. Then
 - All solutions of the differential equation are defined on $(-\infty, \infty)$
 - No solution of the differential equation is defined on $(-\infty, \infty)$
 - The solution of the differential equation satisfying the initial condition $y(x_0) = y_0$, $y_0 > 0$ is defined on $\left(-\infty, x_0 + \frac{1}{y_0}\right)$
 - The solution of the differential equation satisfying the initial condition $y(x_0) = y_0$, $y_0 > 0$, is defined on $\left(-\infty, x_0 - \frac{1}{y_0}\right)$

10. The initial value problem $2x \frac{dy}{dx} = 3(2y-1)$, $y(0) = \frac{1}{2}$ has
(a.) A unique solution
(b.) More than one but only finitely many solutions
(c.) Infinitely many solutions
(d.) No solution
11. The initial value problem $x \frac{dy}{dx} = y + x^2$, $x > 0$; $y(0) = 0$ has
(a.) Infinitely many solutions
(b.) Exactly two solutions
(c.) A unique solution
(d.) No solution
12. If $y(x)$ is the solution of the differential equation $\frac{dy}{dx} = 2(1+y)\sqrt{y}$ satisfying $y(\pi/2) = 1$, then the largest interval (to the right of origin) on which the solution exists is
(a.) $[0, 3\pi/4)$
(b.) $[0, \pi)$
(c.) $[0, 2\pi)$
(d.) $[0, 2\pi/3)$
13. In a sufficiently small neighbourhood around $(x=2)$ the differential equation $\frac{dy}{dx} = \frac{y}{\sqrt{x}}$, $y(2) = 4$ has
(a.) No solution
(b.) A unique solution
(c.) Exactly two solutions
(d.) Infinitely many solutions
14. For the initial value problem (I.V.P.): $y' = f(x, y)$ with $y(0) = 0$ which of the following statements is true
(a.) $f(x, y) = \sqrt{xy}$ satisfies Lipschitz's condition and so I.V.P. has unique solution
(b.) $f(x, y) = \sqrt{xy}$ does not satisfy Lipschitz's condition and so I.V.P. has no solution
(c.) $f(x, y) = |y|$ satisfies Lipschitz's condition and so I.V.P. has unique solution
(d.) $f(x, y) = |y|$ does not satisfy Lipschitz's condition still I.V.P. has unique solution.
15. The function $f(x, y) = |x|y$ satisfies the Lipschitz's condition for (where a & b are fixed constants)
(a.) $-\infty < x < \infty$, $a < y < b$
(b.) $-\infty < x < \infty$, $-\infty < y < \infty$
(c.) $a < x < b$, $-\infty < y < \infty$
(d.) None of these
16. I.V.P. $y' = (3x^2 + 1)\cos^2 y + (x^3 - 2x)\sin 2y$, $y(0) = 0$ on $R = \{(x, y) \mid |x| < \alpha, |y| < \infty, \alpha > 0\}$ has
(a.) Unique solution
(b.) Infinitely many solutions
(c.) Finitely many solutions
(d.) No solution.
17. Consider the I.V.P. $dy = (1 + y^2)dx$; $y(0) = 0$. Then the global interval is
(a.) $-\infty < x < \infty$
(b.) $0 < x \leq \frac{\pi}{2}$
(c.) $-\frac{\pi}{2} < x < \frac{\pi}{2}$
(d.) $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$
18. Largest interval around $(0,0)$ of I.V.P. $y'(x) = x^2 + y^2$, $y(0) = 0$; $R = \{(x, y) \mid |x| \leq 1, |y| \leq 1\}$ by using Picard's existence and uniqueness theorem in which solution exist uniquely is
(a.) $|x| < \frac{1}{4}$
(b.) $|x| < \frac{1}{2}$
(c.) $|x| < \frac{3}{4}$
(d.) $|x| < 1$
19. The initial value problem (I.V.P.) $\frac{dy}{dx} = y^{1/3}$, $y(x_0) = y_0$ has a unique solution in the neighborhood if the point (x_0, y_0)
(a.) Is an arbitrary point in the xy -plane
(b.) Is taken as $(0, 0)$
(c.) Is different from $(0, 0)$

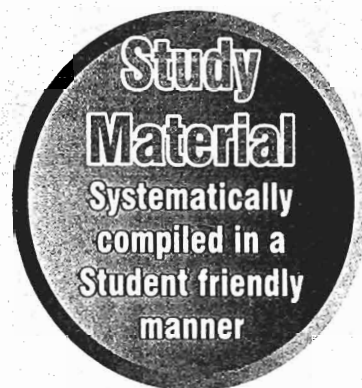
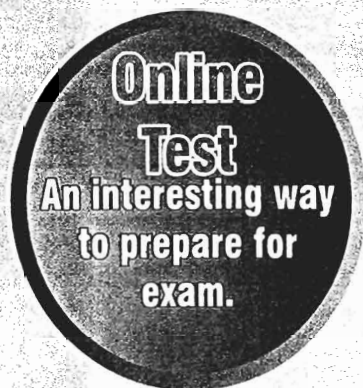
- (d.) Does not lie on the x -axis.
20. For I.V.P. $xy' = 3y, y(1) = 1$
 $R: |x-1| \leq 2, |y-1| \leq 4$.
 (a.) Existence theorem is satisfied hence unique solution exists.
 (b.) Lipschitz's condition is satisfied hence unique solution exists in some region of x
 (c.) Existence theorem & Lipschitz's condition is not satisfied but the problem has unique solution.
 (d.) None of these.
21. The solution of the differential equation $\frac{dy}{dx} = 1 + y^2, y(0) = 1$ exists on the interval
 (a.) $|x| < \frac{\pi}{2}$
 (b.) $-\frac{\pi}{2} < x < \pi$
 (c.) $-\frac{3\pi}{4} < t < \frac{\pi}{4}$
 (d.) $|x| < \pi$
22. The solution to the initial value problem $xy' + 2y = \sin x, y\left(\frac{\pi}{2}\right) = 1$ satisfies
 (a.) $1 + x^2y + x \cos x - \sin x = 0$
 (b.) $-1 + \sin x + x \cos x + x^2y = 0$
 (c.) $x^2y + x \cos x + \sin x - \frac{\pi^2}{4} = 0$
 (d.) $1 + x^2y + x \cos x - \sin x - \frac{\pi^2}{4} = 0$
23. The initial value problem $y' = \sqrt{y}, y(1) = 0$ in a suitable rectangle R has
 (a.) Unique solution
 (b.) Two solutions
 (c.) Infinite number of solutions
 (d.) None of these



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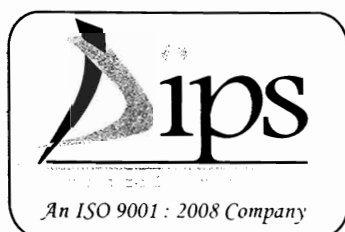
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